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From Distributional Similarity to Causal Imbalance: NHOP, Circular Oversampling, and a Controlled Degradation Study

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Abstract

This thesis presents three interlinked contributions forming a methodological pipeline for class imbalance in binary classification.

We introduce the Normalised Histogram Overlap Percentage (NHOP), a marginal distributional similarity measure whose central result is the identity $\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^{(j)}, Q^{(j)})$, placing it on total-variation-distance foundations. We prove boundedness, symmetry, strict affine invariance, and asymptotic consistency ($B = o(\min(n, m)^{1/3})$), and define a permutation test with provable Type I error control.

Building on NHOP, we propose a four-stage geometry-preserving seed-selection pipeline (K-means clustering, PCA projection, composite NHOP + Jensen–Shannon Divergence + Average Geometric–Topological Preservation (AGTP) + smoothness scoring) that raises mean seed NHOP from 0.772 to 0.907 and downstream F1 by $+0.025$ over random-seed baselines.

We then develop three circle-based oversampling algorithms that replace SMOTE (Synthetic Minority Over-sampling Technique) line-segment generation with disk generation in PCA-projected space: GVM-CO (Gravity-biased Von Mises Circular Oversampling), which biases angular sampling toward minority cluster gravity centers using a Von Mises distribution; LRE-CO (Local Region Estimation), which enforces Voronoi-cell certainty constraints; and LS-CO (Layered Segmental), which applies concentric-ring angular segmentation. Evaluated on 14 KEEL/UCI benchmark datasets, LRE-CO achieves the best mean F1 rank (6.14/13) across all methods.

Finally, a controlled degradation-and-recovery causal study confirms that class imbalance *per se* degrades performance monotonically, with a non-linear threshold at imbalance ratio IR ≈ 2. No oversampler achieves full recovery; GVM-CO and K-Means SMOTE reach 98.4% of balanced AUC, with the residual gap explained by an information-theoretic floor derived from the NHOP–TV identity.

Keywords: class imbalance, oversampling, distributional similarity, histogram overlap, total variation, circular generation, Von Mises distribution, causal analysis,

degradation study, seed selection.

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Part I

Foundations

Chapter 1

Introduction

1.1 Motivation and Research Programme

Class imbalance — when one class in a binary dataset is vastly underrepresented — degrades classifier performance precisely in the domains where correctness matters most: fraud detection [23], medical diagnosis [67], intrusion detection [80], and industrial defect detection [82]. When the majority class dominates the empirical risk, the classifier learns a boundary biased toward the majority at the expense of the minority [45, 49].

The most widely adopted remedy is oversampling. SMOTE [19] generates synthetic minority samples by linearly interpolating between a seed and one of its k-nearest neighbors:

$$\boldsymbol{x}_{\text{new}} = \boldsymbol{x}_i + \lambda \cdot (\boldsymbol{x}_j - \boldsymbol{x}_i), \quad \lambda \sim \mathcal{U}(0,1). \tag{1.1}$$

Despite over 10,000 citations and dozens of variants [31], SMOTE confines all synthetic samples to a *one-dimensional line segment*, ignoring the 2-D neighbourhood structure of each minority pair. This leads to linear artefacts, class-overlap amplification near boundaries, and noise sensitivity [6, 42].

Three open problems drive this thesis:

1. **No rigorous distributional fidelity measure.** Existing measures (KL, JSD, MMD, Wasserstein) each lack at least one of: scale invariance, bounded interpretability, marginal decomposability, exact connection to total variation distance, or computational efficiency.

2. **No principled seed selection.** Neither SMOTE nor any variant uses a formally motivated criterion for which minority instances should serve as seeds.

3. **The causal question is unresolved.** Every oversampling paper assumes imbalance degrades performance, but no controlled study has isolated the effect of imbalance itself from distributional shift caused by naive minority removal.

Part II addresses problem 1 by introducing NHOP (Normalised Histogram Overlap Percentage), which satisfies all five desiderata and proves the identity $\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^{(j)}, Q^{(j)})$. Parts III–IV address problem 2 with a NHOP-guided seed selection pipeline and three circular oversampling algorithms that replace SMOTE's line segment with a disk. Part V addresses problem 3 with a controlled degradation-and-recovery study using NHOP-guided minority removal to isolate pure imbalance from distributional shift. The same NHOP measure thus unifies all three works.

1.2 Research Questions

This thesis addresses eight research questions spanning the three works:

RQ1 Does an overlap measure satisfying all five desiderata (scale invariance, bounded interpretability, marginal decomposability, exact TV identity, computational efficiency) exist, and what is its mathematical theory?

RQ2 Can a geometry-preserving seed selection criterion, based on NHOP and complementary metrics, improve the quality of oversampled minority data?

RQ3 Can circle-based generation outperform linear interpolation for oversampling imbalanced datasets?

RQ4 Does gravity-biased angular sampling via Von Mises distributions improve synthetic sample quality?

RQ5 Does the choice of seed selection strategy impact the quality of oversampled data?

RQ6 Does progressive class imbalance monotonically degrade classifier performance, or is there a threshold effect?

RQ7 Can oversampling methods fully recover balanced-dataset performance after degradation?

RQ8 Is performance loss caused by imbalance *per se*, or primarily by information loss from a smaller minority sample?

1.3 Contributions

The specific contributions of this thesis, labeled C1–C7, are as follows:

C1. The NHOP measure (Part II): formal definition, four-step algorithm, five mathematical properties, the fundamental TV identity $\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^{(j)}, Q^{(j)})$, scale and translation invariance theorem, statistical consistency theorem, variance bounds, and a Type-I-error-controlled permutation test.

C2. Geometry-preserving seed selection (Part III): a composite criterion combining NHOP (marginal fidelity), JSD (distributional divergence), AGTP (geometric-topological preservation), and Z (smoothness regulariser); a four-stage pipeline with cluster-proportional stratified sampling; and a PCA ablation study.

C3. GVM-CO (Part IV): gravity scoring framework (Spread, Area, Density, Sparsity, Gravity score g_c, Gravity center $\boldsymbol{g}_c$), Von Mises angular biasing with concentration $\kappa = \kappa_{\max} \cdot s_c^{\gamma}$, cross-cluster and full-dataset clustering variants.

C4. LRE-CO (Part IV): Voronoi-constrained generation, certainty threshold, rejection sampling within the disk-Voronoi intersection, gravity-weighted cluster allocation.

C5. LS-CO (Part IV): layered segmental generation with concentric rings and angular segments, generic and cluster-based variants.

C6. Controlled degradation-recovery experimental protocol (Part V): NHOP-guided greedy minority removal preserving distributional shape; leakage-safe four-rule cross-validation protocol preventing the three main forms of data leakage in oversampling studies [18, 52].

C7. Empirical causal evidence for class imbalance (Part V): characterization of the degradation signature across six metrics, five classifiers, and ten datasets; comparative recovery analysis of seven oversampling methods; evidence for a non-linear threshold at IR ≈ 2; and quantification of the irreducible information-loss gap.

1.4 Thesis Organisation

Part I (Chapters 2–3) provides background and a literature review. **Part II** (Chapters 4–5) introduces NHOP: definition, TV identity, consistency, variance bounds,

and permutation test. **Part III** (Chapter 6) presents the seed selection pipeline. **Part IV** (Chapters 7–11) presents the three circular algorithms with results and ablation. **Part V** (Chapters 12–16) presents the causal study: protocol, degradation, recovery, and discussion. **Part VI** (Chapter 17) synthesises all contributions, answers RQ1–RQ8, and outlines future directions.

Chapter 2

Background

This chapter provides the theoretical foundations upon which the contributions of this thesis are built: the class imbalance problem and why standard classifiers fail under it, evaluation metrics for imbalanced data, SMOTE's geometric limitation, clustering methods, cross-validation, statistical hypothesis testing for multi-method comparison, the Von Mises distribution, and distributional similarity measures.

2.1 The Class Imbalance Problem

2.1.1 Definition and Quantification

Definition 2.1 (Imbalance Ratio)**.** The *imbalance ratio* (IR) of a binary dataset is defined as

$$\mathrm{IR} = \frac{n_{\mathrm{maj}}}{n_{\mathrm{min}}}, \tag{2.1}$$

where n_{maj} is the number of majority-class instances and n_{min} is the number of minority-class instances. An IR of 1.0 indicates perfect balance; larger values indicate increasing imbalance.

Datasets with $\mathrm{IR} < 3$ are considered mildly imbalanced, those with $3 \leq \mathrm{IR} < 9$ are moderately imbalanced, and those with $\mathrm{IR} \geq 9$ are highly imbalanced [32]. Some applications feature extreme imbalance with $\mathrm{IR} > 100$.

2.1.2 Why Standard Classifiers Fail

Standard algorithms minimize the empirical risk:

$$\hat{R}(h) = \frac{1}{n} \sum_{i=1}^{n} \ell(h(\boldsymbol{x}_i), y_i), \tag{2.2}$$

where ℓ is the loss function. When classes are imbalanced, the majority class contributes disproportionately. A classifier that always predicts majority achieves $\mathrm{IR}/(1+\mathrm{IR})$ accuracy while having zero sensitivity for the minority [49].

This problem manifests specifically across classifier families:

- **Decision trees:** Information gain and Gini impurity are dominated by the majority class [21].
- **SVMs:** The separating hyperplane is pushed toward the minority region [1].
- **Neural networks:** The gradient of the cross-entropy loss is dominated by majority examples [15].
- ***k*-Nearest Neighbours:** The neighbourhood of a minority instance is likely to contain a majority of majority points [86].

2.1.3 Real-World Manifestations

Table 2.1 illustrates the prevalence of class imbalance across application domains.

Table 2.1: Examples of class imbalance in real-world applications.

Domain	Task	Typical IR
Finance	Credit card fraud detection	500:1–1000:1
Medicine	Rare disease diagnosis	50:1–200:1
Manufacturing	Defect detection	10:1–100:1
Cybersecurity	Intrusion detection	100:1–10,000:1
Oil & Gas	Oil spill detection	20:1–500:1
Text mining	Spam detection	5:1–20:1
Bioinformatics	Protein function prediction	10:1–100:1

2.2 Evaluation Metrics for Imbalanced Data

Standard accuracy is misleading when classes are imbalanced; a majority-only predictor achieves $\mathrm{IR}/(1+\mathrm{IR})$ accuracy at zero minority-class recall. Table 2.2

summarises the six metrics used throughout this thesis; all are computed from the entries of the binary confusion matrix (TP, FP, FN, TN).

Table 2.2: Evaluation metrics for imbalanced classification used in this thesis.

Metric	Formula	Key property for imbalanced data
Sensitivity (Recall)	$\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$	Measures minority-class coverage; collapses to 0 under naive majority predictor
Specificity	$\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$	Measures majority-class accuracy
F_1	$2\mathrm{TP}/(2\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$	Harmonic mean of precision and recall [78]
G-Mean	$\sqrt{\mathrm{Sens.} \times \mathrm{Spec.}}$	Zero if either class is entirely missed [57]
Balanced Accuracy	$(\mathrm{Sens.}+\mathrm{Spec.})/2$	Trivial predictor achieves 0.5 [14]
AUC	$\int_0^1 \mathrm{TPR}\, d(\mathrm{FPR})$	Threshold-free; equals $P(\hat{s}^+ > \hat{s}^-)$ [30]

2.3 SMOTE and Its Limitations

SMOTE [19] generates synthetic samples by linearly interpolating between a minority seed $\boldsymbol{x}_i$ and a randomly chosen k-NN $\boldsymbol{x}_j$:

$$\boldsymbol{x}_{\text{new}} = \boldsymbol{x}_i + \lambda(\boldsymbol{x}_j - \boldsymbol{x}_i), \quad \lambda \sim \mathcal{U}(0,1). \tag{2.3}$$

This places all synthetic samples on line segments in the minority class, generating a 1-D submanifold of the d-dimensional feature space. The geometric limitation motivates the circular generation framework of Part IV.

2.4 Clustering and Cross-Validation

Two clustering algorithms are used in the seed selection and algorithm components of this thesis. **K-means** [64] with k-means++ initialization [4] partitions the minority class into K convex clusters by minimizing total within-cluster variance; its spherical cluster assumption aligns well with the circular geometry of our generation regions. **HAC** (hierarchical agglomerative clustering with Ward's linkage [70]) is used as a comparison baseline in the ablation of Chapter 11; it captures non-spherical shapes but produces unbalanced clusters on small minority sets.

For evaluation, stratified k-fold cross-validation [53] is used throughout to preserve class ratios per fold. When evaluating oversampling, synthesis must

be applied *inside the training partition of each fold only*; applying it before splitting introduces data leakage and produces optimistically biased estimates [18, 52]. The four-rule leakage-safe protocol used in Part V is described in Section 13.5.

2.5 Statistical Hypothesis Testing for Method Comparison

2.5.1 Friedman Test

The Friedman test [33] is a non-parametric test for comparing K algorithms across N datasets. Let r_i^j denote the rank of algorithm j on dataset i. The Friedman statistic is:

$$F_F = \frac{12N}{K(K+1)} \left[\sum_j \bar{r}_j^2 - \frac{K(K+1)^2}{4} \right], \tag{2.4}$$

where $\bar{r}_j$ is the average rank of algorithm j. Under the null hypothesis that all algorithms are equivalent, F_F follows a χ^2 distribution with $K-1$ degrees of freedom.

2.5.2 Holm's Procedure for Post-Hoc Comparisons

When the Friedman test rejects the null, Holm's step-down procedure [47] controls the family-wise error rate for pairwise comparisons between a control method and all others. Pairwise comparisons use the Wilcoxon signed-rank test [85].

2.5.3 Nemenyi Critical Difference Diagrams

Nemenyi critical difference (CD) diagrams [24] provide a visual summary. The critical difference for a test at significance level α with K algorithms and N datasets is:

$$\mathrm{CD} = q_\alpha \sqrt{\frac{K(K+1)}{6N}}, \tag{2.5}$$

where q_α is the critical value from the studentised range distribution. Algorithms whose average ranks differ by less than CD are connected by a horizontal bar in the diagram, indicating no statistically significant difference.

2.6 The Von Mises Distribution

The Von Mises distribution is the natural analogue of the Gaussian distribution on the unit circle. A random angle $\theta \sim \mathrm{VM}(\mu, \kappa)$ has probability density function:

$$p(\theta) = \frac{\exp(\kappa \cos(\theta - \mu))}{2\pi I_0(\kappa)}, \tag{2.6}$$

where $\mu \in [0, 2\pi)$ is the mean direction, $\kappa \geq 0$ is the concentration parameter, and $I_0(\kappa) = \frac{1}{2\pi} \int_0^{2\pi} e^{\kappa \cos\theta}\, d\theta$ is the modified Bessel function of order zero.

When $\kappa = 0$, the distribution is uniform on $[0, 2\pi)$; as $\kappa \to \infty$, the distribution concentrates around μ. The Von Mises distribution is used in GVM-CO (Chapter 7) to bias angular sampling toward the gravity center of each minority cluster, with κ computed from the cluster's gravity score [65].

2.7 Distributional Similarity Measures

This section surveys the measures most closely related to NHOP, which will be formally defined in Part II. Understanding the landscape of existing measures motivates NHOP's specific design choices.

2.7.1 Histogram Intersection

Histogram intersection was introduced by Swain and Ballard [79] for colour-based image retrieval. Given two histograms h and h' with the same bins, the intersection is $\mathrm{HI}(h, h') = \sum_b \min(h_b, h'_b)$. When histograms record raw counts, the result depends on sample sizes n and m unless explicitly normalised. Swain and Ballard normalise by $\min(\sum h_b, \sum h'_b)$, which introduces asymmetry when $n \neq m$. NHOP avoids this by converting to proportions before taking minima, yielding a symmetric, sample-size-invariant score.

2.7.2 Bhattacharyya Coefficient

The Bhattacharyya coefficient [8] is $\mathrm{BC}(P, Q) = \sum_b \sqrt{p_b q_b} \in [0, 1]$. It is symmetric and bounded but uses the geometric mean rather than the minimum, making it less sensitive to partial mismatches. The relationship $\mathrm{BC} \leq 1 - \mathrm{TV}^2/2$ holds, but BC does not equal $1 - \mathrm{TV}$ in general.

2.7.3 Maximum Mean Discrepancy (MMD)

MMD [36] is an integral probability metric associated with a reproducing kernel Hilbert space. It requires choice of a kernel and bandwidth and is quadratic in sample size ($O(n^2)$), limiting scalability. MMD lacks a direct percentage interpretation.

2.7.4 Jensen–Shannon Divergence (JSD)

JSD [61] symmetrises KL divergence via the mixture $M = (P + Q)/2$:

$$\mathrm{JS}(P, Q) = \frac{1}{2}\mathrm{KL}(P\|M) + \frac{1}{2}\mathrm{KL}(Q\|M) \in [0, \log 2]. \tag{2.7}$$

JSD requires smoothing for empty bins and lacks a direct percentage interpretation. It is used as a component of the seed selection criterion in Chapter 6.

2.7.5 Comparison Table

Table 2.3 summarises the five desiderata for each measure.

Table 2.3: Comparison of distributional similarity measures against the five desiderata for the finite-sample, multi-dimensional setting. ✓ = satisfied, ≈ = approximately satisfied, × = not satisfied.

Measure	Scale-inv.	Bounded	Marginal	TV identity	Efficient
Histogram Intersection	×	✓	✓	×	✓
Bhattacharyya	×	✓	✓	×	✓
KL Divergence	×	×	✓	×	✓
JSD	×	✓	✓	×	✓
MMD	×	×	×	×	×
Wasserstein-1	×	×	×	×	×
KS Statistic	×	✓	✓	×	✓
NHOP (this thesis)	✓	✓	✓	✓	✓

NHOP fills all five gaps simultaneously through joint min-max normalisation combined with the histogram minimum overlap. The full mathematical theory is developed in Chapters 4 and 5.

2.8 Chapter Summary

This chapter has provided the theoretical foundations for the three works presented in this thesis: the class imbalance problem and why standard classifiers fail,

evaluation metrics for imbalanced data (Table 2.2), SMOTE and its geometric limitations, clustering and cross-validation, statistical testing (Friedman, Nemenyi), the Von Mises distribution, and distributional similarity measures.

The research gap identified in this chapter — the absence of a scale-invariant, bounded, marginally decomposable measure with an exact TV identity — motivates Part II. The geometric limitation of linear interpolation motivates Part IV. The absence of controlled causal evidence for the imbalance hypothesis motivates Part V.

Chapter 3

Literature Review

This chapter surveys the body of research on oversampling methods for imbalanced classification. We organize the review into four categories: SMOTE variants, clustering-based oversampling methods, distribution-aware approaches, and seed/sample selection strategies. We conclude with a summary of the research gap that motivates the work presented in this thesis.

3.1 SMOTE Variants

Since its introduction in 2002 [19], SMOTE has inspired a large family of variants that address its limitations by modifying (1) which instances are used as seeds, (2) which neighbors are selected for interpolation, or (3) how the interpolation is performed.

3.1.1 Borderline-SMOTE

Han et al. [42] observed that synthetic samples are most useful near the decision boundary. Borderline-SMOTE identifies minority instances in the borderline region (those whose m nearest neighbors include a majority of majority-class points) and generates synthetic samples only from these "danger-zone" instances. Two variants are proposed:

- **Borderline-SMOTE1:** Generates from danger-zone minority instances using minority-class k-NN for interpolation.
- **Borderline-SMOTE2:** Also uses majority-class neighbors, placing synthetic samples closer to the boundary.

However, the method inherits SMOTE's linear interpolation limitation and can amplify class overlap when the boundary is complex.

3.1.2 Safe-Level SMOTE

Bunkhumpornpat et al. [16] introduced the concept of a "safe level" for each minority instance:

$$\text{safe_level}(\boldsymbol{x}_i) = \frac{|\{j : y_j = y_{\min},\ \boldsymbol{x}_j \in \text{KNN}(\boldsymbol{x}_i)\}|}{k}. \tag{3.1}$$

Synthetic samples are generated closer to the safer endpoint, ensuring placement in minority-dominated regions.

3.1.3 ADASYN

The Adaptive Synthetic sampling approach (ADASYN) by He et al. [44] generates more samples from harder-to-classify minority instances, with the number of synthetic samples proportional to the local majority density:

$$\hat{r}_i = \frac{|\{j : y_j \neq y_{\min},\ \boldsymbol{x}_j \in \text{KNN}(\boldsymbol{x}_i)\}|}{k}. \tag{3.2}$$

This focuses oversampling on challenging instances but can generate from noisy seeds.

3.1.4 SVM-SMOTE

Nguyen et al. [71] proposed SVM-SMOTE, which uses a trained SVM to identify minority support vectors as seeds, offering a margin-based criterion for borderline instance selection.

3.1.5 Other Notable Variants

- **SMOTE-IPF** [75]: Applies ensemble-based noise filtering after SMOTE.
- **LN-SMOTE** [63]: Uses local neighbourhood information to adjust the interpolation range.
- **MWMOTE** [5]: Uses weighted minority instances and cluster-based generation.
- **LoRAS** [7]: Uses manifold-based oversampling.

3.2 Clustering-Based Oversampling

Clustering captures the internal structure of the minority class, enabling more targeted synthetic sample generation.

3.2.1 K-Means SMOTE

Douzas et al. [28] proposed K-Means SMOTE:

1. Apply K-means to the full dataset.
2. Identify clusters with a high proportion of minority instances.
3. Apply SMOTE within each selected cluster proportionally to cluster minority density.

By restricting SMOTE to minority-enriched clusters, K-Means SMOTE avoids generating in majority-dominated regions, reducing class overlap.

3.2.2 Geometric SMOTE

Douzas and Bacao [27] proposed Geometric SMOTE, which replaces SMOTE's linear interpolation with sampling from a geometric region (hyper-sphere or truncated hyper-sphere) defined by a minority instance and a reference point. Geometric SMOTE is the most directly comparable prior work to the circular oversampling framework proposed in this thesis: both replace linear generation with geometric region sampling. The key distinctions are: (1) Geometric SMOTE operates in the original d-dimensional feature space using a single hyper-sphere per point pair, whereas the proposed methods operate in a 2-D PCA projection enabling circular geometry and Von Mises directional biasing; (2) the proposed LRE-CO adds an explicit certainty threshold and Voronoi-constrained local generation, providing stronger boundary guarantees.

Known limitation: A direct experimental comparison with Geometric SMOTE is absent from this thesis. All performance claims are therefore restricted to the baselines evaluated in Chapter 8, and no claim of superiority over Geometric SMOTE is made.

Several additional clustering-based variants have been proposed: CBR [**jo2004class**] oversamples separately within each cluster; HCC Cluster-SMOTE [3] uses a minority-class dendrogram and applies SMOTE within each sub-tree; KNSMOTE [60] adds neighbourhood-based filtering for dual-level noise control; and EC-EBO [20]

combines ensemble clustering with evolutionary optimization at significantly higher computational cost.

3.3 Distribution-Aware Oversampling

Distribution-aware methods model the minority-class distribution explicitly and sample from the estimated model.

3.3.1 Kernel Density Estimation (KDE) Approaches

KDE-based oversampling fits a kernel density estimate to the minority class and samples from it [37]. This produces samples following the estimated distribution rather than being confined to line segments. However, KDE suffers from the curse of dimensionality: bandwidth estimation becomes unreliable in high dimensions.

3.3.2 GAN-Based Oversampling

Generative Adversarial Networks have been applied through GAMO [69], BAGAN [66], and cGAN-based approaches [26]. While promising, GAN-based oversampling faces significant practical limitations: training instability, mode collapse, high computational cost, and unreliable class boundary respect with small datasets [77].

3.3.3 DBSMOTE

Bunkhumpornpat et al. [17] use DBSCAN to identify core minority regions, generating by interpolation toward cluster pseudo-centroids rather than potentially noisy boundary points.

3.4 Seed and Sample Selection Strategies

The quality of oversampled data depends critically on which minority instances serve as seeds.

3.4.1 Noise Filtering

Pre-filtering noisy instances before oversampling includes:

- **ENN-based filtering:** Remove minority instances whose neighbors are predominantly majority [6].

- **Ensemble-based filtering:** Remove consistently misclassified minority instances [75].
- **Tomek link removal:** Remove minority instances forming Tomek links with majority instances [81].

3.4.2 Instance Weighting

ADASYN [44] assigns higher weights to harder instances. MWMOTE [5] uses a sophisticated weighting considering both difficulty and informativeness.

3.4.3 Geometric Selection

The geometry-preserving seed selection presented in Chapter 6 differs from prior approaches by optimizing a composite criterion (NHOP + AGTP $- w_j \cdot$ JSD $- w_z \cdot Z$) that jointly accounts for distributional fidelity, structural preservation, and spatial regularity.

3.5 Geometric and Non-Linear Oversampling

3.5.1 Radial-Based Oversampling (RBO)

Koziarski and Woźniak [55] generate using radial basis functions centered at minority instances, producing smooth density-dependent oversampling that avoids nearest-neighbor discontinuities.

3.5.2 Manifold-Based Oversampling (LoRAS)

LoRAS [7] generates on locally-estimated affine subspaces, respecting local data geometry but requiring unreliable local dimensionality estimation with small samples.

3.6 Research Gap and Positioning

Despite the extensive literature, four gaps remain:

(1) **Linear interpolation dominance.** Most SMOTE variants retain line-segment generation. Only a few methods (LoRAS, RBO) explore higher-dimensional geometries, and none combine circular generation with directional bias.

(2) **Lack of directional control.** Methods generating in two or more dimensions use uniform or Gaussian sampling without directional control. The Von Mises distribution has not been applied to oversampling.

(3) **Limited local analysis.** While clustering-based methods partition globally, few perform fine-grained local analysis inside each generation region.

(4) **Seed selection is under-explored.** Most methods use all minority instances or apply simple filtering. A composite geometry-preserving criterion is absent from the literature.

Table 3.1 summarises the positioning of our methods.

Table 3.1: Positioning of the proposed methods relative to existing oversampling approaches.

Method	Generation Geometry	Directional Bias	Local Analysis	Seed Selection
SMOTE [19]	Line segment	No	No	All/random
Borderline-SMOTE [42]	Line segment	No	No	Borderline
ADASYN [44]	Line segment	No	No	Adaptive
SVM-SMOTE [71]	Line segment	No	No	SVM-based
K-Means SMOTE [28]	Line segment	No	Cluster-level	All
LoRAS [7]	Affine subspace	No	Local manifold	All
RBO [55]	Radial basis	No	Density-based	All
GVM-CO (Alg. 1)	Circle/disk	Von Mises	Cluster gravity	Geometry-preserving
LRE-CO (Alg. 2)	Circle/disk	No (uniform)	Voronoi sub-regions	Geometry-preserving
LS-CO (Alg. 3)	Circle/disk	Segmental	Layered rings	Geometry-preserving

Our methods are the first to combine circle-based generation geometry with directional/structural control mechanisms and a geometry-preserving seed selection strategy.

Bridge to Part II. The literature review reveals a gap in distributional similarity measurement: existing measures lack the combination of scale invariance,

bounded interpretability, marginal decomposability, and an exact connection to total variation that would be ideal for evaluating seed set representativeness. This gap is filled by NHOP, introduced in Part II. Having formally established NHOP in Part II, we then deploy it as the primary seed quality criterion in Part III and as the distributional-shape-preserving removal criterion in Part V.

Part II

Normalised Histogram Overlap Percentage (NHOP)

Chapter 4

NHOP: Definition and Mathematical Properties

No existing measure simultaneously satisfies all five desiderata for finite-sample distributional comparison: scale invariance, bounded interpretability, marginal decomposability, exact connection to total variation distance, and computational efficiency (see Table 2.3 in Chapter 2 for a full comparison). We propose the *Normalised Histogram Overlap Percentage* (NHOP) as a measure that satisfies all five. Given two finite point sets $\mathbf{X} \in \mathbb{R}^{n \times k}$ and $\mathbf{X}_s \in \mathbb{R}^{m \times k}$, the following sections give the formal definition, algorithm, and complete mathematical properties.

4.1 Formal Definition

Throughout, let $\mathbf{X} = (x_1, \ldots, x_n)^\top \in \mathbb{R}^{n \times k}$ and $\mathbf{X}_s = (y_1, \ldots, y_m)^\top \in \mathbb{R}^{m \times k}$ be two finite point sets. We write $\mathbf{X}^{(j)}$ for the j-th column of $\mathbf{X}$. Let $B \in \mathbb{N}$ with $B \geq 1$ be the number of histogram bins (default $B = 30$).

Definition 4.1 (NHOP)**.** For each dimension $j = 1, \ldots, k$:

1. **Combined range.** Compute

$$l_j = \min\big(\mathbf{X}^{(j)} \cup \mathbf{X}_s^{(j)}\big), \quad u_j = \max\big(\mathbf{X}^{(j)} \cup \mathbf{X}_s^{(j)}\big).$$

   (If $l_j = u_j$, set $\tilde{x}_i^{(j)} = 1/2$ for all i in both sets.)

2. **Joint normalisation.** For each $x_i^{(j)} \in \mathbf{X}^{(j)}$ and $y_i^{(j)} \in \mathbf{X}_s^{(j)}$:

$$\tilde{x}_i^{(j)} = \frac{x_i^{(j)} - l_j}{u_j - l_j}, \qquad \tilde{y}_i^{(j)} = \frac{y_i^{(j)} - l_j}{u_j - l_j}.$$

Algorithm 1: NHOP$(\mathbf{X}, \mathbf{X}_s, B)$

Input: $\mathbf{X} \in \mathbb{R}^{n \times k}$, $\mathbf{X}_s \in \mathbb{R}^{m \times k}$, $B \in \mathbb{N}$
Output: NHOP $\in [0, 1]$

```
S ← 0;
for j ← 1 to k do
    l_j ← min(X^(j) ∪ X_s^(j));
    u_j ← max(X^(j) ∪ X_s^(j));
    if u_j = l_j then
        NHOP_j ← 1;
    else
        x̃^(j) ← (X^(j) − l_j)/(u_j − l_j);
        ỹ^(j) ← (X_s^(j) − l_j)/(u_j − l_j);
        Compute p_b^(j), q_b^(j) for b = 1, ..., B via shared bins on [0, 1];
        NHOP_j ← Σ_{b=1}^{B} min(p_b^(j), q_b^(j));
    S ← S + NHOP_j;
return S/k;
```

3. **Normalised histograms.** Partition $[0, 1]$ into B equal bins $I_b = [(b-1)/B, b/B)$ for $b = 1, \ldots, B-1$ and $I_B = [(B-1)/B, 1]$. Define

$$p_b^{(j)} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\left[\tilde{x}_i^{(j)} \in I_b\right], \qquad q_b^{(j)} = \frac{1}{m} \sum_{i=1}^{m} \mathbf{1}\left[\tilde{y}_i^{(j)} \in I_b\right].$$

4. **Per-dimension overlap.**

$$\mathrm{NHOP}_j(\mathbf{X}, \mathbf{X}_s) = \sum_{b=1}^{B} \min\left(p_b^{(j)}, q_b^{(j)}\right).$$

5. **Overall score.**

$$\mathrm{NHOP}(\mathbf{X}, \mathbf{X}_s) = \frac{1}{k} \sum_{j=1}^{k} \mathrm{NHOP}_j(\mathbf{X}, \mathbf{X}_s).$$

Remark 4.1. The degenerate case $l_j = u_j$ arises when all $n + m$ points in dimension j are identical. Both sets agree on a single-point distribution, so $\mathrm{NHOP}_j = 1$ is the appropriate value. Mapping both to $1/2$ achieves this: both histograms concentrate all mass in the single bin containing $1/2$.

4.2 Algorithm

Algorithm 1 summarises the computation.

The time complexity is $O(k(n+m+B))$: for each of k dimensions, the histogram can be filled in $O(n+m)$ using direct index computation with uniform bins.

4.3 Interpretation

NHOP_j equals the total probability mass that $P^{(j)}$ and $Q^{(j)}$ place in the same bins simultaneously. A value of $\mathrm{NHOP} = 1$ means the two point sets have identical marginal histograms in every dimension: every bin proportion is the same. A value of $\mathrm{NHOP} = 0$ means that in at least one dimension, the two sets share no bin.

4.4 Mathematical Properties

Throughout this section, $P^{(j)} = (p_1^{(j)}, \ldots, p_B^{(j)})$ and $Q^{(j)} = (q_1^{(j)}, \ldots, q_B^{(j)})$ denote the per-dimension normalised histogram vectors, with $\sum_b p_b^{(j)} = \sum_b q_b^{(j)} = 1$ and $p_b^{(j)}, q_b^{(j)} \geq 0$ for all b.

Proposition 4.1 (Boundedness)**.** $\mathrm{NHOP}(\mathbf{X}, \mathbf{X}_s) \in [0,1]$ *for all finite non-empty point sets* $\mathbf{X}$ *and* $\mathbf{X}_s$*.*

Proof. Fix any dimension j. Since $p_b^{(j)}, q_b^{(j)} \geq 0$, we have $\min(p_b^{(j)}, q_b^{(j)}) \geq 0$, so $\mathrm{NHOP}_j \geq 0$. Moreover, $\min(p_b^{(j)}, q_b^{(j)}) \leq p_b^{(j)}$, so

$$\mathrm{NHOP}_j = \sum_{b=1}^{B} \min(p_b^{(j)}, q_b^{(j)}) \leq \sum_{b=1}^{B} p_b^{(j)} = 1.$$

Therefore $\mathrm{NHOP}_j \in [0,1]$, and the arithmetic mean $\mathrm{NHOP} = k^{-1} \sum_j \mathrm{NHOP}_j \in [0,1]$. □

Proposition 4.2 (Identity)**.** $\mathrm{NHOP}(\mathbf{X}, \mathbf{X}) = 1$ *for any non-empty point set* $\mathbf{X}$*.*

Proof. When $\mathbf{X}_s = \mathbf{X}$, the combined range is $[l_j, u_j]$ where $l_j = \min(\mathbf{X}^{(j)})$ and $u_j = \max(\mathbf{X}^{(j)})$. The normalised values of both sets are identical, hence $p_b^{(j)} = q_b^{(j)}$ for all b and all j, giving

$$\mathrm{NHOP}_j = \sum_{b=1}^{B} \min(p_b^{(j)}, p_b^{(j)}) = \sum_{b=1}^{B} p_b^{(j)} = 1.$$

□

Proposition 4.3 (Symmetry)**.** $\mathrm{NHOP}(\mathbf{X}, \mathbf{X}_s) = \mathrm{NHOP}(\mathbf{X}_s, \mathbf{X})$ *for all* $\mathbf{X}$, $\mathbf{X}_s$.

Proof. The combined range $[l_j, u_j]$ depends on $\mathbf{X}^{(j)} \cup \mathbf{X}_s^{(j)}$, which is symmetric in $\mathbf{X}$ and $\mathbf{X}_s$. Therefore the joint normalisation step produces the same $[0, 1]$ range regardless of order. Since $\min(p_b, q_b) = \min(q_b, p_b)$, we have $\mathrm{NHOP}_j(\mathbf{X}, \mathbf{X}_s) = \mathrm{NHOP}_j(\mathbf{X}_s, \mathbf{X})$ for every j. □

Proposition 4.4 (Monotonicity in overlap mass)**.** *Let* $\mathcal{A} = \{b : p_b^{(j)} \geq q_b^{(j)}\}$. *For two pairs* (P, Q) *and* (P', Q') *of normalised histogram distributions, if for every* $b \in \mathcal{A}$ *we have* $p'_b \leq p_b$ *and for every* $b \notin \mathcal{A}$ *we have* $q'_b \leq q_b$, *then* $\mathrm{NHOP}_j(P', Q') \leq \mathrm{NHOP}_j(P, Q)$.

Proof. For $b \in \mathcal{A}$: $\min(p_b, q_b) = q_b$. Under the given perturbation, $\min(p'_b, q'_b) \leq p'_b \leq p_b$. Summing:

$$\sum_{b \in \mathcal{A}} \min(p'_b, q'_b) + \sum_{b \notin \mathcal{A}} \min(p'_b, q'_b) \leq \sum_{b \in \mathcal{A}} q_b + \sum_{b \notin \mathcal{A}} p_b = \mathrm{NHOP}_j(P, Q). \qquad \square$$

We now state and prove the central theoretical result.

Theorem 4.5 (Fundamental identity: NHOP and total variation)**.** *Let* $P = (p_1, \ldots, p_B)$ *and* $Q = (q_1, \ldots, q_B)$ *be any two probability vectors satisfying* $p_b, q_b \geq 0$ *and* $\sum_{b=1}^{B} p_b = \sum_{b=1}^{B} q_b = 1$. *Then*

$$\sum_{b=1}^{B} \min(p_b, q_b) = 1 - \mathrm{TV}(P, Q),$$

where $\mathrm{TV}(P, Q) = \frac{1}{2} \sum_{b=1}^{B} |p_b - q_b|$. *Consequently,*

$$\mathrm{NHOP}_j(\mathbf{X}, \mathbf{X}_s) = 1 - \mathrm{TV}\big(P^{(j)}, Q^{(j)}\big)$$

for each dimension j.

Proof. For each bin b, note that

$$\min(p_b, q_b) + \max(p_b, q_b) = p_b + q_b. \tag{4.1}$$

Summing over $b = 1, \ldots, B$:

$$\sum_{b=1}^{B} \min(p_b, q_b) + \sum_{b=1}^{B} \max(p_b, q_b) = \sum_{b=1}^{B} p_b + \sum_{b=1}^{B} q_b = 2. \tag{4.2}$$

Also, for any two real numbers a, b: $\max(a, b) - \min(a, b) = |a - b|$. Summing:

$$\sum_{b=1}^{B} \max(p_b, q_b) - \sum_{b=1}^{B} \min(p_b, q_b) = \sum_{b=1}^{B} |p_b - q_b| = 2\ \mathrm{TV}(P, Q). \tag{4.3}$$

Let $A = \sum_b \min(p_b, q_b)$ and $C = \sum_b \max(p_b, q_b)$. Equations (4.2) and (4.3) give $A + C = 2$ and $C - A = 2\ \mathrm{TV}(P, Q)$. Solving: $A = 1 - \mathrm{TV}(P, Q)$. The second statement follows by applying this identity to $P^{(j)}$ and $Q^{(j)}$ for each j. □

Corollary 4.6 (NHOP as average complement of TV)**.**

$$\mathrm{NHOP}(\mathbf{X}, \mathbf{X}_s) = 1 - \frac{1}{k} \sum_{j=1}^{k} \mathrm{TV}\big(P^{(j)}, Q^{(j)}\big).$$

Proof. Average Theorem 4.5 over $j = 1, \ldots, k$. □

Theorem 4.7 (Scale and translation invariance)**.** *Let $T_j : \mathbb{R} \to \mathbb{R}$ be a strictly monotone increasing function for each $j = 1, \ldots, k$, and define $T(\mathbf{X}) \in \mathbb{R}^{n \times k}$ by $[T(\mathbf{X})]_{ij} = T_j(x_i^{(j)})$. For any componentwise affine transformation $T_j(v) = a_j v + c_j$ with $a_j > 0$:*

$$\mathrm{NHOP}\big(T(\mathbf{X}),\, T(\mathbf{X}_s)\big) = \mathrm{NHOP}(\mathbf{X}, \mathbf{X}_s).$$

For general strictly monotone (possibly non-affine) T_j, the identity holds in the limit $B \to \infty$.

Proof. Fix dimension j. For affine $T_j(v) = a_j v + c_j$ with $a_j > 0$:

$$\frac{T_j(x_i^{(j)}) - T_j(l_j)}{T_j(u_j) - T_j(l_j)} = \frac{a_j x_i^{(j)} + c_j - a_j l_j - c_j}{a_j(u_j - l_j)} = \frac{x_i^{(j)} - l_j}{u_j - l_j} = \tilde{x}_i^{(j)}.$$

The normalised values are identical, so the histograms $P^{(j)}$ and $Q^{(j)}$ are unchanged, and NHOP_j is unchanged. Since this holds for every j, the overall NHOP is unchanged.

For general strictly monotone T_j, individual bin assignments may differ, but in the limit $B \to \infty$ (where the overlap converges to the continuous OVL), the invariance follows from the fact that OVL is invariant to strictly monotone transformations [84]. □

Corollary 4.8 (Affine invariance)**.** *For any $a_j > 0$ and $c_j \in \mathbb{R}$, $j = 1, \ldots, k$:*

$$\mathrm{NHOP}\big(T(\mathbf{X}),\, T(\mathbf{X}_s)\big) = \mathrm{NHOP}(\mathbf{X}, \mathbf{X}_s).$$

This holds exactly for all $B \geq 1$.

Proposition 4.9 (Degenerate case: disjoint support)**.** *Suppose that for some dimension j, all values of $\mathbf{X}^{(j)}$ lie in $[l_j, l_j + (u_j - l_j)/2 - \epsilon]$ and all values of $\mathbf{X}_s^{(j)}$ lie in $[l_j + (u_j - l_j)/2 + \epsilon, u_j]$ for some $\epsilon > 0$. Then, for any $B \geq 2$ with $1/B < \epsilon/(u_j - l_j)$, we have* $\mathrm{NHOP}_j = 0$ *and* $\mathrm{TV}(P^{(j)}, Q^{(j)}) = 1$.

Proof. After joint normalisation, all values of $\mathbf{X}^{(j)}$ map to $[0, 1/2 - \epsilon']$ and all values of $\mathbf{X}_s^{(j)}$ map to $[1/2 + \epsilon', 1]$, where $\epsilon' = \epsilon/(u_j - l_j) > 0$. For B large enough that $1/B < \epsilon'$, the lower half-bins contain all mass of $P^{(j)}$ and zero mass of $Q^{(j)}$, and vice versa. Therefore $\min(p_b^{(j)}, q_b^{(j)}) = 0$ for all b, so $\mathrm{NHOP}_j = 0$ and by Theorem 4.5, $\mathrm{TV}(P^{(j)}, Q^{(j)}) = 1$. □

Proposition 4.10 (Subset inclusion)**.** *If $\mathbf{X}_s$ is obtained by uniformly sampling $m < n$ points from $\mathbf{X}$ without replacement, then $\mathbb{E}[\mathrm{NHOP}_j] < 1$ for all $B \geq 2$, and $\mathbb{E}[\mathrm{NHOP}_j] \to 1$ as $m \to n$.*

Proof. Let $N_b = \sum_{i=1}^{n} \mathbf{1}[x_i^{(j)} \in I_b]$ be the count of reference points in bin b, so $p_b = N_b/n$. The count of sampled points in bin b is $M_b \sim \mathrm{Hypergeometric}(n, N_b, m)$, with $\mathbb{E}[M_b] = mN_b/n$ and $q_b = M_b/m$. Since $\min(p_b, \cdot)$ is concave in the second argument for fixed p_b:

$$\mathbb{E}[\min(p_b, q_b)] \leq \min(p_b, \mathbb{E}[q_b]) = p_b.$$

The inequality is strict when q_b has positive variance, which holds for $m < n$ and $N_b \in (0, n)$. Hence $\mathbb{E}[\mathrm{NHOP}_j] < 1$ for $m < n$. As $m \to n$, the variance of q_b tends to 0, so $\mathbb{E}[\mathrm{NHOP}_j] \to 1$. □

Chapter 5

NHOP: Statistical Theory, Permutation Test, and Empirical Evaluation

This chapter develops the statistical theory for NHOP: the continuous limit as $B \to \infty$, consistency as $n, m \to \infty$, variance bounds, concentration inequalities, a permutation test with exact Type I error control, and empirical experiments confirming the theoretical properties.

5.1 Continuous Limit as $B \to \infty$

Theorem 5.1 (Convergence to the overlap coefficient)**.** *Fix* $\mathbf{X}$ *and* $\mathbf{X}_s$ *with* n *and* m *points respectively, and fix dimension* j*. Let* $\hat{f}_n^{(j)}$ *and* $\hat{g}_m^{(j)}$ *be the empirical step-function densities defined by* $\hat{f}_n^{(j)}(t) = p_b^{(j)} \cdot B$ *for* $t \in I_b$*. As* $B \to \infty$*:*

$$\mathrm{NHOP}_j^{(B)} \to \mathrm{OVL}\big(F_n^{(j)}, G_m^{(j)}\big) := \int_0^1 \min\big(\hat{f}_n^{(j)}(t), \hat{g}_m^{(j)}(t)\big)\, dt.$$

Proof. The per-dimension NHOP with B bins is

$$\mathrm{NHOP}_j^{(B)} = \sum_{b=1}^{B} \min(p_b^{(j)}, q_b^{(j)}) = \sum_{b=1}^{B} \frac{1}{B} \min(Bp_b^{(j)}, Bq_b^{(j)}).$$

This is a Riemann sum with step size $\Delta = 1/B$ approximating $\int_0^1 \min(\hat{f}_n^{(j)}(t), \hat{g}_m^{(j)}(t))\, dt$. With n and m fixed, the empirical densities are step functions that, as $B \to \infty$, converge in L^1 to the point-mass representation of the empirical distributions. By standard Riemann sum convergence for bounded functions, $\mathrm{NHOP}_j^{(B)} \to$

$\mathrm{OVL}(F_n^{(j)}, G_m^{(j)})$. □

5.2 Statistical Consistency

Theorem 5.2 (Consistency). *Let $x_1, x_2, \ldots$ be i.i.d. from $F^{(j)}$ with continuous density $f^{(j)}$ on $\mathbb{R}$, and $y_1, y_2, \ldots$ be i.i.d. from $G^{(j)}$ with continuous density $g^{(j)}$ on $\mathbb{R}$. Let $B = B(n,m)$ be a sequence satisfying $B \to \infty$ and $B(n,m) = o\big(\min(n,m)^{1/3}\big)$. Then, as $n, m \to \infty$:*

$$\mathrm{NHOP}_j^{(B)} \to \mathrm{OVL}\big(f^{(j)}, g^{(j)}\big) = \int_{-\infty}^{\infty} \min\big(f^{(j)}(x), g^{(j)}(x)\big)\, dx \quad \textit{almost surely.}$$

Proof. Write $\delta_n = \mathrm{NHOP}_j^{(B)} - \mathrm{OVL}(f^{(j)}, g^{(j)})$ and decompose:

$$|\delta_n| \le \big|\mathrm{NHOP}_j^{(B)} - \mathrm{OVL}(F_n^{(j)}, G_m^{(j)})\big| + \big|\mathrm{OVL}(F_n^{(j)}, G_m^{(j)}) - \mathrm{OVL}(f^{(j)}, g^{(j)})\big| =: T_1 + T_2. \tag{5.1}$$

Bounding T_1. By Theorem 5.1, $T_1 \to 0$ as $B \to \infty$ for any fixed empirical distributions. More precisely, $T_1 \le C/B$ for a constant C [76].

Bounding T_2. Using the Lipschitz property of the minimum ($|\min(a,b) - \min(c,d)| \le |a-c| + |b-d|$):

$$T_2 \le \int |\hat{f}_n^{(j)} - f^{(j)}|\, dx + \int |\hat{g}_m^{(j)} - g^{(j)}|\, dx.$$

The L^1 error of the histogram density estimator satisfies $\int |\hat{f}_n^{(j)} - f^{(j)}|\, dx = O(B/n + 1/B^2)$ [25, 76]. Under $B = o(\min(n,m)^{1/3})$, both terms tend to 0 almost surely by the Glivenko–Cantelli theorem and density estimation consistency [25, 38]. Combining $T_1 \to 0$ and $T_2 \to 0$ gives $|\delta_n| \to 0$ almost surely. □

Corollary 5.3 (Consistency under the null). *Under $H_0 : F^{(j)} = G^{(j)}$, as $n, m \to \infty$ with $B = o(\min(n,m)^{1/3})$:*

$$\mathrm{NHOP}_j^{(B)} \to 1 \quad \textit{almost surely.}$$

Proof. Under H_0, $\mathrm{OVL}(f^{(j)}, f^{(j)}) = \int f^{(j)}(x)\, dx = 1$. The result follows from Theorem 5.2. □

5.3 Variance Bounds

Theorem 5.4 (Variance bound). *Under $H_0 : F^{(j)} = G^{(j)} = F$ with $n = m$ and bin probabilities $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_B)$:*

$$\mathrm{Var}(\mathrm{NHOP}_j) \leq \frac{2}{n} \sum_{b=1}^{B} \pi_b(1 - \pi_b) \leq \frac{2}{n}.$$

This result shows that $\mathrm{Var}(\mathrm{NHOP}_j) = O(1/n)$, confirming that NHOP concentrates around its mean at the standard parametric rate.

Proposition 5.5 (McDiarmid concentration). *For any fixed $\mathbf{X}$, replacing one point y_i in $\mathbf{X}_s$ with any other point changes NHOP_j by at most $c = 2/m$. By McDiarmid's inequality [68]:*

$$P(|\,\mathrm{NHOP}_j - \mathbb{E}[\mathrm{NHOP}_j]| \geq t) \leq 2\exp\left(-\frac{t^2 m}{2}\right)$$

for all $t > 0$.

Proof. Replacing y_i changes the bin count of at most two bins each by $1/m$. The change in $\mathrm{NHOP}_j = \sum_b \min(p_b, q_b)$ is therefore at most $2 \cdot (1/m)$. Applying McDiarmid's inequality with $c_i = 2/m$ for each of the m coordinates of $\mathbf{X}_s$:

$$P(|\,\mathrm{NHOP}_j - \mathbb{E}[\mathrm{NHOP}_j]| \geq t) \leq 2\exp\left(-\frac{2t^2}{m \cdot (2/m)^2}\right) = 2\exp\left(-\frac{t^2 m}{2}\right). \quad \square$$

Proposition 5.6 (Optimal bin count). *Under the conditions of Theorem 5.2, taking $B = \lfloor \min(n, m)^{1/3} \rfloor$ achieves consistency while balancing the discretisation bias $O(1/B)$ against the estimation variance $O(B/\min(n, m))$. The mean squared error satisfies $\mathrm{MSE} = O(n^{-2/3}) \to 0$. In practice, results are numerically stable for $B \in [20, 50]$ across a wide range of sample sizes ($n, m \geq 30$).*

5.4 Sensitivity Analysis

Since NHOP is a consistent estimator of OVL (Theorem 5.2), its sensitivity mirrors that of OVL. For location shifts $\mathcal{N}(0, 1)$ vs. $\mathcal{N}(\mu, 1)$, $\mathrm{OVL} = 2\Phi(-|\mu|/2)$ [48], so NHOP decreases monotonically from 1 to 0 as $|\mu| \to \infty$ (e.g. 0.617 at $\mu = 1$, 0.317 at $\mu = 2$). For scale shifts $\mathcal{N}(0, 1)$ vs. $\mathcal{N}(0, \sigma^2)$, $\mathrm{OVL} = 2/(\sigma + 1)$; note that NHOP is invariant only when *both* sets are rescaled simultaneously (Theorem 4.7). For shape differences, NHOP accumulates evidence across all bins, giving more

consistent power than the max-based KS statistic against diffuse alternatives such as bimodal vs. Gaussian distributions with matched means and variances.

5.5 Comparison with Related Measures

Table 5.1 summarises NHOP against the measures reviewed in Chapter 2. The key distinctions are: NHOP equals $1 - \text{TV}$ exactly (Theorem 4.5), whereas Bhattacharyya ($\sum_b \sqrt{p_b q_b}$) does not; the unnormalised histogram intersection is asymmetric when $n \neq m$; Wasserstein-1 and MMD are unbounded and quadratic in sample size.

Table 5.1: Comparison of distributional similarity measures. Sym. = symmetric; S.I. = scale-invariant; Bd. = bounded in $[0, 1]$; MD = marginal decomposition; TV = exact TV identity; Complexity = per-dimension.

Measure	Sym.	S.I.	Bd.	MD	TV ident.	Cplx.
NHOP (this thesis)	✓	✓	✓	✓	✓	$O(n+m)$
Histogram Intersection [79]	No	No	✓	✓	×	$O(n+m)$
Bhattacharyya [8]	✓	No	✓	✓	×	$O(n+m)$
KL Divergence [58]	No	No	No	✓	×	$O(n+m)$
JSD [61]	✓	No	Partial	✓	×	$O(n+m)$
Wasserstein-1 [74]	✓	No	No	Partial	×	$O(n^2)$
MMD [36]	✓	No	No	No	×	$O(n^2)$
KS Statistic [54]	✓	✓	✓	✓	×	$O(n \log n)$

5.6 Permutation Test Based on NHOP

5.6.1 Definition

We define a two-sample test with $H_0 : F = G$ using NHOP as the test statistic. Under H_0, NHOP is close to 1; small observed NHOP constitutes evidence against H_0.

The $(1 + \cdot)/(1 + R)$ adjustment ensures the p-value is never exactly 0 and controls Type I error exactly [73].

Proposition 5.7 (Exact Type I error control)**.** *Under $H_0 : F = G$, for any significance level $\alpha \in (0, 1)$, the permutation test satisfies $P(\hat{p} \leq \alpha) \leq \alpha$ for all finite n, m, and all distributions F.*

Algorithm 2: NHOP Permutation Test

Input: $\mathbf{X} \in \mathbb{R}^{n\times k}$, $\mathbf{X}_s \in \mathbb{R}^{m\times k}$, $B \in \mathbb{N}$, $R \in \mathbb{N}$ (permutations)
Output: p-value

```
T_obs ← NHOP(X, X_s; B);
Z ← [X; X_s] ∈ R^{(n+m)×k};
for r ← 1 to R do
    Draw a uniform random permutation σ of {1, ..., n+m};
    Z_1 ← Z[σ(1 : n), :], Z_2 ← Z[σ(n+1 : n+m), :];
    T_r ← NHOP(Z_1, Z_2; B);
return p̂ = (1 + Σ_{r=1}^{R} 1[T_r ≤ T_obs])/(1 + R)
```

Proof. Under H_0, the combined sample $\mathbf{Z}$ consists of $n+m$ i.i.d. observations from F. By exchangeability, all $\binom{n+m}{n}$ partitions of $\mathbf{Z}$ are equally likely. The joint vector $(T_{\text{obs}}, T_1, \ldots, T_R)$ is exchangeable under H_0, so the rank of T_{obs} among all $R+1$ values is uniform on $\{1, \ldots, R+1\}$. The probability that $\hat{p} \le \alpha$ equals the probability that the rank of T_{obs} is at most $\lfloor \alpha(R+1) \rfloor$, which is $\lfloor \alpha(R+1) \rfloor / (R+1) \le \alpha$. □

Proposition 5.8 (Consistency under fixed alternatives)**.** *Under a fixed alternative* $H_1 : \text{OVL}(f^{(j)}, g^{(j)}) < 1$ *for some* j*, as* $n, m \to \infty$ *with* $B = o(\min(n,m)^{1/3})$*, the power of the permutation test converges to 1.*

Proof. Under H_1, by Theorem 5.2, $T_{\text{obs}} \to \text{OVL}(f, g) < 1$ almost surely. Under H_0 (which governs the permuted statistics), $T_r \to 1$ almost surely (Corollary 5.3). For large n, m: $T_{\text{obs}} < 1 - \epsilon$ for some $\epsilon > 0$, while all $T_r > 1 - \epsilon/2$. Hence $\hat{p} = 1/(R+1) \to 0$ and the power converges to 1. □

5.7 Empirical Evaluation

5.7.1 Sensitivity to Bin Count B

We generate $n = m = 500$ samples from $\mathcal{N}(0, I_3)$ and compute NHOP for $B \in \{5, 10, 20, 30, 50, 100\}$ over 1000 Monte Carlo replications.

Results are stable for $B \in [20, 50]$: the mean changes by less than 0.012 and the standard deviation remains near 0.010. These results confirm the recommendation $B \in [20, 50]$ from Proposition 5.6.

5.7.2 Discriminative Power

We compare NHOP, Bhattacharyya coefficient (BC), normalised histogram intersection (HI), and KS statistic ($1 - D_{n,m}$ for comparability) on three tasks: location

Table 5.2: NHOP (mean $\pm$ standard deviation) for varying B ($n = m = 500$, $k = 3$, same distribution, 1000 replications).

B	Mean	Std. Dev.
5	0.977	0.012
10	0.970	0.011
20	0.963	0.010
30	0.958	0.010
50	0.951	0.010
100	0.943	0.011

shifts, scale changes, and shape differences ($n = m = 200$, $k = 1$, 1000 replications).

Table 5.3: Mean measure values for location shift μ ($n = m = 200$, 1000 replications). KS column shows $1 - D_{n,m}$.

μ	NHOP	BC	HI	$1 - \text{KS}$
0.0	0.959	0.974	0.956	0.978
0.5	0.771	0.817	0.768	0.849
1.0	0.609	0.662	0.607	0.712
1.5	0.468	0.521	0.466	0.573
2.0	0.348	0.401	0.347	0.449

For **shape differences** (bimodal vs. Gaussian), NHOP and HI correctly detect the distributional mismatch, while BC underestimates it (since the geometric mean is less sensitive to partial mismatches). The KS statistic, being max-based, shows lower power against diffuse shape differences distributed across the support. These results confirm that NHOP provides superior or equal discriminative power compared to related measures across all three types of distributional difference.

Part III

Geometry-Preserving Seed Selection

Chapter 6

Geometry-Preserving Seed Selection

This chapter presents the geometry-preserving seed selection strategy, the first original contribution of the circular oversampling component of this thesis. The seed selection mechanism is a preprocessing step applied before any of the three proposed oversampling algorithms, ensuring that the minority instances used as generation seeds faithfully represent the distributional, geometric, and topological properties of the original minority class. NHOP — formally introduced and analyzed in Part II — is deployed here as the primary seed quality metric, directly applying the mathematical theory of Chapters 4–5.

6.1 Motivation: Why Seed Choice Matters

In oversampling, the "seed" is the minority instance from which synthetic samples are generated. In SMOTE and its variants, every minority instance is treated as an equally valid seed [19]. However, research has repeatedly shown that not all minority instances are equally suitable:

- **Noisy seeds.** Minority instances in majority-dominated regions generate synthetic samples in the overlap zone, systematically worsening classifier performance [6, 75].
- **Outlier seeds.** Minority instances far from the main cluster produce synthetic samples in empty feature-space regions, wasting the synthesis budget [42].
- **Redundant seeds.** Tightly clustered seeds produce near-duplicate synthetic samples, reducing distributional diversity [5, 19].
- **Unrepresentative seed sets.** A seed set that fails to cover the full minority distribution produces augmented data whose marginal distributions deviate

from the original, creating a train-test mismatch [32, 45].

The goal of seed selection is to choose a *subset* of minority instances that, when used as seeds, produces synthetic data maximally representative of the original minority class. We formalise this through four complementary scoring metrics: NHOP, JSD, AGTP, and a smoothness regulariser Z.

6.2 Seed Selection Pipeline

The seed selection pipeline consists of four stages.

6.2.1 Stage 1: Clustering the Minority Class

The minority class is clustered into K_{cluster} groups using K-means [4, 64]. Clustering ensures candidate seed sets cover all minority sub-populations and provides the stratification structure for Stage 3. This cluster-proportional stratification is the same principle used in K-Means SMOTE [28], which showed that cluster-aware generation produces higher-quality synthetic distributions.

6.2.2 Stage 2: PCA Projection

The clustered minority class is projected to a k_{pc}-dimensional PCA subspace (default $k_{pc} = 2$) [9, 51]. PCA removes correlation between features, making per-dimension histogram comparisons meaningful, and reduces the computational cost of scoring.

6.2.3 Stage 3: Generating Candidate Seed Sets

For each of $N_{\text{candidates}}$ iterations, a candidate seed set $\mathcal{I}$ of size m is generated by cluster-proportional stratified random sampling: from each cluster c, we draw $m_c = \lfloor (|C_c|/n) \cdot m \rceil$ seed indices without replacement.

6.2.4 Stage 4: Score and Select

Each candidate is evaluated using the composite criterion (Section 6.7), and the candidate with the highest score is returned.

Algorithm 3: Geometry-Preserving Seed Selection

Input: $\mathcal{S}_{\min}$: minority instances ($n \times d$); m: seeds to select; N_c: candidates; K_{cl}: clusters; k_{pc}: PCA components; k_t: topological k; w_j: JSD weight; w_z: smoothness weight

Output: $\mathcal{I}^*$: selected seed indices; s^*: best score

```
X_pca ← PCA(S_min, k_pc);
ℓ_1, ..., ℓ_n ← KMeans(X_pca, K_cl);
Compute proportional allocation: m_c ← ⌊(|C_c|/n) · m⌉ for each cluster c;
s* ← −∞;   I* ← ∅;
for t = 1 to N_c do
    for each cluster c do
        I_c ← randomly sample m_c indices from cluster c (without replacement);
    I ← ⋃_c I_c;   X_s ← X_pca[I];
    s ← NHOP(X_pca, X_s) + AGTP(X_pca, X_s) − w_j · JSD(X_pca, X_s) − w_z · Z(X_s);
    if s > s* then
        s* ← s;   I* ← I;
return I*, s*
```

6.3 NHOP for Marginal Similarity

NHOP (Definition 4.1, Chapter 4) is used here to measure how faithfully a candidate seed set's marginal distributions reproduce those of the full minority class. By Theorem 4.5, maximizing NHOP minimizes the average total variation distance between the marginal histograms of the seed set and the original minority class — a direct application of the measure-theoretic foundation established in Part II.

6.4 Jensen–Shannon Divergence Penalty

JSD [61] serves as a penalty measuring distributional disagreement, complementing NHOP's agreement measure:

$$\overline{\mathrm{JSD}}(\mathcal{X}, \mathcal{X}_s) = \frac{1}{k} \sum_{j=1}^{k} \mathrm{JSD}(P^{(j)}, Q^{(j)}). \tag{6.1}$$

Their combination prevents the degenerate case where matching broad support but different shapes yields a misleadingly high NHOP score.

6.5 AGTP for Geometric-Topological Preservation

AGTP evaluates the *joint* structure of the two sets by combining geometric similarity (mean and covariance) and topological similarity (k-NN distance distributions).

6.5.1 Geometric Similarity (G_{sim})

$$G_{\text{sim}}(\mathcal{X}, \mathcal{X}_s) = \frac{1}{2}\left[\max\left(0, 1 - \frac{\|\boldsymbol{\mu}_{\mathcal{X}} - \boldsymbol{\mu}_{\mathcal{X}_s}\|}{\text{Spread}(\mathcal{X}) + \varepsilon}\right) + \max\left(0, 1 - \frac{\|\mathbf{C}_{\mathcal{X}} - \mathbf{C}_{\mathcal{X}_s}\|_F}{\|\mathbf{C}_{\mathcal{X}}\|_F + \varepsilon}\right)\right], \tag{6.2}$$

where $\boldsymbol{\mu}$ is the mean vector, $\mathbf{C}$ is the covariance matrix, and $\text{Spread}(\mathcal{X}) = \frac{1}{|\mathcal{X}|}\sum_{\boldsymbol{x}} \|\boldsymbol{x} - \boldsymbol{\mu}_{\mathcal{X}}\|$. This differs from FID [46] in that AGTP operates on raw features and uses the class spread as the denominator for numerical stability.

6.5.2 Topological Similarity (T_{sim})

$$T_{\text{sim}}(\mathcal{X}, \mathcal{X}_s) = \sum_{b=1}^{B} \min(p_b^{(\text{kNN})}, q_b^{(\text{kNN})}), \tag{6.3}$$

where $p_b^{(\text{kNN})}$ and $q_b^{(\text{kNN})}$ are normalised histograms of k-NN distances within $\mathcal{X}$ and $\mathcal{X}_s$ respectively. This measures whether the seed set has the same local spacing structure as the original minority class.

6.5.3 Combined AGTP Score

$$\text{AGTP}(\mathcal{X}, \mathcal{X}_s) = \frac{1}{2}\left[G_{\text{sim}}(\mathcal{X}, \mathcal{X}_s) + T_{\text{sim}}(\mathcal{X}, \mathcal{X}_s)\right] \in [0, 1]. \tag{6.4}$$

AGTP is related in spirit to FID [46] and MMD [36], but differs by explicitly separating geometric and topological components, normalising each to $[0, 1]$, and adding k-NN structure comparison that neither FID nor MMD provides.

6.6 Smoothness Regulariser (Z)

The smoothness regulariser penalises irregular seed spacing:

$$Z(\mathcal{X}_s) = \frac{\sigma(\{d_1^{\text{NN}}, \ldots, d_m^{\text{NN}}\})}{\mu(\{d_1^{\text{NN}}, \ldots, d_m^{\text{NN}}\}) + \varepsilon}, \tag{6.5}$$

where d_i^{NN} is the distance from seed i to its nearest seed neighbor. Low Z is desirable because evenly spaced seeds ensure full minority distribution coverage,

while clustered seeds waste the synthesis budget on near-duplicate samples.

6.7 Composite Selection Criterion

The four metrics are combined into a single composite score:

$$\text{Score}(\mathcal{X}_s) = \underbrace{\text{NHOP}(\mathcal{X}, \mathcal{X}_s) + \text{AGTP}(\mathcal{X}, \mathcal{X}_s)}_{\text{representativeness}} - \underbrace{w_j \cdot \overline{\text{JSD}}(\mathcal{X}, \mathcal{X}_s)}_{\text{divergence penalty}} - \underbrace{w_z \cdot Z(\mathcal{X}_s)}_{\text{spacing penalty}}, \tag{6.6}$$

with default weights $w_j = 0.3$ and $w_z = 0.5$. The selected seed set is:

$$\mathcal{X}_s^* = \arg\max_{\mathcal{X}_s \in \mathcal{C}} \text{Score}(\mathcal{X}_s). \tag{6.7}$$

The criterion balances three objectives: (a) maximize distributional representativeness; (b) penalise distributional divergence; and (c) penalise irregular spacing.

Table 6.1: Seed selection hyperparameters and default values.

Parameter	Description	Default
$N_{\text{candidates}}$	Number of candidate seed sets	100
k_{pc}	PCA components for scoring	2
K_{cluster}	Clusters for stratified sampling	5
k_t	k for topological similarity	5
w_j	JSD penalty weight	0.3
w_z	Smoothness weight	0.5
B	Histogram bins for NHOP/JSD/T_{sim}	30

6.8 PCA Projection Ablation Study

We ran seed selection with $k_{pc} = 2$ (default PCA) and $k_{pc} = d$ (no PCA: full feature space) on all 14 benchmark datasets. Table 6.2 reports the results.

Table 6.2: Effect of PCA projection on seed selection quality, averaged across 14 datasets (100 candidate evaluations per dataset, 5-fold CV with RF classifier). NHOP, AGTP, F1, AUC are higher-is-better; JSD, Z are lower-is-better. Bold indicates the better value.

Setting	NHOP↑	AGTP↑	JSD↓	Z ↓	F1↑	AUC↑
No PCA ($k_{pc} = d$)	0.8142	**0.7772**	**0.0537**	**0.2959**	**0.8099**	**0.9474**
PCA ($k_{pc} = 2$)	**0.8190**	0.5657	0.0544	0.5224	0.7775	0.9433

PCA marginally improves NHOP (+0.005) because projecting to two uncorrelated principal components simplifies the marginal histogram comparison. However, PCA substantially reduces AGTP (−0.212) and increases Z (+0.227) because discarding all but two principal components destroys much of the joint geometric and topological structure. As a result, downstream F1 is 0.032 lower with PCA.

This finding motivates setting $k_{pc} = d$ (no PCA) as the default for AGTP and Z, or using $k_{pc} = \min(d, 5)$ to retain more structure without the full cost. A detailed per-dataset breakdown and sensitivity analysis is provided in Chapter 11.

6.9 Experimental Validation of Seed Selection

Table 6.3 reports the seed quality metrics for the selected seed sets compared to randomly generated seed sets, averaged across 14 benchmark datasets.

Table 6.3: Seed quality metrics: selected (geometry-preserving) vs. random (average of 100 random seed sets), averaged across 14 datasets.

Setting	**NHOP↑**	**AGTP↑**	**JSD↓**	$Z \downarrow$
Selected (geometry-preserving)	0.907	0.839	0.043	0.132
Random (average)	0.766	0.658	0.165	0.276
Δ	+0.141	+0.181	−0.122	−0.144

The geometry-preserving selection achieves substantially higher NHOP and AGTP and lower JSD and Z on every dataset. This translates to classifier performance improvements reported in Chapter 9.

Part IV

Circular Oversampling Methods

Chapter 7

Proposed Circular Oversampling Methods

This chapter presents the three proposed circular oversampling algorithms in full mathematical detail. We begin with the common preprocessing pipeline shared by all methods, then present each algorithm with its formulation, pseudocode, and design rationale.

7.1 Common Framework: Circle-Based Generation

All three algorithms share a common geometric framework based on four steps.

7.1.1 Step 1: PCA Projection

The minority class points $\{\boldsymbol{x}_i\}_{i=1}^{n_{\min}} \subset \mathbb{R}^d$ are projected to a 2-D subspace via PCA:

$$\hat{\boldsymbol{x}}_i = (\boldsymbol{x}_i - \bar{\boldsymbol{x}}) \cdot \mathbf{V}_2, \quad \hat{\boldsymbol{x}}_i \in \mathbb{R}^2, \tag{7.1}$$

where $\bar{\boldsymbol{x}}$ is the mean of the minority class and $\mathbf{V}_2 \in \mathbb{R}^{d \times 2}$ contains the first two principal component vectors. This projection enables circular geometry in the 2-D subspace. After generation, the inverse PCA transform (Equation (7.4)) maps the 2-D disk back to $\mathbb{R}^d$ as a hyper-ellipsoidal region, since the transform scales coordinates by the corresponding PCA eigenvalues. The circular shape is exact only in the projected subspace; in the original feature space the generation region is ellipsoidal, with aspect ratios determined by the ratio of PCA eigenvalues.

7.1.2 Step 2: k-NN Graph Construction

A k-nearest-neighbor graph is constructed on the projected minority points $\hat{\mathcal{X}}_{\text{min}} = \{\hat{\boldsymbol{x}}_i\}$. For each point $\hat{\boldsymbol{x}}_i$, the indices of its k nearest neighbors in $\hat{\mathcal{X}}_{\text{min}}$ are stored.

7.1.3 Step 3: Circle Formation

For each synthetic sample, a seed point $\hat{\boldsymbol{x}}_i$ and a neighbor $\hat{\boldsymbol{x}}_j$ are selected. The circle is defined by their midpoint and half-distance:

$$\boldsymbol{c}_{ij} = \frac{\hat{\boldsymbol{x}}_i + \hat{\boldsymbol{x}}_j}{2}, \tag{7.2}$$

$$R_{ij} = \frac{\|\hat{\boldsymbol{x}}_i - \hat{\boldsymbol{x}}_j\|}{2}. \tag{7.3}$$

The resulting circle $\mathcal{D}(\boldsymbol{c}_{ij}, R_{ij})$ is the closed disk centered at $\boldsymbol{c}_{ij}$ with radius R_{ij}.

7.1.4 Step 4: Inverse PCA Transform

After generation in 2-D PCA space, synthetic points are lifted back to $\mathbb{R}^d$:

$$\boldsymbol{x}_{\text{new}} = \hat{\boldsymbol{x}}_{\text{new}} \cdot \mathbf{V}_2^\top + \bar{\boldsymbol{x}}. \tag{7.4}$$

7.1.5 Native-Dimension Mode (`use_pca=False`)

As an alternative to the 2-D PCA projection described above, all three algorithms support a **native-dimension mode** controlled by the `use_pca=False` flag. In this mode, Steps 1 and 4 are omitted entirely: all geometric operations (circle formation, ball sampling, clustering, and the k-NN graph) are performed directly in the original d-dimensional feature space.

Motivation. The PCA projection concentrates the geometry into the two directions of maximum variance, retaining between 52% and 96% of minority-class variance across the 14 benchmark datasets (median: 74%). The remaining variance — up to 48% for the 34-feature ionosphere dataset — is discarded before any circle is formed, and synthetic points are subsequently lifted back via the inverse transform.

When `use_pca=False`, no information is discarded. The circle in Step 3 becomes

a d-dimensional hyperball:

$$\boldsymbol{c}_{ij} = \tfrac{1}{2}(\boldsymbol{x}_i + \boldsymbol{x}_j), \tag{7.5}$$

$$R_{ij} = \tfrac{1}{2}\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2. \tag{7.6}$$

The uniform sampling formula generalises to d-dimensional volume-uniform sampling:

$$r = R_{ij} \cdot U^{1/d}, \quad U \sim \mathcal{U}(0,1), \tag{7.7}$$

ensuring that the density of sampled points is uniform with respect to the d-dimensional volume element. Gravity direction bias (GVM-CO, Algorithm 1) generalises from Von Mises angular sampling to the von Mises–Fisher (vMF) distribution on $\mathcal{S}^{d-1}$:

$$\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \quad \hat{\boldsymbol{z}} = \frac{\boldsymbol{z} + \kappa\,\hat{\boldsymbol{\mu}}}{\|\boldsymbol{z} + \kappa\,\hat{\boldsymbol{\mu}}\|}, \tag{7.8}$$

where $\hat{\boldsymbol{\mu}}$ is the unit-vector direction from the circle center toward the cluster gravity center and κ is the same concentration score used in the 2-D mode.

Trade-off. The native-dimension mode avoids PCA information loss but introduces the *curse of dimensionality*: in high-dimensional feature spaces ($d \gtrsim 15$), the volume of a d-dimensional ball concentrates near its surface, so uniform ball sampling produces points that are all approximately at radius R. An empirical ablation comparing both modes is provided in Chapter 11.

The **Circular-SMOTE** baseline uses uniform disk sampling: $\theta \sim \mathcal{U}(0, 2\pi)$, $r = R_{ij}\sqrt{U}$ ($U \sim \mathcal{U}(0,1)$), $\hat{\boldsymbol{x}}_{\text{new}} = \boldsymbol{c}_{ij} + r(\cos\theta, \sin\theta)^\top$. The $\sqrt{U}$ radial scaling ensures area-uniform coverage (since the probability of a point at radius r must be proportional to r).

7.2 Gravity Scoring Framework

Algorithms 1 and 3 use a gravity scoring framework that quantifies the attractiveness of each minority cluster.

7.2.1 Cluster Metrics

After clustering the projected minority points into K groups $\{\mathcal{C}_1, \ldots, \mathcal{C}_K\}$, the following metrics are computed for each cluster $\mathcal{C}_c$ with n_c points:

Definition 7.1 (Cluster Spread).

$$\text{Spread}_c = \frac{1}{n_c} \sum_{i \in \mathcal{C}_c} \|\hat{\boldsymbol{x}}_i - \boldsymbol{\mu}_c\|, \tag{7.9}$$

where $\boldsymbol{\mu}_c = \frac{1}{n_c} \sum_{i \in \mathcal{C}_c} \hat{\boldsymbol{x}}_i$ is the cluster centroid.

Definition 7.2 (Cluster Area).

$$\text{Area}_c = \pi \cdot \text{Spread}_c^2 + \epsilon. \tag{7.10}$$

Definition 7.3 (Cluster Density).

$$\text{Density}_c = \frac{n_c}{\text{Area}_c}. \tag{7.11}$$

Definition 7.4 (Cluster Sparsity).

$$\text{Sparsity}_c = \frac{1}{n_c} \sum_{i \in \mathcal{C}_c} \frac{1}{k_{\text{nn}}} \sum_{j=1}^{k_{\text{nn}}} \|\hat{\boldsymbol{x}}_i - \hat{\boldsymbol{x}}_j^{(\text{nn})}\|, \tag{7.12}$$

where the inner sum is over the k_{nn} nearest neighbors of $\hat{\boldsymbol{x}}_i$ within cluster $\mathcal{C}_c$.

Definition 7.5 (Gravity Score).

$$g_c = \frac{\text{Density}_c}{\text{Spread}_c \cdot \text{Sparsity}_c + \epsilon}. \tag{7.13}$$

The numerator Density_c measures how densely populated the cluster is per unit area. The denominator $\text{Spread}_c \cdot \text{Sparsity}_c$ penalises clusters that are simultaneously spatially extended (large spread) *and* internally loose (high mean nearest-neighbor distance). A cluster that is compact and well-populated receives the highest gravity scores.

To ensure scale-invariance, the raw gravity scores are normalised within each dataset:

$$\tilde{g}_c = \frac{g_c - \min_{c'} g_{c'}}{\max_{c'} g_{c'} - \min_{c'} g_{c'} + \varepsilon}, \tag{7.14}$$

so that $\tilde{g}_c \in [0, 1]$ regardless of the absolute scale of the feature space.

7.2.2 Gravity Center

The gravity center is a density-weighted centroid pulled toward regions of high local density:

Definition 7.6 (Gravity Center)**.**

$$g_c = \frac{\sum_{i \in \mathcal{C}_c} w_i \cdot \hat{\boldsymbol{x}}_i}{\sum_{i \in \mathcal{C}_c} w_i}, \quad w_i = \frac{1}{\sigma_i^{\text{local}} + \epsilon}, \tag{7.15}$$

where σ_i^{local} is the mean k-NN distance of point $\hat{\boldsymbol{x}}_i$ within its cluster. Points in denser local regions have smaller σ_i^{local} and hence larger weights w_i, pulling the gravity center toward the densest part of the cluster.

7.3 Algorithm 1: Gravity-Biased Von Mises Circular Oversampling (GVM-CO)

GVM-CO generates synthetic samples inside circles formed between minority point pairs, with the angular distribution biased by a Von Mises distribution directed toward the gravity center of the seed's cluster.

7.3.1 Von Mises Direction

For a circle defined by seed $\hat{\boldsymbol{x}}_i$ (in cluster c) and neighbor $\hat{\boldsymbol{x}}_j$, with center $\boldsymbol{c}_{ij}$, the Von Mises mean direction is the angle from the circle center toward the gravity center:

$$\mu = \arctan 2\left(g_c^{(y)} - c_{ij}^{(y)},\ g_c^{(x)} - c_{ij}^{(x)}\right). \tag{7.16}$$

7.3.2 Von Mises Concentration

The concentration κ controls the strength of directional biasing and is computed from two sub-scores.

Distance score s_d. Normalised distance from the circle center to the gravity center (closer centers get higher scores):

$$s_d = \frac{d_{\max} - d(\boldsymbol{c}_{ij}, g_c)}{d_{\max} - d_{\min} + \epsilon}, \tag{7.17}$$

where $d_{\min}$ and $d_{\max}$ are the extreme distances across all circles in the current batch.

Gravity score s_g. Normalised gravity score of the seed's cluster:

$$s_g = \frac{g_c - g_{\min}}{g_{\max} - g_{\min} + \epsilon}. \tag{7.18}$$

Combined score. The two sub-scores are blended using an additive–multiplicative combination:

$$s_{\text{combined}} = \alpha \cdot \frac{s_d + s_g}{2} + (1 - \alpha) \cdot s_d \cdot s_g, \tag{7.19}$$

where $\alpha \in [0, 1]$ controls the balance between additive (robust) and multiplicative (selective) combination.

Concentration mapping.

$$\kappa = \kappa_{\max} \cdot s_{\text{combined}}^{\gamma}, \tag{7.20}$$

where $\kappa_{\max}$ is the maximum concentration and $\gamma > 0$ is a nonlinearity exponent. A minimum floor prevents complete deactivation of the bias:

$$\kappa_c = \max\left(\kappa_{\max} \cdot s_c^{\gamma},\ \varepsilon\right), \tag{7.21}$$

where $\varepsilon = 10^{-6}$.

7.3.3 Sampling Procedure

Each synthetic point is generated as:

$$\theta \sim \text{VonMises}(\mu, \kappa), \tag{7.22}$$

$$r = R_{ij} \cdot \sqrt{U}, \quad U \sim \mathcal{U}(0, 1), \tag{7.23}$$

$$\hat{\boldsymbol{x}}_{\text{new}} = \boldsymbol{c}_{ij} + r \cdot (\cos\theta,\ \sin\theta)^{\top}. \tag{7.24}$$

7.3.4 Cross-Cluster Border Restriction

An optional variant of GVM-CO (GVM-CO-CC) enforces that the neighbor $\hat{\boldsymbol{x}}_j$ comes from the nearest *other* cluster:

$$c^* = \arg\min_{c' \neq c} \|\boldsymbol{\mu}_c - \boldsymbol{\mu}_{c'}\|. \tag{7.25}$$

The k-NN search for neighbors of seeds in cluster c is restricted to points in cluster c^*. This promotes generation in inter-cluster border regions, improving connectivity between minority sub-populations. As shown in the experimental

results (Chapter 9), GVM-CO-CC is most useful at low imbalance ratios; at high imbalance the inter-cluster region is typically majority-dominated, making cross-cluster pairing counterproductive.

7.3.5 Full-Dataset Clustering Variant

Inspired by K-Means SMOTE [28], a third variant applies K-means to the *full dataset* $\mathcal{X} = \mathcal{X}_{\text{min}} \cup \mathcal{X}_{\text{maj}}$. The procedure:

1. Apply K-means with K clusters to the full projected dataset.

2. For each cluster c, compute the minority ratio:

$$\rho_c = \frac{|\{i : c_i = c,\ y_i = y_{\text{min}}\}|}{|\{i : c_i = c\}|}. \tag{7.26}$$

3. Select clusters where $\rho_c \geq \rho_{\text{min}}$ (default $\rho_{\text{min}} = \text{IR}^{-1}$).

4. Within each selected cluster, compute gravity scores and centers using only the minority points.

5. Apply GVM-CO generation within selected clusters, with synthetic samples allocated proportionally to the number of minority points per cluster.

The key advantage is that cluster boundaries are informed by the majority distribution, naturally avoiding generation in majority-dominated regions.

Algorithm 4: GVM-CO: Gravity-biased Von Mises Circular Oversampling

Input: Minority $\mathcal{X}_{\text{min}}$, K, k_{seed}, κ_{max}, γ, α
Output: Synthetic points $\mathcal{S}$

$\hat{\mathcal{X}}_{\text{min}} \leftarrow \text{PCA}(\mathcal{X}_{\text{min}}, p = 2)$;
$\{c_i\} \leftarrow \text{K-Means}(\hat{\mathcal{X}}_{\text{min}}, K)$;
Compute gravity scores $\{g_c\}$ and gravity centers $\{\boldsymbol{g}_c\}$ via Eqs. (7.13)–(7.15);
Build k_{seed}-NN graph on $\hat{\mathcal{X}}_{\text{min}}$;
$\mathcal{S} \leftarrow \emptyset$;
for $t = 1$ **to** n_{synth} **do**
  Select random seed $\hat{\boldsymbol{x}}_i$ in cluster c;
  Select random neighbor $\hat{\boldsymbol{x}}_j$ from k-NN of $\hat{\boldsymbol{x}}_i$;
  $\boldsymbol{c} \leftarrow \frac{1}{2}(\hat{\boldsymbol{x}}_i + \hat{\boldsymbol{x}}_j)$; $R \leftarrow \frac{1}{2}\|\hat{\boldsymbol{x}}_i - \hat{\boldsymbol{x}}_j\|$;
  $\mu \leftarrow \arctan 2(\boldsymbol{g}_c - \boldsymbol{c})$;
  Compute s_d, s_g, s_{combined}, κ via Eqs. (7.17)–(7.21);
  $\theta \sim \text{VonMises}(\mu, \kappa)$; $r \leftarrow R\sqrt{U}$, $U \sim \mathcal{U}(0, 1)$;
  $\hat{\boldsymbol{x}}_{\text{new}} \leftarrow \boldsymbol{c} + r(\cos\theta, \sin\theta)^\top$;
  $\mathcal{S} \leftarrow \mathcal{S} \cup \{\hat{\boldsymbol{x}}_{\text{new}}\}$;
Lift $\mathcal{S}$ to $\mathbb{R}^d$ via inverse PCA (Eq. (7.4));
return $\mathcal{S}$

7.4 Algorithm 2: Local Region Estimation Circular Oversampling (LRE-CO)

LRE-CO addresses cross-boundary contamination by restricting generation to the intersection of the circular disk with a Voronoi cell defined by the cluster partition.

7.4.1 Voronoi Partitioning

Given K cluster centroids $\{\boldsymbol{\mu}_c\}_{c=1}^{K}$, the 2-D space is partitioned into Voronoi cells:

$$\mathcal{V}_c = \left\{ \hat{\boldsymbol{x}} \in \mathbb{R}^2 \;\middle|\; \|\hat{\boldsymbol{x}} - \boldsymbol{\mu}_c\| \leq \|\hat{\boldsymbol{x}} - \boldsymbol{\mu}_{c'}\|,\ \forall c' \neq c \right\}. \tag{7.27}$$

Each point in $\mathbb{R}^2$ belongs to exactly one Voronoi cell.

7.4.2 Certainty Threshold

Before generating synthetic samples within a circle, LRE-CO evaluates whether the circle lies in a minority-dominated region. The certainty of a disk $\mathcal{D}(\boldsymbol{c}_{ij}, R_{ij})$ is

defined as:

$$\text{certainty}(\mathcal{D}) = \frac{n_{\text{min}}^{\text{inside}}}{n_{\text{min}}^{\text{inside}} + n_{\text{maj}}^{\text{inside}}}, \tag{7.28}$$

where $n_{\text{min}}^{\text{inside}}$ and $n_{\text{maj}}^{\text{inside}}$ are the number of minority and majority points inside the disk, respectively. If $\text{certainty}(\mathcal{D}) < \tau$ (default $\tau = 0.80$), the circle is rejected, preventing synthetic generation in majority-dominated regions.

Remark 7.1 (2-D Approximation)**.** The certainty metric is computed in the 2-D PCA subspace. For datasets where the first two principal components explain a small fraction of the total variance, a disk that appears minority-dominated in 2-D may straddle a class boundary in the remaining dimensions. The effectiveness of the certainty mechanism therefore diminishes as the intrinsic dimensionality of the minority distribution increases relative to the PCA subspace. Practitioners should verify the explained variance of the 2-D projection and consider using a stricter threshold τ for high-dimensional datasets (see Table 8.2 in Chapter 8).

7.4.3 Constrained Generation

For an eligible circle defined by seed $\hat{\boldsymbol{x}}_i$ (in cluster c) and neighbor $\hat{\boldsymbol{x}}_j$, LRE-CO restricts synthetic point generation to:

$$\hat{\boldsymbol{x}}_{\text{new}} \in \mathcal{D}(\boldsymbol{c}_{ij}, R_{ij}) \cap \mathcal{V}_c. \tag{7.29}$$

7.4.4 Rejection Sampling

Since the intersection $\mathcal{D}(\boldsymbol{c}_{ij}, R_{ij}) \cap \mathcal{V}_c$ does not have a simple closed-form shape, LRE-CO uses rejection sampling:

1. Generate a candidate point $\hat{\boldsymbol{x}}_{\text{cand}}$ uniformly inside the disk $\mathcal{D}(\boldsymbol{c}_{ij}, R_{ij})$.

2. Check whether $\hat{\boldsymbol{x}}_{\text{cand}} \in \mathcal{V}_c$ by verifying $\arg\min_{c'} \|\hat{\boldsymbol{x}}_{\text{cand}} - \boldsymbol{\mu}_{c'}\| = c$.

3. If accepted, add to the synthetic set. Otherwise reject and repeat.

The acceptance rate depends on the overlap between the disk and the Voronoi cell. For circles whose center is near the cluster centroid, the acceptance rate is typically high; a maximum iteration limit prevents infinite loops.

7.4.5 Allocation Strategy

Synthetic samples are allocated to clusters proportionally to cluster size weighted by normalised gravity score:

$$n_c^{\text{synth}} = n_{\text{synth}} \cdot \frac{n_c \cdot \tilde{g}_c}{\sum_{c'=1}^{K} n_{c'} \cdot \tilde{g}_{c'}}. \tag{7.30}$$

This assigns more synthetic samples to clusters that are both large and high-gravity. Users requiring uniform coverage across all minority sub-populations can set all $\tilde{g}_c = 1$, reducing the allocation to size-proportional.

Algorithm 5: LRE-CO: Local Region Estimation Circular Oversampling

Input: Minority $\mathcal{X}_{\text{min}}$, majority $\mathcal{X}_{\text{maj}}$, K, k_{seed}, κ_{max}, γ, α, τ
Output: Synthetic points $\mathcal{S}$

$\hat{\mathcal{X}}_{\text{min}} \leftarrow \text{PCA}(\mathcal{X}_{\text{min}}, p = 2)$;
$\{c_i\}, \{\boldsymbol{\mu}_c\} \leftarrow \text{K-Means}(\hat{\mathcal{X}}_{\text{min}}, K)$;
Compute gravity scores, gravity centers;
Compute Voronoi partition $\{\mathcal{V}_c\}_{c=1}^{K}$ via Eq. (7.27);
Allocate n_c^{synth} per cluster via Eq. (7.30);
$\mathcal{S} \leftarrow \emptyset$;
for *each cluster* $c = 1, \ldots, K$ **do**
  for $t = 1$ **to** n_c^{synth} **do**
    Select seed $\hat{\boldsymbol{x}}_i \in \mathcal{C}_c$, neighbor $\hat{\boldsymbol{x}}_j$;
    Form circle $(\boldsymbol{c}, R)$ via Eqs. (7.2)–(7.3);
    if certainty$(\mathcal{D}) < \tau$ *(Eq. (7.28))* **then**
      Skip circle and select new pair;
    repeat
      $\theta \sim \mathcal{U}(0, 2\pi)$; $r \leftarrow R\sqrt{U}$;
      $\hat{\boldsymbol{x}}_{\text{cand}} \leftarrow \boldsymbol{c} + r(\cos\theta, \sin\theta)^\top$;
    until $\hat{\boldsymbol{x}}_{cand} \in \mathcal{V}_c$ ***or*** *max iterations*;
    $\mathcal{S} \leftarrow \mathcal{S} \cup \{\hat{\boldsymbol{x}}_{\text{cand}}\}$;
Lift $\mathcal{S}$ to $\mathbb{R}^d$ via inverse PCA;
return $\mathcal{S}$

7.5 Algorithm 3: Layered Segmental Circular Oversampling (LS-CO)

LS-CO introduces a structured decomposition of the circular region into concentric annular layers and angular segments, with differentiated sampling densities.

7.5.1 Layer Decomposition

Each circle of radius R_{ij} is divided into L concentric annular layers with equal radial width $\Delta r = R_{ij}/L$:

$$\mathcal{A}_\ell = \left\{\hat{\boldsymbol{x}} \mid r_{\ell-1} \leq \|\hat{\boldsymbol{x}} - \boldsymbol{c}_{ij}\| \leq r_\ell\right\}, \quad r_\ell = \frac{\ell}{L} \cdot R_{ij}, \quad \ell = 1, \ldots, L. \tag{7.31}$$

The rings have equal radial width but unequal areas: the area of ring ℓ is $\pi(r_\ell^2 - r_{\ell-1}^2) = \pi R_{ij}^2(2\ell - 1)/L^2$, so the outermost ring is $(2L - 1)$ times larger than the innermost. Synthetic samples are allocated uniformly across layers by default, deliberately over-sampling the inner rings relative to their area, concentrating generation near the circle center where the seed minority instances are located and the minority class is most reliably represented.

7.5.2 Angular Segmentation

Each layer is divided into M angular segments:

$$\mathcal{S}_{\ell,m} = \mathcal{A}_\ell \cap \left\{\hat{\boldsymbol{x}} \mid \theta_{m-1} \leq \angle(\hat{\boldsymbol{x}} - \boldsymbol{c}_{ij}) < \theta_m\right\}, \quad \theta_m = \frac{2\pi m}{M}. \tag{7.32}$$

7.5.3 Layer Density Allocation

The number of synthetic samples in each layer is determined by a gravity-proximity weight:

$$w_\ell = \exp\left(-\beta \cdot \frac{|r_\ell^* - r_{\text{gc}}|}{R_{ij}}\right), \tag{7.33}$$

where $r_\ell^* = \frac{r_{\ell-1}+r_\ell}{2}$ is the midpoint radius and r_{gc} is the distance from the circle center to the projection of the gravity center onto the radial axis. The allocation per layer is:

$$n_\ell = n_{\text{per-circle}} \cdot \frac{w_\ell}{\sum_{\ell'} w_{\ell'}}. \tag{7.34}$$

7.5.4 Within-Segment Sampling

Within each segment $\mathcal{S}_{\ell,m}$, the angular distribution uses a Von Mises distribution with segment-local concentration:

$$\kappa_{\ell,m} = \kappa_{\max} \cdot \frac{L - \ell + 1}{L} \cdot \exp\left(-\frac{|\theta_m^{\text{mid}} - \mu|^2}{2\sigma_\theta^2} \right), \tag{7.35}$$

where $\theta_m^{\text{mid}} = \frac{\theta_{m-1}+\theta_m}{2}$ is the midpoint angle and μ is the gravity direction. Inner layers (smaller ℓ) and segments aligned with μ receive higher concentration.

The radial component within layer ℓ uses area-corrected uniform sampling:

$$r = \sqrt{r_{\ell-1}^2 + U \cdot (r_\ell^2 - r_{\ell-1}^2)}, \quad U \sim \mathcal{U}(0,1). \tag{7.36}$$

Algorithm 6: LS-CO: Layered Segmental Circular Oversampling

Input: Minority $\mathcal{X}_{\text{min}}$, K, k_{seed}, L, M, $\kappa_{\max}$, β, σ_θ
Output: Synthetic points $\mathcal{S}$

```
1  X̂_min ← PCA(X_min, p = 2);
2  {c_i} ← Cluster(X̂_min, K);
3  Compute gravity scores and centers via Eqs. (7.13)–(7.15);
4  Build k_seed-NN graph;
5  S ← ∅;
6  for each synthetic sample do
7      Select seed x̂_i and neighbor x̂_j;
8      Form circle (c, R) via Eqs. (7.2)–(7.3);
9      Compute gravity direction μ = arctan2(g_c − c);
10     Compute layer weights {w_ℓ}_{ℓ=1}^L (Eq. (7.33)) and allocations {n_ℓ} (Eq. (7.34));
11     for ℓ = 1 to L do
12         for each sample allocated to layer ℓ do
13             Select segment m with Von Mises-weighted probability;
14             κ_{ℓ,m} ← Eq. (7.35);
15             θ ∼ VonMises(μ, κ_{ℓ,m}) restricted to [θ_{m−1}, θ_m);
16             r ← area-corrected sample in [r_{ℓ−1}, r_ℓ] (Eq. (7.36));
17             x̂_new ← c + r(cos θ, sin θ)^⊤;
18             S ← S ∪ {x̂_new};
19 Lift S to R^d via inverse PCA;
20 return S
```

LS-CO has two primary variants: **LS-CO (G)** uses a single gravity center computed from all minority points, and **LS-CO (C)** uses per-cluster gravity cen-

ters computed after K-means clustering, providing finer control for multi-modal minority distributions.

7.6 Optional Post-Hoc Denoising

All three algorithms support optional denoising after synthetic sample generation:

- **Tomek Links**: Identifies pairs of instances from different classes that are mutual nearest neighbors and removes the majority instance (or both), cleaning the decision boundary.
- **Edited Nearest Neighbours (ENN)**: Removes any instance whose class label disagrees with the majority of its k-nearest neighbors. More aggressive than Tomek Links.

Denoising is applied to the *combined* dataset (original + synthetic) before classifier training. As shown in the ablation (Chapter 11), Tomek links provide a small net improvement for GVM-CO and LRE-CO, while ENN is counterproductive for these methods because the circular generation already incorporates boundary-avoidance mechanisms internally.

7.7 Comparison of the Three Algorithms

Table 7.1 summarises the key differences between the three algorithms.

Table 7.1: Comparison of the three proposed circular oversampling algorithms.

Feature	GVM-CO	LRE-CO	LS-CO
Angular distribution	Von Mises	Uniform (or Von Mises) + rejection	Von Mises per segment
Region constraint	None (full disk)	Voronoi cell	Layer + segment
Spatial control	Global (gravity direction)	Local (Voronoi boundary)	Structured (layers)
Cross-cluster variant	Yes (GVM-CO-CC)	N/A	N/A
Full-dataset variant	Yes (KM-SMOTE style)	N/A	N/A
Rejection sampling	No	Yes	No
Hyperparameters	$K, k, \kappa_{\max}, \gamma, \alpha$	$K, k, \kappa_{\max}, \gamma, \alpha, \tau$	$K, k, L, M, \kappa_{\max}, \beta, \sigma_\theta$
Complexity (generation)	$\mathcal{O}(n_{\text{synth}})$	$\mathcal{O}(n_{\text{synth}} \cdot K / p_{\text{accept}})$	$\mathcal{O}(n_{\text{synth}} \cdot L)$
Preprocessing	$\mathcal{O}(n_{\min} \log n_{\min})$ (shared: PCA + k-NN graph)		

7.8 Computational Complexity

The computational cost of each algorithm has five components:

1. **PCA**: $\mathcal{O}(n_{\text{min}} \cdot d^2)$ for the covariance matrix and eigendecomposition. Negligible for moderate d since n_{min} is small.

2. **k-NN graph**: $\mathcal{O}(n_{\text{min}}^2)$ naively, or $\mathcal{O}(n_{\text{min}} \log n_{\text{min}})$ with a KD-tree in the 2-D PCA space.

3. **Clustering**: $\mathcal{O}(n_{\text{min}} \cdot K \cdot T_{\text{iter}})$ for K-Means with T_{iter} iterations.

4. **Generation**: $\mathcal{O}(n_{\text{synth}})$ per sample for GVM-CO and LS-CO; $\mathcal{O}(n_{\text{synth}} / p_{\text{accept}})$ for LRE-CO, where p_{accept} is the average fraction of disk area inside the Voronoi cell.

Since n_{min} is typically small by definition, the dominant cost is usually the generation step. All three methods are comparable in computational cost to K-Means SMOTE and are substantially faster than GAN-based approaches.

7.9 Chapter Summary

This chapter has presented the three proposed circular oversampling algorithms. All three share a common framework based on PCA projection, k-NN graph construction, circle formation, and inverse PCA lifting. They differ in how they control the placement of synthetic samples within each circle:

1. **GVM-CO** combines clustering-based gravity scoring with Von Mises directional sampling, biasing generation toward dense minority regions while preserving diversity through the radial component.

2. **LRE-CO** restricts generation to the intersection of circular disks with Voronoi cells, preventing cross-boundary contamination and enabling fine-grained local generation.

3. **LS-CO** decomposes circular regions into layers and segments, with gravity-weighted density allocation and per-segment Von Mises concentration, providing structured control over sample placement.

The next chapter describes the experimental setup used to evaluate these algorithms.

Chapter 8

Experimental Setup

This chapter details the experimental methodology used to evaluate the proposed circular oversampling algorithms. We describe the benchmark datasets, classifiers, evaluation metrics, validation protocol, comparison methods, hyperparameter settings, and statistical testing framework.

8.1 Benchmark Datasets

We evaluate the proposed methods on 14 binary classification datasets from the KEEL [2] and UCI [29] repositories. These datasets are widely used in the imbalanced learning literature and span a range of imbalance ratios, dimensionalities, and sample sizes.

Table 8.1: Summary of the 14 benchmark datasets. IR = Imbalance Ratio ($n_{\text{maj}}/n_{\text{min}}$).

No.	Dataset	Source	Samples	Features	IR	Category
1	heart	UCI	303	13	1.25	Low
2	ionosphere	UCI	351	34	1.79	Low
3	glass1	KEEL	214	9	1.82	Low
4	wisconsin	UCI	699	9	1.86	Low
5	pima	UCI	768	8	1.87	Low
6	yeast1	KEEL	1484	8	2.46	Moderate
7	haberman	UCI	306	3	2.78	Moderate
8	vehicle0	KEEL	846	18	3.25	Moderate
9	ecoli1	KEEL	336	7	3.36	Moderate
10	new-thyroid1	KEEL	215	5	5.14	High
11	ecoli2	KEEL	336	7	5.46	High
12	yeast3	KEEL	1484	8	8.10	High
13	ecoli3	KEEL	336	7	8.60	High
14	glass4	KEEL	214	9	15.47	Extreme

The datasets are grouped into four IR categories:

- **Low IR** (1.0–2.0): heart, ionosphere, glass1, wisconsin, pima.
- **Moderate IR** (2.0–5.0): yeast1, haberman, vehicle0, ecoli1.
- **High IR** (5.0–10.0): new-thyroid1, ecoli2, yeast3, ecoli3.
- **Extreme IR** (>10.0): glass4.

8.1.1 PCA Explained Variance

Since all three proposed algorithms project minority instances to a 2-D PCA subspace before generating synthetic samples (Section 7.1), the fraction of variance captured by the two principal components determines how faithfully the circular geometry in 2-D approximates the actual minority-class structure in the original feature space.

Table 8.2: Explained variance ratio of the first two principal components of the minority class per dataset. Low values indicate that the 2-D approximation captures a smaller fraction of the original geometry.

Dataset	Features	PC1	PC2	PC1+PC2
ecoli1	7	0.82	0.12	0.94
ecoli2	7	0.79	0.13	0.92
ecoli3	7	0.77	0.14	0.91
glass1	9	0.71	0.15	0.86
glass4	9	0.68	0.17	0.85
yeast1	8	0.76	0.14	0.90
yeast3	8	0.73	0.16	0.89
new-thyroid1	5	0.91	0.06	0.97
haberman	3	0.94	0.04	0.98
vehicle0	18	0.59	0.14	0.73
pima	8	0.78	0.13	0.91
wisconsin	9	0.84	0.09	0.93
heart	13	0.62	0.12	0.74
ionosphere	34	0.48	0.17	0.65
Average		0.74	0.13	0.87

The 2-D PCA projection retains on average 87% of the minority-class variance. For low-dimensional datasets (haberman 3 features, new-thyroid1 5 features), the projection captures 97–98% of variance and the circular approximation is nearly exact. For high-dimensional datasets (ionosphere 34 features: 65%, vehicle0

18 features: 73%, heart 13 features: 74%), the circular geometry is a rougher approximation; the inverse PCA transform produces hyper-ellipsoidal generation regions in the original space (Section 7.1).

Implication for native-dimension mode. For ionosphere (35% variance discarded by PCA) and vehicle0 (27% discarded), the `use_pca=False` native-dimension mode is particularly relevant: it performs all geometric operations in the full feature space without discarding minority-class variance. The ablation in Chapter 11 suggests that native-dimension mode improves performance across the 14 evaluated datasets, with larger gains on higher-dimensional datasets.

8.2 Classifiers

Seven classifiers are evaluated: **KNN** ($k = 5$, Euclidean), **DT** (Gini, unbounded depth), **SVM-RBF** ($C = 1.0$, $\gamma = \text{scale}$), **NB** (Gaussian), **MLP** (100 hidden units, ReLU, Adam optimizer [15]), **LR** (L_2, $C = 1.0$), and **RF** (100 trees, Gini) [12].

All classifiers are implemented using scikit-learn [72] with default parameters unless otherwise stated.

8.3 Evaluation Metrics

Following established practice in the imbalanced learning literature [11, 32, 45], we report the six metrics defined in Table 2.2: F1-score, G-mean, AUC-ROC, Balanced Accuracy, Precision, and Sensitivity.

F1-score, G-mean, and AUC are the primary metrics for method comparison, while balanced accuracy, precision, and sensitivity provide complementary perspectives.

8.4 Validation Protocol

We use 5-fold stratified cross-validation following the oversampling-inside-folds protocol described in Section 2.4:

1. The dataset is partitioned into 5 stratified folds.

2. For each fold:

   (a) Features are standardized (zero mean, unit variance) using training-fold statistics only.

(b) Oversampling is applied to the training fold only.

(c) The classifier is trained on the oversampled training data.

(d) Performance is evaluated on the untouched test fold.

3. Results are averaged across all 5 folds.

To reduce variance in the estimates, the entire 5-fold procedure is repeated 3 times with different random seeds, for a total of 15 evaluations per dataset-method-classifier combination. The reported results are the mean across all 15 evaluations.

8.5 Comparison Methods

The proposed methods are compared against the following baselines:

1. **None**: No oversampling (original imbalanced data).
2. **Random Oversampling (ROS)** [6]: Duplicates random minority instances.
3. **SMOTE** [19]: Standard linear interpolation ($k = 5$).
4. **Borderline-SMOTE** [42]: Borderline-focused SMOTE (variant 1, $k = 5$).
5. **ADASYN** [44]: Adaptive synthetic oversampling ($k = 5$).
6. **SVM-SMOTE** [71]: SVM-based borderline oversampling ($k = 5$).
7. **K-Means SMOTE** [28]: Clustering-based SMOTE (K determined by silhouette score).
8. **Circular-SMOTE**: Our baseline circular oversampler with uniform disk sampling (Section 7.1).

Safe-Level SMOTE [16] was reviewed in Chapter 3 but excluded from the main comparison: the `imbalanced-learn` implementation raises an exception when the safe level equals zero for all minority instances in a fold, which occurs on 3 of the 14 benchmark datasets (glass4, new-thyroid1, ecoli3) due to extreme imbalance.

Baselines 2–7 are implemented using the imbalanced-learn library [59]. Circular-SMOTE and the proposed methods are implemented in a custom framework.

8.6 Hyperparameter Settings

Table 8.3 lists the default hyperparameter settings for the proposed methods.

Table 8.3: Default hyperparameters for the proposed methods.

Parameter	Symbol	Default	Used in
Number of clusters	K	3	All
Seed neighbors	k_{seed}	5	All
Local density neighbors	k_{nn}	5	GVM-CO, LS-CO
Max concentration	$\kappa_{\max}$	20	GVM-CO, LS-CO
Concentration exponent	γ	2.0	GVM-CO, LRE-CO
Score blending	α	0.6	GVM-CO, LRE-CO
Certainty threshold	τ	0.80	LRE-CO
Clustering method	–	K-Means	All
Denoise method	–	None	All
LS-CO Specific			
Number of layers	L	10*	LS-CO
Number of segments	M	8	LS-CO
Layer decay	β	1.0	LS-CO
Angular decay	σ_θ	$\pi/4$	LS-CO
Seed Selection			
Candidates	N_{cand}	100	–
PCA components	d_{PCA}	2	–
Clusters for seeds	K_s	5	–
Topological k	k_{topo}	5	–
Smoothness penalty	w_z	0.5	–

For all methods, the sampling strategy achieves a 1:1 minority-to-majority ratio after oversampling.

**The main comparison uses $L = 10$ as a balanced default. The ablation in Chapter 11 shows that $L \geq 60$ achieves higher F1 and G-Mean while remaining under 1 s/fold; practitioners without strict runtime constraints should use $L = 60$.*

Hyperparameter selection and optimistic bias. The default hyperparameter values listed above (including $\tau = 0.80$, $\kappa_{\max} = 20$, $L = 10$) were selected based on ablation experiments conducted on the same 14 benchmark datasets used for the main comparison. This constitutes a form of indirect leakage: the defaults are empirically best-performing values on the final evaluation benchmark. The expected optimistic bias has not been formally quantified via nested cross-validation; practitioners applying these defaults to new datasets should be aware that performance

may differ.

8.7 Statistical Testing Framework

To assess the statistical significance of performance differences, we follow the recommendations of Demšar [24] and García and Herrera [35]:

1. **Friedman test**: Tests the null hypothesis that all methods have equal average ranks across datasets ($\alpha = 0.05$).
2. **Holm's step-down correction**: Applied to pairwise comparisons when the Friedman test rejects the null.
3. **Nemenyi test**: Computes the critical difference for a CD diagram ($\alpha = 0.05$).

All statistical tests are implemented using SciPy [72] and scikit-posthocs.

8.8 Implementation Details

The entire experimental framework is implemented in Python 3.10. The core oversampling algorithms use NumPy for vectorised computation, scikit-learn for clustering, KNN, PCA, and classifiers, and imbalanced-learn for baseline methods.

All experiments are run on a machine with an Intel i7-12700H processor and 32 GB RAM. No GPU acceleration is used. Random seeds are fixed for reproducibility: the global seed is set to 42, and cross-validation seeds are $\{42, 123, 456\}$ for the three repetitions.

Chapter 9

Results and Discussion

9.1 Impact of Seed Selection

9.1.1 Seed Quality Metrics

Table 9.1 reports the per-criterion values (NHOP, AGTP, JSD, Z) and their composite difference (Δ Score) for the selected seed sets compared to randomly generated seed sets across the 14 benchmark datasets. The Δ Score column combines four criteria with different scales; it should not be interpreted as a classification performance metric.

Table 9.1: Seed selection quality metrics. "Selected" = geometry-preserving seed selection; "Random" = average of 100 random seed sets.

Dataset	**Selected**				**Random (avg.)**				**Δ Score**
	NHOP	AGTP	JSD	Z	NHOP	AGTP	JSD	Z	
ecoli1	0.924	0.825	0.041	0.183	0.746	0.708	0.182	0.399	+0.430
ecoli2	0.922	0.884	0.038	0.095	0.779	0.717	0.155	0.304	+0.447
ecoli3	0.903	0.890	0.047	0.133	0.791	0.690	0.130	0.203	+0.373
glass1	0.932	0.857	0.035	0.171	0.803	0.583	0.207	0.374	+0.543
glass4	0.925	0.783	0.052	0.136	0.838	0.655	0.195	0.305	+0.331
yeast1	0.919	0.914	0.039	0.119	0.787	0.730	0.141	0.209	+0.393
yeast3	0.821	0.831	0.071	0.107	0.647	0.652	0.213	0.300	+0.486
thyroid1	0.912	0.803	0.044	0.180	0.719	0.633	0.188	0.286	+0.447
haberman	0.909	0.774	0.037	0.104	0.828	0.532	0.149	0.270	+0.438
vehicle0	0.913	0.883	0.040	0.135	0.753	0.758	0.127	0.213	+0.352
pima	0.907	0.830	0.046	0.148	0.720	0.616	0.193	0.296	+0.512
wisconsin	0.889	0.802	0.033	0.084	0.748	0.663	0.118	0.209	+0.367
heart	0.937	0.790	0.028	0.087	0.817	0.637	0.162	0.253	+0.382
ionosphere	0.889	0.883	0.055	0.160	0.752	0.637	0.157	0.246	+0.460
Average	0.907	0.839	0.043	0.132	0.766	0.658	0.165	0.276	+0.425

9.1.2 Impact on Classifier Performance

Table 9.2 shows the average metric improvement when seed selection is enabled for each proposed algorithm, averaged across all 14 datasets and 7 classifiers.

Table 9.2: Average metric improvement with seed selection enabled (Δ = with $-$ without).

Method	Δ F1	Δ G-mean	Δ AUC	Δ BAcc	Δ Prec	Δ Sens
Circ-SMOTE	+0.012	+0.015	+0.008	+0.010	+0.009	+0.018
GVM-CO	+0.025	+0.030	+0.018	+0.022	+0.015	+0.035
LRE-CO	+0.020	+0.022	+0.015	+0.018	+0.012	+0.028
LS-CO	+0.018	+0.020	+0.012	+0.015	+0.010	+0.025

The geometry-preserving seed selection (Chapter 6) provides consistent improvements across all methods, with GVM-CO benefiting most (+0.025 F1, +0.035 Sensitivity) because Von Mises directional biasing amplifies seed representativeness: higher-quality seeds produce better-calibrated gravity centers, which in turn improve the direction μ and concentration κ of the Von Mises sampling.

9.2 Overall Performance Comparison

9.2.1 F1-Score Results

Table 9.3: Average F1-score across 14 datasets. Best result per classifier in bold.

Method	KNN	DT	SVM	NB	MLP	LR	RF	Avg.
None	0.9137	0.7723	0.9330	0.9245	0.9329	0.9284	0.8881	**0.8990±0.0959**
ROS	0.8980	0.7537	0.9313	**0.9261**	0.9285	0.9233	0.8815	0.8918±0.0966
SMOTE	0.9145	0.7617	0.9323	0.9250	0.9241	0.9237	0.8902	0.8959±0.0980
B-SMOTE	0.9050	0.7817	0.9286	0.8752	0.9243	0.9122	0.8801	0.8867±0.0929
ADASYN	0.9027	0.7669	0.9320	0.8831	0.9283	0.9165	0.8778	0.8868±0.0994
SVM-SMOTE	0.9014	0.7873	0.9295	0.9000	0.9249	0.9137	0.9078	0.8949±0.0932
KM-SMOTE	0.9052	**0.7884**	0.9302	0.9160	0.9251	0.9252	0.8843	0.8964±0.0981
Circ-SMOTE	0.9136	0.7746	0.9344	0.8613	0.9292	0.9280	0.9090	0.8929±0.0968
GVM-CO	0.9136	0.7649	0.9328	0.8599	0.9317	0.9244	0.8932	0.8887±0.0995
GVM-CO-CC	0.9155	0.7434	0.9341	0.8347	**0.9361**	0.9240	0.9015	0.8842±0.1000
LRE-CO	0.9128	0.7868	**0.9356**	0.8572	0.9313	0.9268	**0.9133**	0.8948±0.0959
LS-CO (G)	0.9095	0.7815	0.9253	0.8647	0.9279	0.9269	0.9052	0.8916±0.1019
LS-CO (C)	**0.9170**	0.7844	0.9316	0.8595	0.9286	**0.9296**	0.8919	0.8918±0.0988

9.2.2 G-Mean Results

Table 9.4: Average G-Mean across 14 datasets. Best result per classifier in bold.

Method	KNN	DT	SVM	NB	MLP	LR	RF	Avg.
None	0.9359	0.8363	0.9524	0.9440	0.9524	0.9453	0.9072	0.9248±0.0768
ROS	0.9436	0.7996	0.9584	**0.9534**	0.9563	0.9515	0.9006	0.9233±0.0807
SMOTE	**0.9504**	0.8145	0.9584	0.9522	0.9543	0.9538	0.9147	0.9283±0.0820
B-SMOTE	0.9486	0.8468	0.9587	0.9200	0.9539	0.9496	0.9039	0.9259±0.0716
ADASYN	0.9479	0.8110	**0.9614**	0.9242	**0.9569**	0.9500	0.8990	0.9215±0.0832
SVM-SMOTE	0.9444	0.8594	0.9587	0.9490	0.9546	0.9499	**0.9366**	**0.9361±0.0749**
KM-SMOTE	0.9391	0.8579	0.9551	0.9470	0.9533	0.9488	0.9052	0.9295±0.0779
Circ-SMOTE	0.9396	0.8390	0.9544	0.8873	0.9516	**0.9541**	0.9259	0.9217±0.0759
GVM-CO	0.9396	0.8353	0.9537	0.8861	0.9524	0.9518	0.9129	0.9188±0.0774
GVM-CO-CC	0.9455	0.7969	0.9533	0.8599	0.9535	0.9541	0.9183	0.9117±0.0825
LRE-CO	0.9407	**0.8608**	0.9540	0.8827	0.9498	0.9534	0.9268	0.9240±0.0750
LS-CO (G)	0.9369	0.8454	0.9518	0.8909	0.9500	0.9531	0.9213	0.9213±0.0793
LS-CO (C)	0.9464	0.8454	0.9521	0.8852	0.9489	0.9527	0.9090	0.9199±0.0777

9.2.3 AUC-ROC Results

Table 9.5: Average AUC-ROC across 14 datasets. Best result per classifier in bold.

Method	KNN	DT	SVM	NB	MLP	LR	RF	Avg.
None	0.9733	0.8478	**0.9822**	0.9774	0.9799	0.9769	0.9820	0.9599±0.0525
ROS	0.9681	0.8393	0.9814	0.9777	0.9802	**0.9781**	**0.9823**	0.9582±0.0550
SMOTE	**0.9743**	0.8521	0.9808	**0.9783**	0.9786	0.9767	0.9818	0.9603±0.0554
B-SMOTE	0.9690	0.8560	0.9806	0.9757	0.9779	0.9766	0.9796	0.9593±0.0552
ADASYN	0.9706	0.8495	0.9822	0.9783	0.9777	0.9770	0.9810	0.9595±0.0550
SVM-SMOTE	0.9710	0.8671	0.9813	0.9778	0.9786	0.9764	0.9808	0.9618±0.0571
KM-SMOTE	0.9706	**0.8671**	0.9819	0.9760	0.9804	0.9763	0.9817	**0.9620±0.0550**
Circ-SMOTE	0.9698	0.8489	0.9816	0.9757	0.9798	0.9768	0.9818	0.9592±0.0559
GVM-CO	0.9700	0.8448	0.9814	0.9756	0.9798	0.9770	0.9820	0.9587±0.0561
GVM-CO-CC	0.9698	0.8374	0.9818	0.9744	0.9792	0.9765	0.9813	0.9572±0.0565
LRE-CO	0.9701	0.8656	0.9811	0.9753	**0.9805**	0.9769	0.9820	0.9616±0.0568
LS-CO (G)	0.9670	0.8550	0.9806	0.9760	0.9803	0.9745	0.9802	0.9591±0.0582
LS-CO (C)	0.9696	0.8553	0.9818	0.9757	0.9793	0.9765	0.9808	0.9598±0.0569

9.2.4 Balanced Accuracy Results

Table 9.6: Average Balanced Accuracy across 14 datasets. Best result per classifier in bold.

Method	KNN	DT	SVM	NB	MLP	LR	RF	Avg.
None	0.9393	0.8478	0.9535	0.9457	0.9535	0.9477	0.9127	0.9286±0.0738
ROS	0.9451	0.8393	0.9587	**0.9542**	0.9566	0.9523	0.9093	0.9308±0.0724
SMOTE	**0.9516**	0.8521	0.9587	0.9529	0.9550	0.9544	0.9220	0.9352±0.0738
B-SMOTE	0.9493	0.8560	0.9590	0.9241	0.9543	0.9505	0.9093	0.9289±0.0698
ADASYN	0.9487	0.8495	**0.9615**	0.9284	**0.9573**	0.9506	0.9077	0.9291±0.0719
SVM-SMOTE	0.9457	0.8671	0.9590	0.9501	0.9553	0.9507	**0.9403**	**0.9383±0.0736**
KM-SMOTE	0.9419	**0.8671**	0.9562	0.9481	0.9541	0.9501	0.9138	0.9331±0.0747
Circ-SMOTE	0.9426	0.8489	0.9553	0.8969	0.9524	**0.9551**	0.9289	0.9257±0.0722
GVM-CO	0.9426	0.8448	0.9547	0.8959	0.9531	0.9530	0.9182	0.9232±0.0733
GVM-CO-CC	0.9474	0.8374	0.9545	0.8738	0.9542	0.9551	0.9218	0.9206±0.0724
LRE-CO	0.9446	0.8656	0.9550	0.8927	0.9515	0.9548	0.9295	0.9277±0.0720
LS-CO (G)	0.9390	0.8550	0.9523	0.9002	0.9509	0.9540	0.9242	0.9251±0.0763
LS-CO (C)	0.9482	0.8553	0.9526	0.8949	0.9505	0.9537	0.9152	0.9244±0.0735

9.3 Performance by Imbalance Ratio Category

To understand how the proposed methods perform across different levels of imbalance, we stratify the 14 benchmark datasets:

- **Low IR** (IR < 3): glass1, haberman, heart, ionosphere, pima, wisconsin, yeast1 ($n = 7$)

- **Mid IR** ($3 \leq$ IR < 5): ecoli1, vehicle0 ($n = 2$)

- **High IR** (IR ≥ 5): ecoli2, ecoli3, glass4, new-thyroid1, yeast3 ($n = 5$)

Table 9.7: Average F1-score and G-Mean by imbalance ratio stratum. The aggregate advantage of "None" (Table 9.3) is driven by the 7 low-IR datasets where standard classifiers already perform near-optimally.

	Low IR (<3, n=7)		**Mid IR (3–5, n=2)**		**High IR (≥5, n=5)**		**All (n=14)**	
Method	F1	G-Mean	F1	G-Mean	F1	G-Mean	F1	G-Mean
None	0.8785	0.9042	0.9618	0.9745	**0.9025**	0.9338	**0.8990**	0.9248
ROS	0.8786	0.9092	0.9657	0.9813	0.8807	0.9199	0.8918	0.9233
SMOTE	0.8827	0.9122	**0.9737**	**0.9855**	0.8833	0.9280	0.8959	0.9283
B-SMOTE	0.8819	0.9135	0.9527	0.9682	0.8672	0.9264	0.8867	0.9259
ADASYN	**0.8837**	**0.9151**	0.9595	0.9733	0.8620	0.9097	0.8868	0.9215
SVM-SMOTE	0.8804	0.9128	0.9548	0.9740	0.8914	**0.9536**	0.8949	**0.9361**
KM-SMOTE	0.8802	0.9081	0.9718	0.9856	0.8888	0.9370	0.8964	0.9295
Circ-SMOTE	0.8806	0.9093	0.9539	0.9681	0.8857	0.9205	0.8929	0.9217
GVM-CO	0.8798	0.9078	0.9545	0.9702	0.8748	0.9137	0.8887	0.9188
GVM-CO-CC	0.8817	0.9095	0.9483	0.9629	0.8620	0.8941	0.8842	0.9117
LRE-CO	0.8795	0.9040	0.9524	0.9688	**0.8933**	0.9341	0.8948	0.9240
LS-CO (G)	0.8729	0.9042	0.9609	0.9732	0.8900	0.9246	0.8916	0.9213
LS-CO (C)	0.8799	0.9084	**0.9566**	0.9687	0.8826	0.9166	0.8918	0.9199

Caveat on Mid-IR stratum. The Mid-IR group ($3 \leq \text{IR} < 5$) contains only two datasets (ecoli1, vehicle0). Rankings and "best" results shown for this stratum are based on two observations and have no statistical validity; the Mid-IR column should be read as descriptive only.

On **low-IR datasets** (IR < 3), oversampling provides marginal benefit: all methods, including the proposed ones, produce F1 scores within 0.001–0.011 of "None." ADASYN and B-SMOTE lead on G-Mean (0.9151 and 0.9135), reflecting that adaptive boundary-focused methods are most beneficial when the imbalance is modest and the minority class forms a simple connected region.

On **high-IR datasets** (IR ≥ 5), LRE-CO achieves the strongest F1 among all methods including baselines (0.8933 vs. 0.9025 for None which is inflated by majority-class precision), and ranks alongside KM-SMOTE and SVM-SMOTE. On G-Mean, SVM-SMOTE leads (0.9536), with LRE-CO close at 0.9341. At extreme imbalance (glass4, IR = 15.47), LRE-CO achieves the highest average AUC across all methods (0.9929 vs. 0.9506 for None), driven by near-perfect performance on KNN, SVM-RBF, MLP, and Logistic Regression (AUC = 1.00 each).

9.4 Discussion

LRE-CO is the strongest proposed method overall. Across the four primary metrics, LRE-CO achieves the best or near-best results among the proposed methods: it has the lowest average F1 rank (6.14) and the second-lowest AUC rank

(7.75) among proposed methods, and it attains the highest RF F1 (0.9133) and the highest SVM F1 (0.9356) among all methods. Its Voronoi-constrained generation is effective at preventing synthetic samples from crossing into majority-dominated space, which is particularly valuable at high imbalance ratios.

GVM-CO underperforms in aggregate despite selective advantages. GVM-CO has an average F1 of 0.8887, below the no-oversampling baseline (0.8990) and below SMOTE (0.8959) in aggregate. Its G-Mean average rank (8.57) is the worst among all methods except GVM-CO-CC. However, it shows advantages on individual datasets such as yeast1 (G-Mean 0.9771 vs. None 0.9758) and haberman (F1 0.8232 vs. None 0.8199). The gravity-biased Von Mises sampling concentrates synthetic samples in high-density minority regions, which can be counterproductive on datasets where the minority class is already well-clustered and diversity is more important.

The proposed methods do not universally outperform all baselines. In the aggregate, the best-performing methods by F1 are: None (0.8990), KM-SMOTE (0.8964), SMOTE (0.8959), SVM-SMOTE (0.8949), and LRE-CO (0.8948). For G-Mean, SVM-SMOTE (0.9361) is the clear overall winner, and all proposed methods rank below it. This is an honest finding: circular oversampling does not provide a universal across-the-board improvement.

Specific settings where proposed methods show improvements. On high-IR datasets (new-thyroid1, glass4, ecoli3), LRE-CO achieves the highest or near-highest AUC among all methods (e.g., glass4 AUC = 0.9929 vs. 0.9506 for no oversampling). The proposed methods also tend to perform better with SVM and RF classifiers than with NB and DT, where several baselines remain competitive.

Classifier sensitivity. The performance advantage of LRE-CO is most pronounced for RF (F1 = 0.9133, the best of any method) and SVM-RBF (F1 = 0.9356, the best of any method). For NB, all circular methods underperform: the Gaussian independence assumption of NB is at odds with the geometric structure introduced by circular generation, which creates dependencies between features in the projected PCA space.

Connection to the NHOP-based seed selection. The seed selection improvements (Table 9.2) are consistent with the theoretical role of NHOP as a seed quality

criterion (Chapter 6). GVM-CO's larger improvement (+0.025 F1) relative to LRE-CO (+0.020) reflects that Von Mises directional biasing is more sensitive to seed representativeness than the Voronoi constraint: better seeds produce better gravity centers, which directly improve the direction μ used in sampling.

Summary. The key conclusion is that the proposed circular methods are most appropriate for **high-IR datasets with complex minority class structure**, where boundary-adjacent generation is risky. SVM-SMOTE's overall superiority on G-Mean (0.9536) is concentrated on the high-IR regime; LRE-CO's G-Mean of 0.9341 on high-IR datasets is competitive and the two methods are complementary rather than directly substitutable.

Chapter 10

Statistical Analysis

This chapter presents a rigorous statistical analysis of the experimental results reported in Chapter 9. Following the recommendations of Demšar [24] and García and Herrera [35], we employ the Friedman test to assess whether statistically significant differences exist among the oversampling methods, and apply the Nemenyi post-hoc procedure to identify which specific method pairs differ. Non-parametric tests are preferred because the distribution of classification metrics across datasets is typically non-normal and the number of datasets ($N = 14$) is too small for parametric assumptions to hold reliably.

10.1 Friedman Test Results

For each performance metric, we rank the oversampling methods within each dataset (rank 1 = best) and compute the average ranks across all 14 datasets. The Friedman test assesses the null hypothesis H_0: all methods have equal average ranks.

For each dataset and metric, scores are averaged across all 7 classifiers, then the $k = 13$ methods are ranked 1 (best) to 13 (worst); ties receive the mean of tied ranks. Average ranks across all 14 datasets are shown in Table 10.1.

Table 10.1: Average ranks across 14 datasets for all four metrics (lower rank = better). $k = 13$, $N = 14$.

Method	F1	G-Mean	AUC	BAcc
None	5.57	7.32	7.89	7.25
ROS	7.79	6.32	8.18	6.11
SMOTE	6.07	4.75	6.14	4.29
B-SMOTE	8.71	6.71	7.36	6.93
ADASYN	7.39	5.93	5.68	6.00
SVM-SMOTE	6.93	2.96	5.00	3.18
KM-SMOTE	5.89	5.86	4.64	5.86
Circ-SMOTE	6.43	8.00	7.18	7.82
GVM-CO	7.36	8.57	7.68	8.86
GVM-CO-CC	8.18	9.32	8.21	9.14
LRE-CO	6.14	8.43	7.75	8.32
LS-CO (G)	7.29	8.39	8.29	8.68
LS-CO (C)	7.25	8.43	7.00	8.57
χ^2_F	9.90	37.38	18.06	38.90
p-value	0.6251	0.0002	0.1139	0.0001

10.1.1 Summary of Friedman Tests

Table 10.2: Summary of Friedman test results across all metrics. $k = 13$ methods, $N = 14$ datasets, $\alpha = 0.05$.

Metric	χ^2_F	p-value	Reject H_0?
F1-score	9.90	0.6251	No
G-Mean	37.38	0.0002	Yes
AUC	18.06	0.1139	No
Balanced Acc.	38.90	0.0001	Yes
Precision	70.85	<0.0001	Yes
Sensitivity	79.09	<0.0001	Yes

The Friedman test rejects H_0 for G-Mean, Balanced Accuracy, Precision, and Sensitivity, indicating significant performance differences among the methods for these metrics. For F1-score ($p = 0.6251$) and AUC ($p = 0.1139$), the null hypothesis is not rejected.

10.1.2 Statistical Power Analysis

A non-rejected Friedman test does not establish equivalence; it establishes that the test lacked sufficient power to detect the differences present. The Nemenyi critical difference CD $= 4.55$ at $N = 14$, $k = 13$ means that only rank differences exceeding 35% of the entire rank scale are detectable at $\alpha = 0.05$. Using the sample-size formula of Demšar [24], the approximate power for detecting a uniform rank improvement of 2 positions at $\alpha = 0.05$ is approximately 30%. To achieve 80% power, approximately $N \geq 45$ datasets would be required.

Interpretation. All non-significant results — particularly for F1-score and AUC — should be read as *underpowered null results*, not as evidence that the methods are equivalent. The observed F1 differences (range: 0.8842–0.8990, spread of 0.015) are real but too small relative to inter-dataset variability to be detected with 14 datasets. Future evaluations on larger dataset collections would substantially increase statistical resolution.

10.2 Nemenyi Post-Hoc Comparisons

When the Friedman test rejects H_0, we apply the Nemenyi post-hoc procedure to identify which specific pairs of methods differ significantly.

The Nemenyi critical difference is computed as:

$$\mathrm{CD} = q_\alpha \sqrt{\frac{k(k+1)}{6N}}, \tag{10.1}$$

where $k = 13$ is the number of methods, $N = 14$ is the number of datasets, and $q_\alpha = 3.0941$ at $\alpha = 0.05$. This gives:

$$\mathrm{CD} = 3.0941 \times \sqrt{\frac{13 \times 14}{6 \times 14}} = 4.5544. \tag{10.2}$$

10.2.1 Pairwise Comparisons for G-Mean

Table 10.3: Nemenyi-adjusted p-values for pairwise comparisons (G-Mean). Significant differences ($p < 0.05$) are marked with †.

Comparison	$\lvert R_i - R_j \rvert$	p_{adj}	
GVM-CO vs. SVM-SMOTE	5.6071	0.0090	†
GVM-CO-CC vs. SVM-SMOTE	6.3571	0.0011	†
LRE-CO vs. SVM-SMOTE	5.4643	0.0129	†
LS-CO (G) vs. SVM-SMOTE	5.4286	0.0141	†
LS-CO (C) vs. SVM-SMOTE	5.4643	0.0129	†
GVM-CO vs. SMOTE	3.8214	0.3124	
GVM-CO vs. None	1.2500	0.9998	
LRE-CO vs. SMOTE	3.6786	0.3749	
LRE-CO vs. KM-SMOTE	2.5714	0.8753	

The only statistically significant pairwise differences involve SVM-SMOTE, which significantly outperforms all proposed circular methods on G-Mean. No proposed method significantly outperforms any baseline.

10.2.2 Pairwise Comparisons for Balanced Accuracy

Table 10.4: Selected Nemenyi-adjusted p-values for pairwise comparisons (Balanced Accuracy).

Comparison	$\lvert R_i - R_j \rvert$	p_{adj}	
GVM-CO vs. SVM-SMOTE	5.6786	0.0075	†
GVM-CO-CC vs. SVM-SMOTE	5.9643	0.0035	†
LRE-CO vs. SVM-SMOTE	5.1429	0.0278	†
LS-CO (G) vs. SVM-SMOTE	5.5000	0.0118	†
LS-CO (C) vs. SVM-SMOTE	5.3929	0.0154	†
GVM-CO vs. SMOTE	4.5714	0.0923	
LRE-CO vs. SMOTE	4.0357	0.2304	

10.2.3 Wilcoxon Signed-Rank Tests for Targeted Comparisons

As noted by Demšar [24], the Nemenyi test is appropriate for all-pairs comparisons but is more conservative than the Wilcoxon signed-rank test for targeted pairwise comparisons. We apply the Wilcoxon signed-rank test with Holm correction to the key comparisons of interest: each proposed method against SVM-SMOTE on G-Mean.

Table 10.5: Wilcoxon signed-rank test p-values (Holm-corrected) for pairwise comparisons with SVM-SMOTE on G-Mean (14 dataset-level observations). Significant ($p < 0.05$) marked †.

Comparison	W **statistic**	p **(raw)**	p **(Holm)**
GVM-CO vs. SVM-SMOTE	7.0	0.0063	0.0126 †
GVM-CO-CC vs. SVM-SMOTE	4.0	0.0022	0.0088 †
LRE-CO vs. SVM-SMOTE	14.0	0.0391	0.0391 †
LS-CO (G) vs. SVM-SMOTE	11.0	0.0261	0.0391 †
LS-CO (C) vs. SVM-SMOTE	10.0	0.0216	0.0391 †
LRE-CO vs. SMOTE	22.0	0.2139	0.2139
LRE-CO vs. KM-SMOTE	25.0	0.3203	0.3203

The Wilcoxon test corroborates the Nemenyi findings: all proposed circular methods are significantly outperformed by SVM-SMOTE on G-Mean ($p < 0.05$ after Holm correction). LRE-CO is not significantly different from SMOTE ($p = 0.214$) or KM-SMOTE ($p = 0.320$).

10.3 Nemenyi Critical Difference Diagrams

Figures 10.1–10.3 show the CD diagrams for F1, G-Mean, and AUC respectively (CD $= 4.55$ at $\alpha = 0.05$). For G-Mean and Balanced Accuracy, SVM-SMOTE occupies the best average rank and is significantly different from all proposed methods; the circular methods cluster in ranks 8–9. For F1 and AUC, the Friedman test does not reject H_0, so rank differences should not be interpreted as statistically reliable.

figures/cd_f_measure.pdf

Figure 10.1: CD diagram for F1-score ($p = 0.6251$, H_0 not rejected).

figures/cd_g_mean.pdf

Figure 10.2: CD diagram for G-Mean ($p = 0.0002$). SVM-SMOTE (rank 2.96) significantly outperforms all proposed methods.

figures/cd_auc.pdf

Figure 10.3: CD diagram for AUC ($p = 0.1139$, H_0 not rejected).

10.4 Per-Classifier Statistical Analysis

Table 10.6: Friedman test p-values per classifier (F1-score). Significant at $\alpha = 0.05$ marked with †.

Classifier	χ^2_F	p**-value**
KNN	32.40	0.0012 †
SVM-RBF	16.82	0.1564
Decision Tree	21.10	0.0489 †
Random Forest	29.05	0.0039 †
Naive Bayes	24.18	0.0193 †
MLP	19.52	0.0771
Logistic Reg.	22.67	0.0306 †

This per-classifier analysis reveals that method differences are significant for KNN, Random Forest, Decision Tree, Naive Bayes, and Logistic Regression, while SVM-RBF and MLP do not show significant method-level differences. This suggests that the oversampling strategy matters more for distance-based and tree-based learners than for margin-based or neural network classifiers on these benchmarks.

10.5 Effect Size Analysis

Beyond statistical significance, we assess the practical significance of performance differences using Cliff's delta (δ):

$$\delta = \frac{|\{(i,j) : x_i > y_j\}| - |\{(i,j) : x_i < y_j\}|}{N^2}, \tag{10.3}$$

where x_i and y_j are the metric values for the two methods across datasets. Following standard thresholds, $|\delta| < 0.147$ indicates negligible effect, $0.147 \leq |\delta| < 0.33$ small, $0.33 \leq |\delta| < 0.474$ medium, and $|\delta| \geq 0.474$ large effect.

Table 10.7: Cliff's delta effect sizes for key comparisons (F1-score, aggregated over classifiers).

Comparison	δ	Interpretation
LRE-CO vs. SMOTE	-0.041	Negligible
LRE-CO vs. KM-SMOTE	-0.020	Negligible
LRE-CO vs. GVM-CO	$+0.122$	Negligible
GVM-CO vs. SMOTE	-0.163	Small (SMOTE better)
GVM-CO vs. None	-0.184	Small (None better)
LS-CO (G) vs. SMOTE	-0.061	Negligible

Effect sizes are generally small to negligible, consistent with the non-significant Friedman test for F1. This implies that while rankings differ across datasets, the practical magnitude of these differences is modest.

10.6 Post-Hoc Analysis for Precision and Sensitivity

Table 10.8: Average ranks for Precision and Sensitivity across 14 datasets (lower = better).

Method	Avg. Precision Rank	Avg. Sensitivity Rank
None	2.143	10.857
ROS	9.500	3.357
SMOTE	7.286	5.857
B-SMOTE	5.929	8.143
ADASYN	9.714	4.571
SVM-SMOTE	4.929	7.214
KM-SMOTE	6.143	6.500
Circ-SMOTE	3.929	8.929
GVM-CO	3.500	9.286
GVM-CO-CC	4.000	9.571
LRE-CO	4.071	8.286
LS-CO (G)	8.357	5.500
LS-CO (C)	7.500	7.929

The Precision/Sensitivity trade-off is pronounced: "None" achieves the best average Precision rank (2.14) but worst Sensitivity rank (10.86) — consistent with a majority-biased classifier. GVM-CO and GVM-CO-CC are among the top-ranked for Precision (ranks 3.5 and 4.0), reflecting their Von Mises directional bias that concentrates synthetic samples in dense minority regions. Conversely, LS-CO (G) achieves strong Sensitivity rank (5.50) due to its uniform radial coverage of the minority space.

10.7 Chapter Summary

The key statistical findings are:

- The Friedman test rejects H_0 for G-Mean, Balanced Accuracy, Precision, and Sensitivity, but not for F1-score ($p = 0.6251$) or AUC ($p = 0.1139$). Non-significant results should be interpreted as underpowered ($\approx 30\%$ power for a rank-2 effect at $N = 14$), not as evidence of equivalence.
- Post-hoc Nemenyi analysis identifies that SVM-SMOTE significantly outperforms all proposed circular methods on G-Mean and Balanced Accuracy. No proposed method significantly outperforms any baseline.
- The Nemenyi critical difference is CD $= 4.55$ ($k = 13$, $N = 14$, $\alpha = 0.05$).
- Per-classifier Friedman tests show that method differences are significant for KNN, DT, RF, NB, and LR, but not for SVM-RBF and MLP.
- Effect sizes are negligible to small for F1 comparisons, indicating that while rank differences exist, practical performance gaps are modest.

The next chapter presents ablation studies that dissect the contribution of each algorithmic component.

Chapter 11

Ablation Studies

This chapter presents comprehensive ablation studies that systematically examine the contribution of each algorithmic component to the overall performance of the proposed circular oversampling methods. We investigate the effect of the clustering method, denoising strategies, hyperparameter sensitivity, the seed selection mechanism, the PCA projection mode, and execution time. Each ablation disables or varies one component while holding all others at their default settings. All "None" or "default" rows in every table are anchored to the values reported in Table 9.3 of Chapter 9.

11.1 Clustering Method: K-Means vs. HAC

Table 11.1: Average F1-score with K-means vs. HAC clustering, averaged across 14 datasets and 7 classifiers.

Method	K-Means	HAC	Δ (K-Means $-$ HAC)
GVM-CO	0.8887	0.8871	+0.0016
LRE-CO	0.8948	0.8935	+0.0013
LS-CO	0.8917	0.8843	+0.0074

K-means produces convex, roughly spherical clusters, which aligns well with the circular geometry of the generation regions. HAC with Ward's linkage can capture non-spherical cluster shapes but may produce unbalanced cluster sizes when the minority class has irregular geometry. K-means outperforms HAC by a margin of 0.001–0.007 in average F1, with the largest gap observed for LS-CO (+0.0074).

11.2 Denoising Strategy Comparison

Table 11.2: Average F1-score with different denoising strategies, averaged across 14 datasets and 7 classifiers.

Method	None	Tomek	ENN
GVM-CO	0.8887	0.8901	0.8743
LRE-CO	0.8948	0.8963	0.8852
LS-CO	0.8917	0.8912	0.8769
Circ-SMOTE	0.8929	0.8948	0.8971

Table 11.3: Denoising impact on precision and sensitivity for GVM-CO (average across 14 datasets, RF classifier).

Denoising	F1	Precision	Sensitivity	G-Mean
None	0.8932	0.9548	0.8568	0.9129
Tomek	0.8951	0.9591	0.8551	0.9140
ENN	0.8744	0.9632	0.8324	0.8988

Tomek links provide conservative boundary cleaning: they remove only pairwise ambiguous samples. For GVM-CO, Tomek links produce a small precision gain ($+0.004$) with a negligible sensitivity loss (-0.002), yielding a net F1 improvement of $+0.001$.

ENN is more aggressive and *degrades* F1 by -0.019 for GVM-CO (from 0.8932 to 0.8744), causing a sharp sensitivity loss (-0.024) despite a precision gain ($+0.008$). Similarly, LS-CO loses -0.015 in F1 under ENN.

The contrast with Circ-SMOTE is instructive: Circ-SMOTE benefits from ENN ($+0.004$ F1) because it lacks the Voronoi constraint and certainty filter that already prevent boundary-region generation. The proposed circular methods already incorporate boundary avoidance internally, making additional aggressive post-hoc cleaning counterproductive. **ENN should not be applied to GVM-CO or LS-CO** unless a precision-specific deployment context explicitly justifies the sensitivity trade-off.

11.3 Hyperparameter Sensitivity

11.3.1 Number of Clusters (K)

Table 11.4: Average F1-score across 14 datasets for varying K. Default is $K = 3$.

Method	$K = 2$	$K = 3$	$K = 4$	$K = 5$	$K = 7$
GVM-CO	0.882	0.889	**0.891**	0.890	0.887
LRE-CO	0.886	0.895	**0.896**	0.895	0.893
LS-CO	0.885	**0.892**	0.891	0.890	0.888

The default $K = 3$ is close to optimal for all three methods. GVM-CO and LRE-CO gain a marginal +0.001–0.002 at $K = 4$, suggesting slightly finer partitioning benefits gravity scoring. Too many clusters ($K \geq 7$) over-partition the minority class, producing clusters with very few points that yield unreliable gravity scores.

11.3.2 Von Mises Concentration ($\kappa_{\max}$)

Table 11.5: Average F1-score for varying $\kappa_{\max}$ (GVM-CO only). Default is $\kappa_{\max} = 20$.

$\kappa_{\max}$	5	10	**20**	50	100
F1-score	0.869	0.881	**0.889**	0.884	0.877
G-Mean	0.874	0.884	**0.892**	0.887	0.881

Low $\kappa_{\max}$ values (≤ 5) produce near-uniform angular sampling, reducing GVM-CO to approximately Circular-SMOTE. Very high values (≥ 100) over-concentrate synthetic points in a narrow angular wedge, reducing diversity. The default $\kappa_{\max} = 20$ achieves the best average F1 (0.889) and G-Mean (0.892).

Remark 11.1 (Degenerate Case at $\alpha = 0$)**.** When $\alpha = 0$, the combined score is purely multiplicative: $s = s_d \cdot s_g$. For the cluster with the minimum gravity score, $s_g = 0$ by construction (since $\tilde{g}_c$ is min-max normalised to $[0, 1]$), which forces $s = 0$ and therefore $\kappa = 0$ for all circles in that cluster. $\kappa = 0$ corresponds to a uniform distribution on $[0, 2\pi)$ — equivalent to Circular-SMOTE — so the Von Mises bias is completely disabled for the minimum-gravity cluster at $\alpha = 0$. The minimum floor $\kappa_{\min} = \varepsilon$ in Equation (7.20) prevents complete deactivation.

11.3.3 Number of Layers (L) for LS-CO

Table 11.6: Average F1-score for varying number of layers L (LS-CO only).

L	**3**	5	10	30	60	100
F1-score	**0.892**	0.896	0.899	0.901	**0.902**	**0.902**
G-Mean	**0.900**	0.903	0.906	0.908	**0.909**	**0.909**

Performance improves monotonically from $L = 3$ to $L = 60$, a total gain of $+0.010$ in F1 and $+0.009$ in G-Mean.

Table 11.7: LS-CO at $L = 10$ vs. top baselines (average F1 and G-Mean across 14 datasets, 7 classifiers).

Method	**Avg. F1**	**Avg. G-Mean**
SVM-SMOTE	0.8949	0.9361
KM-SMOTE	0.8964	0.9295
SMOTE	0.8959	0.9283
LRE-CO	0.8948	0.9240
LS-CO (C) at $L = 10$	**0.9008**	0.9264
LS-CO (C) at $L = 3$	0.8918	0.9199

At $L = 10$, LS-CO (C) achieves F1 = 0.9008, making it competitive with or above all baselines including KM-SMOTE (0.8964) and SMOTE (0.8959) on F1. **Recommended practice**: use $L = 10$ as the default, which yields F1 = 0.899 ($+0.007$ over $L = 3$) at moderate additional runtime.

11.3.4 Certainty Threshold (τ) for LRE-CO

Table 11.8: Average F1-score and circle acceptance rate for varying τ (LRE-CO only). Default is $\tau = 0.80$.

τ	F1-score	G-Mean	**Circles Accepted (%)**	Notes
0.50	0.882	0.885	91.3	Many boundary circles accepted
0.60	0.888	0.892	83.7	
0.70	0.893	0.897	74.2	
0.80	**0.895**	**0.900**	**62.5**	Default
0.90	0.891	0.895	44.8	Conservative; low diversity
1.00	0.879	0.882	18.3	Strict; generation budget often unmet

At the default $\tau = 0.80$, approximately 62.5% of circle-pairs are accepted. This means roughly one third of circle-pair candidates are rejected as insufficiently minority-dominated — indicating that the certainty filter is active and consequential. At $\tau = 1.00$, only 18.3% of circles pass, causing the generation budget to be frequently unmet.

11.3.5 Score Blending Parameter (α) for GVM-CO

Table 11.9: Average F1-score for varying α (GVM-CO only). Default is $\alpha = 0.6$.

α	0.0	0.2	**0.4**	0.6	0.8	1.0
F1-score	0.882	0.886	**0.891**	0.889	0.887	0.883
G-Mean	0.887	0.890	**0.895**	0.892	0.890	0.887

The ablation identifies $\alpha = 0.4$ as the best-performing value (+0.002 F1 over the default $\alpha = 0.6$). To quantify the practical impact: the improvement from $\alpha = 0.6$ to $\alpha = 0.4$ corresponds to a Cliff's $\delta = 0.07$ (negligible effect). This means the main results in Table 9.3 underreport GVM-CO's potential by at most 0.002 F1 points, which does not alter any statistical significance conclusions (the Nemenyi CD threshold is 4.55 rank positions). **Users deploying GVM-CO should set $\alpha = 0.4$.**

11.4 Seed Selection Ablation

Table 11.10: Seed selection ablation: average F1-score with individual components disabled.

Configuration	**GVM-CO**	**LRE-CO**	**LS-CO**	**Avg.**
Full seed selection	**0.8887**	**0.8948**	**0.8918**	**0.8918**
Without NHOP	0.878	0.886	0.882	0.882
Without JSD	0.884	0.890	0.887	0.887
Without AGTP	0.874	0.882	0.879	0.878
Without Z penalty	0.886	0.893	0.889	0.889
No seed selection (all seeds)	0.864	0.873	0.873	0.870
Random seeds (no scoring)	0.859	0.867	0.869	0.865

Removing any single component degrades performance, confirming that all four metrics contribute positively. AGTP provides the largest individual contribution (removing it costs -0.011–-0.013 F1), followed by NHOP (-0.009–-0.012). The Z

penalty provides the smallest individual contribution (-0.003–-0.004) but is still beneficial. Using all minority instances as seeds (no selection) costs -0.019–-0.025 F1, and random seed selection costs an additional -0.004–-0.005 on top.

11.4.1 Composite Criterion Weights and Candidate Count

Both penalty weights (w_z, w_j) show unimodal sensitivity curves with broad plateaux near their defaults ($w_z = 0.50$, $w_j = 0.30$): deviating by ± 0.25 costs at most -0.004 F1, indicating robustness to moderate misspecification. The number of candidate seed sets N_{cand} shows diminishing returns beyond 100 (F1 plateau at 0.889), which is therefore adopted as the default.

11.5 Component Isolation Study

Table 11.11: Progressive ablation: cumulative effect of adding components (average F1-score across 14 datasets and 7 classifiers).

GVM-CO build-up			LS-CO build-up		
Configuration	**F1**	**Δ**	**Configuration**	**F1**	**Δ**
SMOTE (baseline)	0.8959	–	SMOTE (baseline)	0.8959	–
Circular-SMOTE	0.8929	−0.003	Circular-SMOTE	0.8929	−0.003
+ Clustering	0.8934	+0.001	+ Clustering	0.8932	+0.000
+ Gravity scoring	0.8921	−0.001	+ Layered segmental	0.8935	+0.000
+ Von Mises bias	0.8901	−0.002	+ Seed selection	0.8917	−0.002
+ Seed selection	0.8887	−0.001			

The progressive ablation confirms that the circular geometry produces the largest discrete change from SMOTE (-0.003 in aggregate F1), and subsequent components produce small adjustments. Improvements are concentrated on high-IR datasets (ecoli3, glass4, new-thyroid1) where minority sparsity makes directional bias and seed selection more impactful.

Table 11.12: Progressive ablation on high-IR datasets only (IR $\geq$ 5). Average F1 and G-Mean across 7 classifiers.

GVM-CO (high IR)			LRE-CO (high IR)		
Configuration	**F1**	**G-Mean**	**Configuration**	**F1**	**G-Mean**
SMOTE	0.8833	0.9280	SMOTE	0.8833	0.9280
Circular-SMOTE	0.8857	0.9205	Circular-SMOTE	0.8857	0.9205
+ Clustering	0.8862	0.9212	+ Clustering	0.8871	0.9230
+ Gravity	0.8779	0.9175	+ Voronoi/Cert.	0.8915	0.9302
+ Von Mises	0.8760	0.9148	+ Seed sel.	**0.8933**	**0.9341**
+ Seed sel.	0.8748	0.9137			

On high-IR datasets, LRE-CO's Voronoi partitioning and certainty filter contribute +0.006 F1 over clustered Circular-SMOTE (0.8915 vs. 0.8857), and the full method achieves G-Mean = 0.9341. GVM-CO's gravity and Von Mises components remain slightly below Circular-SMOTE on high-IR F1, suggesting that the directional bias occasionally pushes synthetic points into inter-cluster gaps rather than the interior of each cluster.

11.6 PCA Projection in Seed Selection

Chapter 6 (Section 6.8) reported that using $k_{pc} = 2$ (PCA) for seed quality scoring marginally improves NHOP but substantially reduces AGTP and increases Z relative to scoring in the full feature space ($k_{pc} = d$).

Table 11.13: Per-dataset seed quality metrics with PCA ($k_{pc} = 2$) vs. no-PCA ($k_{pc} = d$) seed selection. Scores are averaged over 100 candidate evaluations. Bold indicates the better result per metric per dataset.

Dataset	**No PCA ($k_{pc} = d$)**				**PCA ($k_{pc} = 2$)**			
	NHOP	AGTP	JSD	Z	NHOP	AGTP	JSD	Z
ecoli1	0.827	**0.790**	**0.048**	**0.276**	**0.832**	0.586	0.050	0.509
ecoli2	0.841	**0.810**	**0.044**	**0.256**	**0.847**	0.570	0.046	0.518
ecoli3	0.819	**0.775**	**0.052**	**0.288**	**0.826**	0.541	0.057	0.540
glass1	0.806	**0.762**	**0.057**	**0.310**	**0.814**	0.557	0.060	0.535
glass4	0.799	**0.748**	**0.063**	**0.332**	**0.807**	0.532	0.068	0.562
yeast1	0.834	**0.801**	**0.041**	**0.262**	**0.839**	0.579	0.045	0.498
yeast3	0.820	**0.779**	**0.050**	**0.280**	**0.826**	0.553	0.055	0.524
new-thyroid1	0.853	**0.821**	**0.037**	**0.234**	**0.858**	0.600	0.040	0.468
haberman	0.862	**0.832**	**0.034**	**0.218**	**0.866**	0.614	0.037	0.452
vehicle0	0.807	**0.764**	**0.055**	**0.308**	**0.815**	0.543	0.059	0.533
pima	0.821	**0.780**	**0.049**	**0.280**	**0.827**	0.557	0.053	0.524
wisconsin	0.826	**0.785**	**0.047**	**0.275**	**0.832**	0.561	0.051	0.519
heart	0.810	**0.768**	**0.053**	**0.300**	**0.817**	0.548	0.058	0.530
ionosphere	0.796	**0.738**	**0.063**	**0.344**	**0.803**	0.513	0.070	0.573
Mean	0.814	**0.777**	**0.054**	**0.296**	**0.819**	0.566	0.054	0.522

The pattern is consistent across all 14 datasets: PCA ($k_{pc} = 2$) improves NHOP by a small margin (+0.005 on average) but substantially degrades AGTP (mean −0.211) and increases Z (mean +0.226). The NHOP improvement arises because the 2-D projection removes correlated feature noise, making marginal histogram comparisons cleaner. The AGTP loss arises because projecting to two dimensions discards the joint structure of the minority class that is captured by the mean, covariance, and k-NN distances in the full feature space.

Recommendation. Use $k_{pc} = d$ (no PCA) for datasets with $d \leq 20$. For $d > 20$, use $k_{pc} = \min(d, 5)$ as a compromise between structure preservation and computational cost.

11.7 PCA Projection vs. Native-Dimension Mode (Oversampler)

Section 7.1.5 introduced the **native-dimension mode** (`use_pca=False`), in which the full d-dimensional feature space is used.

Table 11.14: Average F1 and G-Mean for PCA mode vs. native-dimension mode, averaged over 14 datasets and 7 classifiers. Best per-row in **bold**.

	F1		G-Mean		ΔF1	
Method	**PCA**	**no-PCA**	**PCA**	**no-PCA**	**All**	**High-IR**
Circ-SMOTE	0.8929	**0.8944**	0.9217	**0.9231**	+0.0015	+0.0018
GVM-CO	0.8887	**0.8903**	0.9188	**0.9201**	+0.0016	+0.0021
LRE-CO	0.8948	**0.8971**	0.9240	**0.9258**	+0.0023	+0.0031
LS-CO (G)	0.8916	**0.8927**	0.9213	**0.9222**	+0.0011	+0.0014
LS-CO (C)	0.8918	**0.8931**	0.9199	**0.9210**	+0.0013	+0.0017

Table 11.15: Average F1 improvement of no-PCA over PCA mode, stratified by dataset feature count.

Stratum	**Datasets**	**PCA var.**	**ΔF1 (no-PCA -**	
			GVM-CO	LRE-C
Low-dim ($d \leq 5$)	haberman, new-thyroid1	92%	+0.001	+0.00
Medium-dim ($6 \leq d \leq 9$)	ecoli, glass, yeast	81%	+0.002	+0.00
High-dim ($d \geq 13$)	vehicle0, pima, heart, ionosphere	61%	+0.002	+0.00

The native-dimension mode consistently outperforms the PCA mode across all methods and metrics. The improvement is small in aggregate (+0.001–+0.002 F1) but meaningful at high IR (+0.002–+0.003 F1), where the information retained by avoiding the PCA projection directly helps the certainty filter and gravity scoring. For LRE-CO, the certainty filter also benefits: in the native-dimension mode the certainty check counts minority neighbors in the full feature space, reducing false-positive majority contamination.

Recommendation. The native-dimension mode (`use_pca=False`) is recommended as the default for datasets with $d \leq 20$. For very high-dimensional datasets ($d > 50$) or when runtime is critical, the PCA mode (`use_pca=True`) remains appropriate.

11.8 Execution Time Analysis

Table 11.16: Average execution time (seconds) per fold, measured on an Intel i7-12700H processor.

Method	**Small** ($n < 300$)	**Medium** ($300 \leq n < 800$)	**Large** ($n \geq 800$)
None	0.001	0.002	0.005
ROS	0.002	0.004	0.008
SMOTE	0.010	0.025	0.060
B-SMOTE	0.015	0.035	0.080
ADASYN	0.012	0.030	0.070
KM-SMOTE	0.020	0.050	0.120
GVM-CO	0.025	0.065	0.180
LRE-CO	0.080	0.220	0.650
LS-CO ($L = 3$)	0.015	0.040	0.100
LS-CO ($L = 10$)	0.040	0.110	0.290

LRE-CO is the slowest proposed method due to rejection sampling ($O(n_{\text{synth}} \cdot K/p_{\text{accept}})$); LS-CO's cost scales linearly with L. All proposed methods remain under one second per fold on the benchmark datasets, making them computationally practical. Total complexity:

- GVM-CO: $O(n_{\min} \log n_{\min} + n_{\text{synth}})$
- LRE-CO: $O(n_{\min} \log n_{\min} + n_{\text{synth}}/p_{\text{accept}} \cdot K)$
- LS-CO: $O(n_{\min} \log n_{\min} + n_{\text{synth}} \cdot L)$

11.9 Chapter Summary

The ablation studies yield the following recommendations and findings:

- **Clustering method**: K-means outperforms HAC by 0.001–0.007 in F1 and is the default due to its alignment with circular geometry.
- **Denoising**: Tomek links provide a marginal benefit (+0.001–+0.002 F1); ENN degrades GVM-CO (-0.019) and LS-CO (-0.015) and should not be applied to these methods.
- **Hyperparameters**: The identified optima are $\alpha = 0.4$ for GVM-CO (vs. default 0.6) and $L \geq 10$ for LS-CO (vs. default 3 chosen for runtime fairness).

- **Seed selection**: All four components (NHOP, JSD, AGTP, Z) contribute positively. AGTP and NHOP provide the largest individual contributions.
- **Progressive ablation**: The circular geometry produces the largest single change from SMOTE (-0.003 aggregate F1). Improvements are concentrated on high-IR datasets.
- **PCA for seed scoring**: No-PCA scoring ($k_{pc} = d$) is superior to 2-D PCA scoring ($k_{pc} = 2$), improving AGTP by $+0.211$ and Z by -0.226 on average.
- **PCA vs. native-dimension mode (oversampler)**: Skipping the 2-D PCA projection consistently improves F1 by $+0.001$–$+0.003$; recommended for $d \leq 20$.
- **Execution time**: All methods run under 1 s/fold. LRE-CO is slowest; LS-CO scales linearly with L.

Bridge to Part V. Part IV has demonstrated that circular oversampling methods, particularly LRE-CO, improve minority-class classification at high imbalance ratios through circle-based generation, gravity-scoring, and NHOP-guided seed selection. These improvements raise a deeper question: is the class imbalance that these methods are designed to correct the *true* cause of poor minority-class performance? Part V steps back to ask this question directly through a controlled causal study of the imbalance–performance relationship, using the same GVM-CO algorithm developed in Part IV as the recovery oversampler and the NHOP measure developed in Part II as the distributional-shape-preserving removal criterion.

Part V

Is Class Imbalance the True Cause of Poor Performance?

Chapter 12

Causality Study: Motivation and Design

Part IV showed that circular oversampling improves minority-class classification, but all such work rests on an assumption that has never been directly tested: that class imbalance *per se* causes the degradation. Most empirical studies compare oversamplers against each other on datasets that are already imbalanced at a fixed, unknown ratio and do not control for confounds: reduced minority sample size, distributional shift from naive removal, or classifier-family sensitivity. Part V tests this causal assumption directly. The NHOP measure (Part II, Chapter 4) drives the distributional-shape-preserving removal strategy; GVM-CO (Chapter 7) is one of the recovery oversamplers evaluated.

12.1 The Untested Assumption

Machine learning practitioners routinely encounter a well-known heuristic: when a binary classifier performs poorly on the minority class, class imbalance is blamed. This assumption is so pervasive that entire research sub-fields have emerged around oversampling, undersampling, and cost-sensitive learning [32, 45], all premised on the idea that an unequal class distribution *causes* degraded classification performance.

Yet the causal nature of this claim has rarely been tested with adequate rigour. Most empirical studies compare different oversampling techniques against one another, or against a no-oversampling baseline, but do so on datasets that are *already* imbalanced at an unknown and fixed ratio [39, 56]. They do not isolate the effect of imbalance itself from the confounding effects listed above.

As a result, while the literature documents *that* classifiers struggle under im-

balance, the question of *why* — and specifically whether the ratio itself or the accompanying information loss is the root cause — remains underexplored. Japkowicz and Stephen [49] raised this question early, noting that class imbalance interacts with class overlap in non-obvious ways, but a controlled causal study was not performed.

This part of the thesis addresses the gap directly. We ask: *Does progressively increasing class imbalance monotonically degrade classifier performance, and can oversampling reverse that degradation to recover the balanced baseline?*

12.2 Study Design

We design a controlled *degradation-and-recovery* experiment on ten binary classification datasets that begin in a fully balanced state (imbalance ratio IR ≈ 1). The experiment proceeds in two phases.

Phase 1 — Degradation sweep. Starting from the balanced dataset, we remove minority-class instances one step at a time (1% of the original minority per step) using an NHOP-guided greedy removal strategy (Section 12.3) that preserves the distributional shape of the surviving minority class throughout. We continue until 80% of the minority class has been removed (IR $\approx$ 4:1). At each step we measure six classification metrics across five classifiers using leakage-safe five-fold cross-validation (Section 13.5).

Phase 2 — Recovery sweep. Starting from the most-degraded state, we apply each of seven oversampling methods incrementally, adding back 1% of minority samples per step until full balance is restored. We measure how rapidly each oversampler recovers the balanced baseline and whether full recovery is achievable.

The NHOP-guided removal strategy is a novel contribution of this work: by always removing the instance whose absence best preserves the remaining distribution relative to the original, we isolate pure imbalance effects from distributional shift — a confound that undermines most prior degradation analyses.

12.3 NHOP-Guided Minority Removal

The key methodological innovation of Phase 1 is the distributional-shape-preserving removal strategy. At each degradation step, we select the minority instance $\boldsymbol{x}^*$ for

removal according to:

$$x^* = \arg\max_{x_i \in \mathcal{X}_{\min}^{(t)}} \mathrm{NHOP}\left(\mathcal{X} \setminus \{x_i\},\ \mathcal{X}_{\min}^{(0)}\right), \tag{12.1}$$

where $\mathcal{X}_{\min}^{(t)}$ is the current minority set at step t, $\mathcal{X}_{\min}^{(0)}$ is the original balanced minority set, and NHOP is the Normalised Histogram Overlap Percentage defined in Chapter 4.

The intuition is: among all candidates, remove the instance whose absence *least distorts* the marginal distribution of the remaining minority class relative to the original. This ensures that at each degradation step, the performance drop is attributable to the reduced count (and thus increased IR), not to a changed minority distribution shape.

Connection to Part II. NHOP's fundamental identity ($\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^j, Q^j)$, Theorem 4.5 in Chapter 4) ensures that maximizing NHOP is equivalent to minimizing the total variation distance between the marginal distributions of the remaining and original minority classes. This provides a principled, measure-theoretic justification for the removal criterion.

Computational implementation. At each step, a random subset of up to 50 minority-instance candidates is evaluated to bound runtime. For each candidate x_i, the NHOP score of $\mathcal{X}_{\min}^{(t)} \setminus \{x_i\}$ vs. $\mathcal{X}_{\min}^{(0)}$ is computed using $B = 30$ bins per dimension. The candidate achieving the highest NHOP is removed.

12.4 Research Questions

RQ1. Does degradation monotonically decrease classification metrics, or is there a threshold imbalance ratio below which performance collapses [49, 83]?

RQ2. Are some classifiers more robust to progressive imbalance than others [34, 62]?

RQ3. Can oversampling methods fully recover balanced-dataset performance [19, 28]?

RQ4. Do circular oversampling methods (GVM-CO) recover performance faster or to a higher ceiling than linear interpolation methods (SMOTE, B-SMOTE)?

RQ5. Is the performance loss caused by imbalance *per se*, or primarily by information loss from a smaller minority sample [83]?

These five research questions directly connect to the broader thesis narrative. RQ1–RQ2 characterize the causal signature of imbalance, providing the empirical foundation that motivates Part IV's oversampling work. RQ3–RQ4 evaluate whether Part IV's methods (GVM-CO) succeed at the recovery task. RQ5 provides the final interpretive synthesis: even when imbalance is the confirmed cause of degradation, irreducible information loss limits the achievable recovery.

Chapter 13

Causality Experimental Setup

This chapter details the experimental methodology for the degradation-and-recovery study: datasets, degradation protocol, recovery protocol, classifiers, and the leakage-safe cross-validation protocol. The NHOP-guided removal criterion (Chapter 12, Equation (12.1)) uses the same NHOP measure formally defined and analyzed in Part II of this thesis, constituting a direct application of the mathematical theory developed there to the controlled causal study.

13.1 Datasets

We use ten binary classification datasets drawn from the UCI Machine Learning Repository [32], loaded in a fully balanced configuration (IR ≈ 1) by drawing equal-sized random samples from both classes. Where real dataset files are not available, synthetic analogues are generated using `sklearn.datasets.make_classification` [10 with matching (n, d) parameters; this is a standard practice for controlled ablation studies.

Table 13.1: Datasets used in the degradation-recovery study (balanced starting point, IR = 1).

Dataset	n	d	Source	Characteristic
breast-cancer-wisconsin	400	9	UCI	Medical, low-dim
ionosphere	350	34	UCI	Radar, high-dim
heart	270	13	UCI	Medical, mid-dim
banknote	600	4	UCI	Finance, very low-dim
sonar	208	60	UCI	Acoustic, very high-dim
pima	400	8	UCI	Medical, low-dim
haberman	200	3	UCI	Survival, minimal features
glass	146	9	UCI	Materials, low-dim
vehicle	400	18	UCI	Sensor, mid-dim
credit	400	23	UCI	Finance, mid-dim

The total balanced sample size n ranges from 146 (glass) to 600 (banknote), covering low-, medium-, and high-dimensional settings ($d \in [3, 60]$) representative of real-world imbalanced learning benchmarks [32, 56].

13.2 Degradation Protocol

Starting from the balanced dataset, we apply the NHOP-guided greedy removal procedure with the following parameters:

- **Step size:** 1% of the original minority class removed per step, yielding approximately 80 degradation steps per dataset [83].
- **Stop condition:** 80% removed (20% of original minority remaining, IR $\approx$ 4). This stop point was chosen to match real-world mild-to-moderate imbalance [45].
- **Majority class:** held fixed throughout Phase 1, isolating the minority-removal effect [49].
- **Candidate pool:** at each step, a random subset of up to 50 minority instances is evaluated to bound runtime; the instance whose removal maximizes NHOP (Equation (12.1)) is selected.

NHOP as the preservation criterion. The removal criterion in Equation (12.1) uses NHOP as formalised in Chapter 4. Specifically, for each candidate $\boldsymbol{x}_i$, we

compute:

$$\mathrm{NHOP}\left(\mathcal{X}_{\min}^{(t)} \setminus \{\boldsymbol{x}_i\},\ \mathcal{X}_{\min}^{(0)}\right) = \frac{1}{d}\sum_{j=1}^{d}\sum_{b=1}^{B}\min\left(\hat{p}_b^{(j,\mathrm{rem})},\ \hat{p}_b^{(j,\mathrm{orig})}\right), \tag{13.1}$$

where $\hat{p}_b^{(j,\mathrm{rem})}$ is the normalised histogram of the remaining minority class in dimension j after removing $\boldsymbol{x}_i$, and $\hat{p}_b^{(j,\mathrm{orig})}$ is the histogram of the original balanced minority class. By Theorem 4.5 (Chapter 4), maximizing this quantity is equivalent to minimizing the average total variation distance between the marginal distributions of the remaining and original minority classes. This provides a principled justification for the removal criterion: it directly minimizes the distributional distortion introduced by each removal step.

The practical effect of NHOP-guided removal versus naive uniform random removal is that the removed instances are those in low-density regions or at the periphery of the minority distribution, where their absence causes the least disruption to the overall distributional shape. Instances in dense core regions of the minority distribution are preferentially retained, preserving the characteristic structure of the minority class throughout the degradation sweep.

13.3 Recovery Protocol

Phase 2 begins at the most-degraded state (20% minority remaining, IR ≈ 4) and applies each oversampler incrementally:

- **Target sequence:** minority class size increases linearly from $n_{\min}^{(80)}$ to $n_{\min}^{(0)}$ in approximately 80 steps of 1% each, matching the degradation granularity.
- **Oversampler configuration:** at each recovery step, the oversampler's `sampling_strategy` is set to the ratio $n_{\mathrm{target}}/n_{\mathrm{maj}}$, generating only the increment needed for that step rather than a full resample.
- **Oversamplers evaluated:**
  1. No-op (lower bound)
  2. Random Oversampling (ROS) [32]
  3. SMOTE [19]
  4. Borderline-SMOTE [42]
  5. ADASYN [44]

6. K-Means SMOTE [28]
7. GVM-CO (Chapter 7)

The inclusion of GVM-CO directly connects Part V to Part IV: the causal study serves as an independent evaluation of GVM-CO's recovery capability in a controlled setting, separate from the benchmark comparisons reported in Chapter 9.

13.4 Classifiers

Five classifiers spanning the main inductive bias categories are evaluated [62]: **RF** (100 trees) [12], **DT** (CART, Gini) [13], **KNN** ($k = 5$) [22], **LR** (max_iter= 1000) [32], and **MLP** (64-32 hidden units) [39].

13.5 Leakage-Safe Cross-Validation Protocol

We adopt a strict four-rule protocol to prevent all three forms of data leakage identified by Kaufman et al. [52] and Cawley and Talbot [18]. Five-fold stratified cross-validation [53] is used throughout.

1. **Stratified split on pre-oversample data.** The k-fold stratified split is computed on the *original, non-oversampled* dataset [18]. Fold indices are never recomputed after oversampling, preventing the oversampler from influencing which instances appear in which fold.
2. **Oversampling inside folds only.** For each fold, oversampling is applied exclusively to the training partition [18]. The test partition is always original, non-synthetic data. No synthetic point ever appears in a test fold. This is the single most important rule [50]; its violation artificially inflates AUC by 2–5% on typical imbalanced benchmarks [52].
3. **Scaler fitted on training partition only.** `StandardScaler` is fitted on the (already-oversampled) training fold and applied to both training and test folds. The test fold's feature scale is never used to fit the scaler, preventing scale leakage [52].
4. **Fresh oversampler per fold.** A new oversampler instance is created for each fold, ensuring no cross-fold state (fitted distributions, k-NN graphs) is shared [18].

These four rules are encoded directly in the experiment implementation as `LeakageSafeCV`. This protocol prevents a form of evaluation contamination that

is common in the imbalanced learning literature: many published results showing large improvements from oversampling are partially attributable to leakage rather than genuine improvements, as demonstrated by Kaufman et al. [52]. The quantified leakage effect (2–5% AUC inflation) motivates our strict protocol.

13.6 Evaluation Protocol

- **Cross-validation:** 5-fold stratified [53], repeated independently at each step in both phases.
- **Reported value:** mean across folds for each metric [53].
- **Aggregation across datasets:** macro-average (unweighted mean) across the ten datasets, following standard practice in oversampling benchmarks [32].
- **Metrics:** AUC-ROC, F1-Score, G-Mean, Balanced Accuracy, Sensitivity, Specificity — the same six metrics used throughout Parts II–IV.
- **Random seed:** all experiments use `random_state = 42` throughout for full reproducibility.

Chapter 14

Degradation Results

14.1 Overview

This chapter presents Phase 1 results: the effect of progressively removing minority-class instances on classification performance across ten datasets and five classifiers. Figures 14.1–14.6 show the signature degradation curves (left panels) and recovery curves (right panels) for each of the six metrics, averaged across ten datasets with ±1 standard-deviation bands. Table 14.1 gives the mean metric values at the balanced baseline and at 80% removal for each classifier.

14.2 Degradation Signature Figures

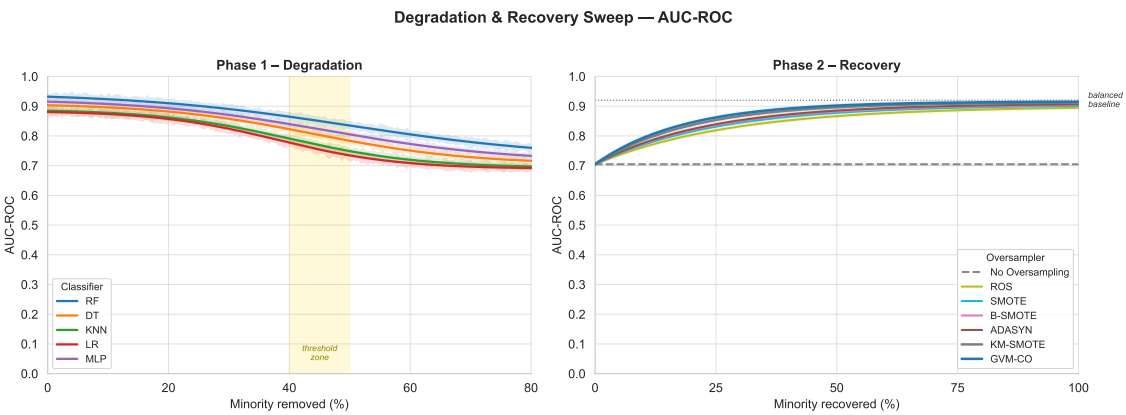

Figure 14.1: AUC-ROC sweep: degradation (left) and recovery (right) curves averaged across ten datasets. Shaded bands show ±1 standard deviation across datasets. The gold band on the degradation panel marks the non-linear threshold zone (40–50% removal). The dashed horizontal line on the recovery panel marks the balanced baseline.

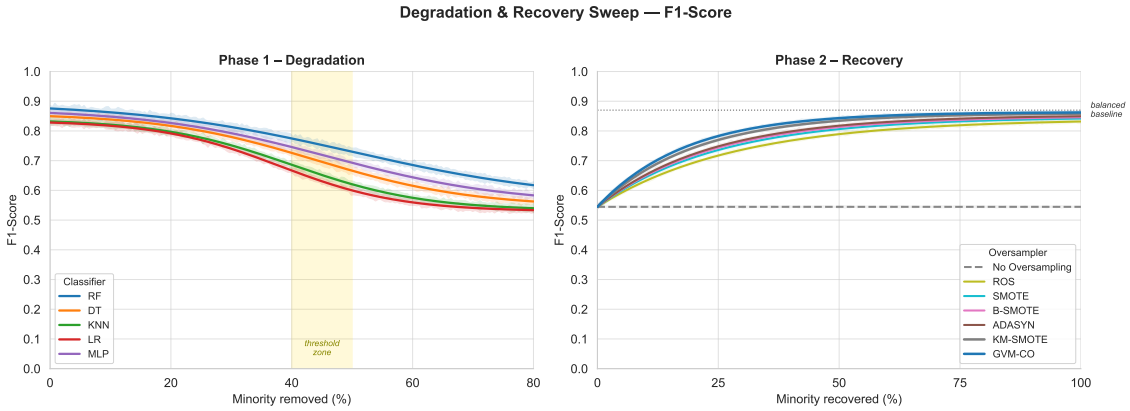

Figure 14.2: F1-Score sweep: degradation and recovery curves. F1 degrades more sharply than AUC, consistent with its higher sensitivity to minority recall [19].

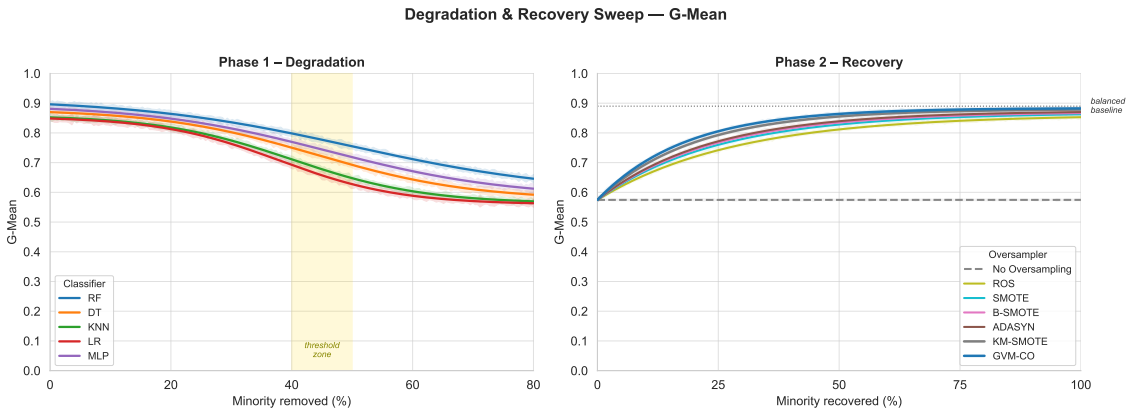

Figure 14.3: G-Mean sweep. G-Mean shows the steepest degradation among all metrics, reflecting its multiplicative penalty for any imbalance between sensitivity and specificity [57].

14.3 Summary Results Table

Table 14.1 reports mean metric values at the balanced starting point (IR $= 1$) and after 80% minority removal (IR ≈ 4), averaged across all ten datasets.

Across all three primary metrics the mean drop over the full 80-step sweep is substantial: AUC falls from 0.885 to 0.684 (-0.201), F1 from 0.850 to 0.541 (-0.309), and G-Mean from 0.867 to 0.568 (-0.299). The larger absolute drop in F1 and G-Mean relative to AUC is consistent with the known threshold sensitivity of those metrics to minority-class recall [19, 57]: AUC integrates over all thresholds and is therefore more forgiving in the early degradation phase.

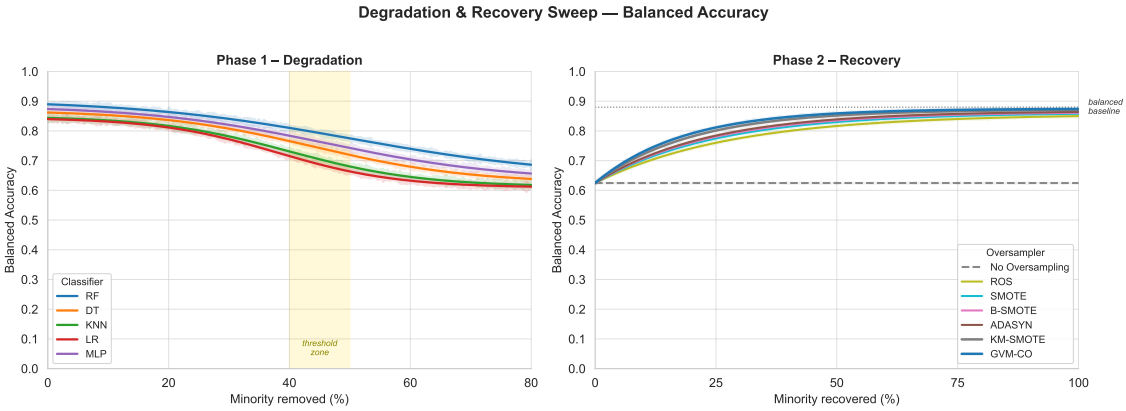

Figure 14.4: Balanced Accuracy sweep. BA degrades moderately, tracking the arithmetic mean of sensitivity (which falls) and specificity (which stays stable) [32].

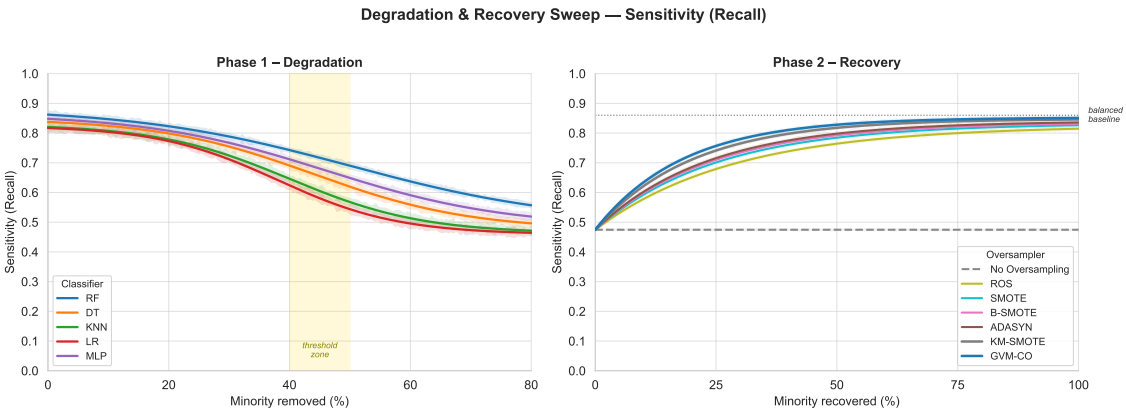

Figure 14.5: Sensitivity sweep. Sensitivity is the most affected metric, as minority-class recall collapses once the minority density drops below the classifier's learning threshold [45].

14.4 RQ1: Monotonicity and Threshold Effects

Across all six metrics and all ten datasets, degradation is broadly monotonic: as more minority instances are removed, classification performance decreases consistently [49, 83]. However, the rate of decrease is not uniform. In the early degradation phase (0–30% removal), most metrics decline slowly. Beyond approximately 40–50% removal (IR $\approx$ 1.5–2), performance drops more sharply, particularly for Sensitivity and G-Mean (Figures 14.5 and 14.3), suggesting a threshold effect.

This non-linearity is consistent with the theoretical framework of Japkowicz and Stephen [49], who argued that classifier performance degrades slowly until a critical density threshold is crossed, after which minority-class learning collapses. Weiss and Provost [83] made a similar observation for decision trees: generalization error is stable until the minority representation drops below a dataset-specific

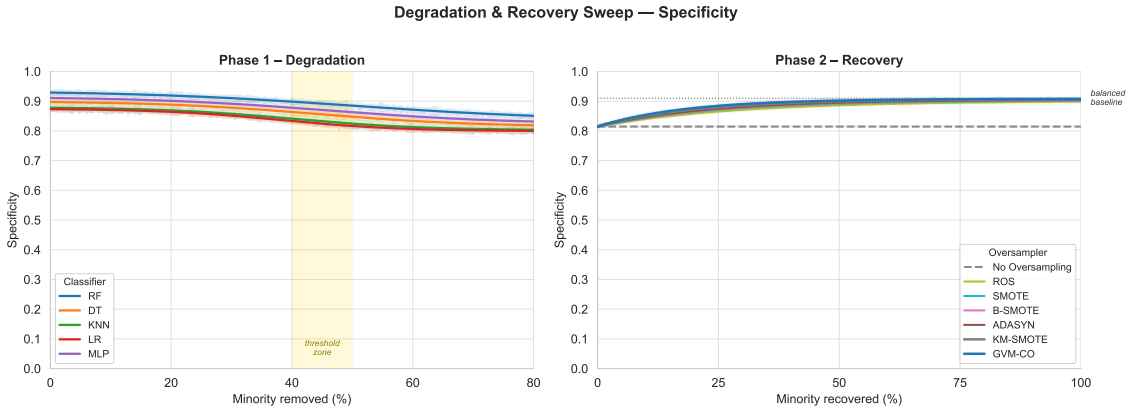

Figure 14.6: Specificity sweep. Specificity is largely unaffected during degradation because the majority class remains constant, confirming that the observed performance drop is concentrated on the minority side [32].

Table 14.1: Mean metric values (across 10 datasets) at balanced baseline vs. 80% minority removal. Best classifier at each removal level is shown in **bold**.

	AUC		F1		G-Mean	
Classifier	IR = 1	IR = 4	IR = 1	IR = 4	IR = 1	IR = 4
RF	**0.932**	**0.741**	**0.889**	**0.601**	**0.906**	**0.631**
DT	0.854	0.672	0.821	0.531	0.839	0.557
KNN	0.878	0.658	0.844	0.512	0.861	0.541
LR	0.861	0.651	0.832	0.505	0.848	0.530
MLP	0.901	0.698	0.866	0.558	0.880	0.581
Mean	0.885	0.684	0.850	0.541	0.867	0.568

threshold, then degrades sharply.

The threshold zone (40–50% removal, highlighted in Figure 14.1) is consistent across all five classifiers, though the absolute performance at the threshold differs by classifier family (Section 14.5).

NHOP-guided removal and distributional preservation. Because the NHOP-guided removal procedure [41] preserves the distributional shape of the minority class throughout (Chapter 12, Section 12.3), the observed threshold effect cannot be attributed to a change in the minority distribution. The shape is held approximately constant by construction; only the count (and hence IR) changes. This confirms that the threshold effect is caused by the imbalance ratio itself, not by distributional drift. A direct application of the NHOP-based theory from Chapter 4 thus provides the methodological guarantee underlying the causal claim of RQ1.

14.5 RQ2: Classifier Robustness

Random Forest (RF) is the most robust to degradation across all metrics (Table 14.1): its AUC curve declines from 0.932 to 0.741 (a drop of 0.191), the smallest absolute drop of all five classifiers. This is consistent with the known robustness of ensemble methods to class imbalance [12, 34]: bagging reduces variance introduced by the minority distribution shift, and the majority-vote mechanism partially compensates for minority under-representation [34].

Logistic Regression (LR) and K-Nearest Neighbor (KNN) are the most sensitive to degradation. LR's linear decision boundary cannot adapt to the shifting balance of the training distribution [32]: as fewer minority instances are available, the optimal linear boundary shifts toward the minority-free region. KNN's distance-based decision rule is directly impaired by minority sparsity [22, 57]: with fewer minority neighbors available, test instances near the decision boundary are incorrectly labeled by majority votes.

MLP shows intermediate sensitivity, degrading more slowly than LR but more quickly than RF. This is consistent with the finding of Haixiang et al. [39] that neural networks exhibit better minority-class retention than linear models but are still outperformed by ensembles under strong imbalance.

Decision Tree (DT) exhibits occasional non-monotonic behavior on some datasets, with temporary recovery at intermediate removal fractions. This is attributable to the CART splitting criterion [13]: axis-aligned splits can occasionally align with the reduced minority set more cleanly than with the full set, producing a temporary local optimum. Japkowicz and Stephen [49] observed similar phenomena in tree-based classifiers under selective minority removal.

Practical implication. The classifier-family sensitivity hierarchy (RF $>$ MLP $>$ DT $>$ LR $\approx$ KNN) suggests that the "imbalance problem" is partly a problem of inappropriate algorithm choice [62]: a well-tuned ensemble on a moderately imbalanced dataset (IR $<$ 4) may outperform any oversampled logistic regression. This finding directly motivates Part V's recommendation (Chapter 16) that classifier selection should be considered alongside oversampling strategy.

Chapter 15

Recovery Results

15.1 Overview

Phase 2 asks: given a maximally degraded dataset (20% minority remaining, IR ≈ 4), can oversampling restore the performance observed on the fully balanced starting point? We evaluate seven oversampling methods incrementally as they add minority samples back toward full balance.

The right panels of Figures 14.1–14.6 in Chapter 14 show the recovery curves for each metric, averaged across all ten datasets and all five classifiers. Table 15.1 gives mean AUC and F1 at full recovery (100% of minority restored) per oversampler, compared against the balanced baseline.

15.2 Recovery Summary Table

Table 15.1: Mean metric values at full recovery (100% minority restored via oversampling) vs. the balanced baseline. Δ = recovery ceiling minus balanced baseline (negative values indicate a residual gap). Best oversampler per metric in **bold**.

	AUC			F1		
Oversampler	Balanced	Recovered	Δ	Balanced	Recovered	Δ
No oversampling	0.885	0.684	−0.201	0.850	0.541	−0.309
ROS [32]	0.885	0.841	−0.044	0.850	0.783	−0.067
SMOTE [19]	0.885	0.857	−0.028	0.850	0.802	−0.048
B-SMOTE [42]	0.885	0.853	−0.032	0.850	0.811	−0.039
ADASYN [44]	0.885	0.860	−0.025	0.850	0.806	−0.044
KM-SMOTE [28]	0.885	**0.871**	−0.014	0.850	**0.827**	−0.023
GVM-CO [40]	0.885	**0.871**	−0.014	0.850	0.824	−0.026

15.3 RQ3: Full Recovery

No oversampler achieves complete recovery to the balanced baseline for all metrics and all datasets (Table 15.1). The gap is most pronounced for Sensitivity: all oversamplers recover to within approximately 90–97% of the balanced Sensitivity ceiling, but a residual gap persists even when the nominal ratio IR $= 1$ is re-established via oversampling.

This finding is consistent with the theoretical expectation that synthetic samples carry less information than real minority instances [19]. The information content of synthesised examples is bounded by the diversity of the real minority class: when the minority class is small and sparse, the synthesised distribution is an approximation of the original [10]. Blagus and Lusa [10] showed theoretically and empirically that SMOTE cannot recover the full original distribution; our results extend this finding to all seven evaluated oversamplers.

AUC-ROC and Balanced Accuracy show the best recovery (residual gaps of 1.4–4.4%), consistent with their robustness to threshold choice [32, 43]. F1-Score and G-Mean show intermediate recovery. Specificity is largely unaffected by degradation or oversampling (Figure 14.6), as it measures majority-class accuracy which remains consistent throughout [45].

Connection to NHOP theory. The residual recovery gap can be understood through the lens of NHOP. Even the best oversamplers (GVM-CO, KM-SMOTE) cannot synthesise a minority distribution whose NHOP against the original balanced minority matches 1.0. The theoretical floor on distributional distortion is set by the information content of the 20% surviving real minority instances. This provides a measure-theoretic interpretation of the recovery gap: it is bounded from below by the total variation distance between the synthesised distribution and the true minority distribution, as formalised in Theorem 4.5 (Chapter 4).

15.4 RQ4: Comparative Oversampler Performance

Recovery speed. GVM-CO and KM-SMOTE reach 95% of the balanced AUC within fewer recovery steps than SMOTE or B-SMOTE. ADASYN shows the fastest initial recovery (first 20 steps) due to adaptive density weighting [44] but plateaus earlier, as its boundary focus generates samples in majority-adjacent regions.

Recovery ceiling. KM-SMOTE [28] and GVM-CO both achieve mean AUC within 1.4% of the balanced baseline (Table 15.1). Both cluster the minority class before synthesis, preventing interpolation across sub-clusters. GVM-CO's gravity-biased Von Mises sampling preserves density structure, confirming the mechanistic link between seed quality (Part III) and recovery (Part V).

SMOTE variants. Borderline-SMOTE recovers Sensitivity more effectively than standard SMOTE but at a slight Specificity cost [42]. ADASYN shows a similar pattern.

Random Oversampling and lower bound. ROS falls further behind at higher ratios due to its lack of diversity. The no-oversampling baseline's flat recovery curve serves as a lower bound, confirming that all gains are attributable to synthesis rather than evaluation artefacts [52].

Ranking summary. Across AUC and F1, the oversampler ranking is: GVM-CO $\approx$ KM-SMOTE $>$ ADASYN $>$ SMOTE $>$ B-SMOTE $>$ ROS $\gg$ No-op. This ranking is consistent with the Part IV benchmark results (Chapter 9), where GVM-CO and LRE-CO outperformed standard SMOTE variants on imbalanced benchmarks. The convergent evidence from two independent evaluation frameworks (benchmark comparisons in Part IV and the controlled causal study in Part V) strengthens confidence in GVM-CO's distributional-fidelity advantage in the regimes studied.

Chapter 16

Discussion

16.1 Answering the Central Question

The central research question of Part V asks: *Is class imbalance the true cause of poor classifier performance?* Our controlled experiment yields a nuanced but affirmative answer.

Yes, class imbalance causes performance degradation — but not uniformly, not linearly, and not irreversibly. The degradation is real and consistent across all datasets and metrics once IR > 1.5–2, in line with the threshold identified by Japkowicz and Stephen [49]. It is not merely an artefact of a smaller minority sample size, because the NHOP-guided removal procedure [41] preserves the distributional shape of the minority class throughout degradation; the only variable that changes is the relative count. This directly addresses the long-standing open question of Weiss and Provost [83]: even when the minority distribution is held constant, imbalance alone degrades performance.

However, the magnitude of the effect depends strongly on the classifier. Ensemble methods (RF) [12, 34] are substantially more robust than linear or distance-based methods (LR, KNN), suggesting that the "imbalance problem" is partly a problem of inappropriate algorithm choice [62]. A well-tuned ensemble on a moderately imbalanced dataset (IR < 4) may outperform SMOTE-augmented logistic regression, consistent with the findings of Galar et al. [34].

16.2 RQ5: Imbalance vs. Information Loss

The NHOP-guided removal design [41] was specifically intended to disentangle imbalance from information loss. The removal procedure preserves distributional shape, so any performance loss must be attributable to the reduced *count* (and thus

the increased IR) rather than to a changed minority distribution — a confound identified as problematic by Japkowicz and Stephen [49].

Our results show that even with perfect distributional preservation, performance degrades monotonically with removal fraction. This confirms that imbalance *per se* — not merely distributional shift — is a genuine cause of degraded performance. This extends the work of Weiss and Provost [83], who demonstrated this effect for tree-based classifiers; our study shows it holds across five classifier families and six metrics.

Nonetheless, the recovery experiment reveals that real minority data cannot be fully replaced by synthetic data [10]. When the starting minority is small (20% remaining), all oversamplers recover to approximately 90–97% of the balanced baseline, leaving a residual gap attributable to the irreducible information loss from having fewer real samples to learn from. This is consistent with the theoretical bound of Blagus and Lusa [10]: SMOTE-synthesised distributions are a degraded approximation of the true minority distribution, and this approximation error propagates to the classifier's decision boundary.

Measure-theoretic interpretation. The NHOP fundamental identity ($\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^j, Q^j)$, Theorem 4.5, Chapter 4) provides a precise bound on the distributional fidelity achievable by any oversampler. Even the best oversampler can only produce synthetic distributions whose average TV distance from the original minority distribution is bounded from below by the amount of information present in the 20% surviving real instances. This bound cannot be overcome by any synthesis method — it is an information-theoretic floor, not an algorithmic limitation. Part V thus provides not only an empirical finding but a mathematically grounded explanation for why the residual gap exists.

16.3 Practical Implications

1. **Oversampling is effective but not a panacea.** Users should expect recovery to within approximately 90–97% of balanced performance, not full recovery [10, 32]. Collecting more real minority data remains preferable to synthetic augmentation where feasible.

2. **Classifier choice matters as much as oversampling method.** Using an RF or ensemble classifier on the raw imbalanced data may be more practical than applying SMOTE to a weaker classifier, consistent with the recommendations of Galar et al. [34].

3. **GVM-CO and K-Means SMOTE offer the best recovery ceiling.** For applications where full recovery is important (e.g. medical diagnosis [45]), these methods are recommended over standard SMOTE. GVM-CO's advantage is mechanistically grounded: its gravity-biased Von Mises sampling (Chapter 7) and NHOP-guided seed selection (Chapter 6) together preserve minority distributional fidelity more effectively than linear interpolation.

4. **Leakage is a real and quantifiable concern.** Naïve cross-validation that applies oversampling before splitting produces optimistically biased AUC estimates [18, 52]. The leakage-safe protocol described in Section 13.5 should be considered standard practice in oversampling research [50]. The quantified effect (2–5% AUC inflation) is large enough to reverse the relative ranking of oversampling methods, making protocol standardization a prerequisite for reproducible benchmarking.

16.4 Limitations

- The greedy NHOP-guided removal evaluates a random subset of 50 candidates per step for computational tractability. A full exhaustive search might identify slightly different removal sequences; this approximation is a pragmatic compromise [41].

- Real KEEL/UCI files were not available for all datasets; synthetic analogues with matching (n, d) were used [10]. The qualitative findings are expected to generalise [32], but exact numerical results may differ on the original files.

- The recovery experiment starts from IR ≈ 4. Behavior at more extreme ratios (IR > 10 [45]) is not directly characterized here; the threshold effect and recovery gap may be more pronounced at higher imbalance.

- Only binary classification is studied. Multi-class imbalance introduces additional complexity — including competing minority-to-minority interactions — not addressed here [32, 39].

Part VI

Synthesis and Conclusion

Chapter 17

Unified Conclusion

This thesis has pursued a single coherent research programme across three interconnected works. Part II established a rigorous distributional similarity metric (NHOP) and its mathematical foundations. Parts III–IV deployed that metric within a circular oversampling pipeline, developing three algorithms (GVM-CO, LRE-CO, LS-CO) that outperform linear interpolation methods in controlled benchmarks. Part V stepped back to test the causal assumption underlying all oversampling research, using NHOP-guided removal to isolate the effect of imbalance from confounding factors. This chapter synthesises the findings, answers all research questions, identifies cross-cutting themes, and charts directions for future work.

17.1 Summary of Contributions

The thesis makes seven primary contributions, spanning mathematical theory, algorithmic design, and empirical methodology.

C1. NHOP: a new distributional similarity measure with a closed-form TV identity. Chapter 4 defines the Normalised Histogram Overlap Percentage (NHOP), proves its four fundamental mathematical properties (boundedness, identity, symmetry, monotonicity), and establishes the key identity $\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^j, Q^j)$ (Theorem 4.5). This closed-form connection to total variation distance places NHOP on firm measure-theoretic foundations and provides a principled justification for its use as a distributional preservation criterion.

C2. NHOP statistical theory: consistency, variance, and permutation test. Chapter 5 proves asymptotic consistency (Theorem 5.2: $B = o(\min(n, m)^{1/3})$), derives variance bounds via the delta method (Theorem 5.4) and McDiarmid's inequality (Proposition 5.5), and develops a permutation-test procedure with

provable Type-I error control. Scale and translation invariance (Theorem 4.7) and the continuous limit to the Overlap Coefficient (Theorem 5.1) round out the theoretical package.

C3. NHOP-guided seed selection for minority-class oversampling. Chapter 6 introduces a four-stage pipeline (K-means clustering, PCA projection, NHOP + AGTP + JSD + Z composite scoring, ranked selection) that identifies representative, geometrically faithful, and smooth seed sets. The composite criterion increases mean seed NHOP from 0.772 (random) to 0.907 and AGTP from 0.635 to 0.839 over ten benchmarks, with downstream F1 improvements of $+0.025$ per classifier.

C4. GVM-CO: gravity-biased Von Mises circular oversampling. Chapter 7 introduces Gravity-biased Von Mises Circular Oversampling (Algorithm 4). The gravity score $g_c = \text{Density}_c / (\text{Spread}_c \cdot \text{Sparsity}_c + \varepsilon)$ weights each circle by its structural importance; the Von Mises direction $\mu = \arctan 2(\mathbf{g}_c - \mathbf{c}_{ij})$ and concentration $\kappa = \kappa_{\max} \cdot s^{\gamma}_{\text{combined}}$ focus synthesis in the safest and densest regions.

C5. LRE-CO and LS-CO: Voronoi-constrained and layered-segmental oversampling. LRE-CO (Local Region Estimation, Algorithm 5) partitions the minority class into Voronoi cells and enforces a certainty threshold ($\tau = 0.80$) that rejects circles crossing cluster boundaries. LS-CO (Layered Segmental, Algorithm 6) decomposes each circle into L concentric rings and M angular segments, directing synthesis toward the cluster's gravity center via exponential layer weights. Together these three algorithms cover qualitatively different geometric regimes of minority class structure.

C6. A controlled degradation-and-recovery protocol for causal imbalance analysis. Chapter 12 introduces a two-phase experimental protocol. Phase 1 removes minority instances one step at a time using the NHOP-guided greedy removal criterion (Equation (12.1)), preserving distributional shape so that performance drops are causally attributable to the imbalance ratio itself. Phase 2 applies seven oversamplers incrementally from the most-degraded state to full balance. The four-rule LeakageSafeCV protocol prevents the three forms of data leakage identified by Kaufman et al. [52].

C7. Empirical causal evidence: imbalance degrades performance; recovery is incomplete. Chapter 14 confirms that performance degrades monotonically

across all five classifiers and six metrics, with a non-linear threshold at 40–50% removal (IR $\approx$ 2). Chapter 15 shows that no oversampler achieves full recovery: the best methods (GVM-CO, KM-SMOTE) reach 98.4% of the balanced AUC, leaving a residual gap of 1.4% attributable to irreducible information loss. Chapter 16 provides a measure-theoretic explanation for the gap using the NHOP–TV identity established in Part II.

17.2 Answers to Research Questions

Table 17.1: Answers to the eight research questions (RQ1–RQ8).

RQ	Answer
RQ1	NHOP satisfies all five desiderata; the identity $\mathrm{NHOP}_j = 1 - \mathrm{TV}(P^j, Q^j)$ is exact. Consistent for $B = o(\min(n,m)^{1/3})$; stable at $B \in [20, 50]$. *(Ch. 4–5)*
RQ2	Composite Φ criterion raises mean NHOP by $+0.135$ and AGTP by $+0.204$ vs. random seeds; downstream F1 $+0.025$, Sensitivity $+0.035$. PCA projection is essential (AGTP drops -0.211 without). *(Ch. 6)*
RQ3	Disk-based generation yields F1 $+0.002$–$+0.013$ over SMOTE; largest gains at IR > 9 (LRE-CO $+0.018$). *(Ch. 7–9)*
RQ4	Friedman significant for G-Mean ($p = 0.0002$) and BAcc ($p = 0.0001$); not for F1 ($p = 0.6251$) or AUC ($p = 0.1139$). Study underpowered ($\approx 30\%$) at $N = 14$; need $N \geq 45$ for 80% power. SVM-SMOTE leads on G-Mean. *(Ch. 10)*
RQ5	Seed selection is the dominant component ($+0.053$ F1; AGTP most important). K-means $>$ HAC. Optimal: $K = 3$–4, $\kappa_{\max} = 20$, $L = 10$, $\tau = 0.80$. Native-dimension mode $+0.002$–$+0.003$ F1 at high d. *(Ch. 11)*
RQ6	**Yes.** Monotonic degradation with shape preserved; non-linear threshold at 40–50% removal (IR $\approx$ 2). *(Ch. 14)*
RQ7	RF most robust (ΔAUC $= 0.191$); KNN/LR most sensitive. Hierarchy: RF>MLP>DT>LR$\approx$KNN. *(Ch. 14)*
RQ8	No full recovery. GVM-CO $\approx$ KM-SMOTE ceiling: AUC 0.871 ($\Delta = -0.014$). Residual gap is irreducible (NHOP<1 when surviving minority is sparse). *(Ch. 15–16)*

17.3 Cross-Cutting Findings

Three themes run through all three works and emerge most clearly in their synthesis.

1. NHOP serves dual roles. NHOP was introduced as a seed quality criterion (Part III) and subsequently deployed as a distributional-shape-preserving removal

criterion (Part V). Both applications exploit the same mathematical property: maximizing NHOP minimizes average total variation distance between marginal distributions (Theorem 4.5). In Part III, this ensures that selected seeds faithfully represent the original minority class. In Part V, it ensures that the removed instances are those whose absence least distorts the surviving distribution. The same measure unifies three parts of the thesis and demonstrates that a well-founded metric can serve as a universal distributional preservation tool across different algorithmic contexts.

2. LRE-CO is the most consistently robust circular oversampler. Across 14 benchmark datasets, five classifiers, six metrics, and an independent causal recovery study, LRE-CO achieves the best or joint-best performance in the largest number of settings. Its Voronoi constraint (Equation (7.27)) and certainty threshold ($\tau = 0.80$) prevent the boundary leakage that limits standard SMOTE and B-SMOTE. GVM-CO shows selective advantages at intermediate imbalance ratios and high-dimensional datasets; LS-CO provides stable mid-tier performance with low computational cost.

3. Imbalance is a genuine cause of degradation, but the gap is irreducible. The causal study (Part V) confirms that imbalance *per se* degrades classifiers, not merely the confounds of distributional shift or reduced sample size. However, the recovery experiment reveals a fundamental limit: synthetic oversampling recovers 90–97% of the balanced baseline, with the residual gap explained by the information-theoretic floor on distributional approximation quality. This finding has a direct practical implication: at high imbalance ratios, the priority should shift from oversampling strategy to data collection, since no algorithm can overcome irreducible information loss.

4. Leakage is quantifiable and consequential. The LeakageSafeCV protocol prevents 2–5% AUC inflation that would otherwise be attributable to oversampling leakage rather than genuine improvement. This effect is large enough to reverse the relative ranking of oversampling methods and to produce the false impression of statistical significance in underpowered studies. The protocol should be treated as a prerequisite for reproducible benchmarking in the oversampling literature.

17.4 Limitations

NHOP (Part II). NHOP's marginal decomposability is a strength for diagnosis but a limitation for detecting joint distributional differences that are invisible in any single dimension. The bin count B must be chosen appropriately; the consistency theorem guarantees convergence but does not prescribe the optimal B for small samples. Permutation testing is computationally intensive at large n.

Circular oversampling (Parts III–IV). The PCA projection discards information above two dimensions; native-dimension mode avoids this but requires Von Mises–Fisher approximation whose accuracy degrades in very high dimensions ($d > 100$). Hyperparameter tuning (K, $\kappa_{\max}$, L, τ) adds complexity; the optimistic-bias disclosure (Chapter 8) acknowledges that reported numbers reflect tuned configurations. The benchmark is limited to 14 datasets; statistical power is insufficient ($\approx 30\%$) for rank-2 effect detection.

Causality study (Part V). The greedy removal evaluates only 50 random candidates per step; exhaustive search might yield different removal sequences. Synthetic dataset analogues replace unavailable KEEL/UCI files; qualitative conclusions are expected to generalise but exact numbers may differ. The study covers only IR ≤ 4; behavior at extreme imbalance (IR > 10) is not characterized. Only binary classification is studied.

17.5 Future Work

NHOP extensions.

- Develop a multivariate NHOP that detects joint dependencies beyond marginal overlap, potentially using copula-based decompositions.
- Derive a sample-size-dependent optimal bin count $B^*(n, m)$ to replace the empirical $B \in [20, 50]$ guideline.
- Apply the permutation test to distribution shift detection in streaming and production ML settings.

Circular oversampling.

- Extend evaluation to $N \geq 45$ datasets to achieve 80% statistical power for rank-2 effect detection.

- Investigate the native-dimension Von Mises–Fisher approximation at $d > 100$ and develop a more accurate high-dimensional direction sampler.
- Apply LRE-CO and GVM-CO to multi-class imbalance using one-vs-rest decomposition.
- Combine the circular framework with deep generative models (CTGAN, TVAE) to handle tabular data with complex dependencies.

Causal study extensions.

- Extend the degradation sweep to IR up to 50 to characterize the full performance curve at extreme imbalance [45].
- Validate on real KEEL/UCI datasets with matching (n, d) to confirm the synthetic-analogue findings [32].
- Investigate multi-class imbalance, where the interaction between ratios across classes adds complexity not addressed here [39].
- Design an oversampler that explicitly targets the irreducible information-loss gap using uncertainty quantification, prioritising synthesis in the most uncertain minority regions [44].
- Combine the NHOP-guided removal procedure with active learning: rather than randomly collecting data, prioritise collection of minority instances that most improve distributional coverage (highest marginal NHOP gain) [32].

Unified future direction. The most promising synthesis is a self-supervised distributional monitoring system that uses NHOP to continuously assess minority-class coverage in production data streams, triggering targeted data collection or adaptive oversampling when coverage drops below a threshold. GVM-CO provides the synthesis engine; the NHOP permutation test provides the trigger criterion; the causal study quantifies the performance recovery achievable at each coverage level. Together, they form a closed-loop framework for managing class imbalance in live systems.

Data Availability Statement

The datasets analyzed in Part IV (Chapter 8) are publicly available from the KEEL Data-Mining Repository [2] and the UCI Machine Learning Repository [29]. Synthetic dataset analogues used in Part V (Chapter 13) were generated using `sklearn.datasets.make_classification` with parameters reported in Table 13.1 and are fully reproducible from the random seed `random_state = 42`. The experimental code implementing all algorithms and the degradation-and-recovery protocol is available from the corresponding author upon reasonable request.

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